

Diffuse Bunching with Frictions: Theory and Estimation*

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Abstract

We incorporate a general model of frictions into the bunching-based elasticity estimator. The new estimator replaces bunching-window bounds with a “lumpiness parameter” and uses fewer parameters than the conventional approach while delivering additional economically-interpretable quantities such as the size of frictions and unobserved adjustment costs. The model matches rich observed bunching patterns such as sharp-peaked diffuse bunching around tax kinks and depressed density in notch-adjacent dominated regions. We apply the estimator to canonical settings including the U.S. EITC kink, and to administrative tax data from South Africa, where we uncover novel insights on unobserved VAT compliance costs and kink misperceptions. *JEL* codes: H30, J20, O12

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1 Introduction

The elasticity of taxable income is among the most central parameters in public economics, appearing as an input in many economic forecasts. It is a key statistic in models of optimal taxation, governing the optimal asymptotic top tax rate on high earners as well as the revenue-maximizing tax rate. One influential estimation strategy, proposed by Saez (2010), seeks to quantify this elasticity by measuring the amount of excess bunching mass in the income distribution around tax bracket thresholds where there is a change in the marginal tax rate (a “kink”) or tax level (a “notch”).¹

The standard bunching estimator computes the excess mass in an income density around a tax kink or notch relative to a counterfactual density that would arise if the tax were linear. This excess mass is converted into an elasticity using a formula which Saez (2010) derives using a frictionless model that predicts an atom of excess mass at the kink. In contrast, bunching seen in empirical distributions is typically diffuse, blurring the distinction between bunching induced by tax incentives and fluctuations in the underlying productivity distribution that would appear even under a linear tax. A now-conventional solution, originally proposed by Chetty et al. (2011), is to measure diffuse excess bunching mass as the difference between the observed density and a smooth function fitted to the observed density outside of a visually specified “bunching window” around the kink; Kleven and Waseem (2013) extend this approach to notches.² These approaches abstract from the frictions that produce diffusion, leaving open the possibility that they do not successfully recover the (potentially friction-affected) structural elasticity of taxable income. Moreover, by compressing density distortions around the tax bracket threshold into a single excess mass statistic, they risk discarding useful information about taxpayers’ behavioral responses to kinks and notches.

In this paper, we present a new bunching estimator, based on a flexible yet parsimonious model of income adjustment frictions, which we label the sparsity-based approach. Rather than choosing their income frictionlessly from a continuum, we model taxpayers that choose their utility-maximizing option from a sparse set of income opportunities generated by a Poisson process. We relate this structure to a range of microfoundations commonly used in the literature on income adjustment: search and matching environments in which opportunities arrive as Poisson events over a decision horizon (in the tradition of Mortensen

¹For examples, see Chetty et al. (2011), Kleven and Waseem (2013), Mortenson and Whitten (2020), Rees-Jones (2018) and others reviewed in Kleven (2016). Bunching estimation has also been applied to domains of retirement incentives (Manoli and Weber, 2016), mobile phone services (Grubb and Osborne, 2015), and educational test scores for both students (Diamond and Persson, 2016; Dee et al., 2019) and teachers (Brehm, Imberman and Lovenheim, 2017), marathon times (Allen et al., 2017), and home sales (Andersen et al., 2022).

²To account for non-zero density at “dominated incomes” above the notch, Kleven and Waseem (2013) assume a fraction of taxpayers are unresponsive, leaving unmodeled the source of diffusion in (one-sided) excess mass.

and Pissarides, 1994) generate exactly the same income distributions as the sparsity-based frictions model, while nonsequential search models in which taxpayers compare a fixed number of independent draws (Burdett and Judd, 1983; Wildenbeest, 2011) converge rapidly to this model as the number of draws grows. A benchmark case of this model—in which the opportunity intensity is homogeneous and indirect utility is locally quadratic—admits a tractable closed form, which we call the “uniform sparsity model.” We show that the single friction parameter in this model has a clear money-metric interpretation as the average distance between discrete income opportunities; the approximation error away from this tractable benchmark is negligible for economically realistic calibrations.

The uniform sparsity model generates precise predictions about the shape of bunching distortions around tax kinks and notches. Tax kinks induce tent-shaped (i.e., sharp-peaked and fat-tailed) symmetric bunching, as the many taxpayers who would exactly bunch under the frictionless model select their income opportunity nearest to the kink. Tax notches, in contrast, produce asymmetric bunching in the model, with diffuse excess mass on the desirable side of the notch threshold and a discontinuous drop to a depressed, yet still positive, density in dominated incomes on the undesirable side. Intuitively, although such incomes are dominated by earning at the threshold—where effort is lower and post-tax income is higher—some taxpayers’ next-best opportunities are sufficiently far away that an opportunity in the dominated region is preferred. We show that other natural models of frictions—including one-draw models in which taxpayers target an income but realize a noisy version of it, and fixed-fraction models in which some taxpayers are entirely unresponsive, as in Kleven and Waseem (2013)—produce distinct bunching patterns that are empirically distinguishable from the sparsity-based frictions model. In practice, the distinctive features of the uniform sparsity model appear to be present for many bunching patterns empirically documented around tax kinks (see Saez, 2010; Mortenson and Whitten, 2020) and notches (see Kopczuk and Munroe, 2015; Best and Kleven, 2018).

To demonstrate the usefulness of this new approach, we apply it to four different settings. Two of these settings reconsider classic results using data from prior literature, and two generate new insights from newly-utilized administrative data. In all four cases, our model is able to accurately match the empirically observed bunching distributions.

First, we re-analyze data on bunching at the Earned Income Tax Credit (EITC) kink studied in Saez (2010) and more recently in Mortenson and Whitten (2020). Our model predicts a peaked, tent-shaped pattern of bunching around this threshold that closely matches empirical data. This contrasts with the standard approach, which produces only a smoothed counterfactual density and an excess mass estimate, but no positive prediction about the pattern of that excess mass around the threshold. We show that accounting for income

adjustment frictions leads to estimated elasticities that are nearly twice as large as in previous work. These results are consistent with simulations reported in the Appendix, where we show that under a simulated model of sparsity-based frictions the standard estimators tend to substantially underestimate the true elasticities. This finding is consistent with recent work demonstrating that conventional bunching-based estimators produce lower elasticity estimates than other methods even when applied to the same data (Einav, Finkelstein and Schrimpf, 2017; He, Peng and Wang, 2021).

Second, we re-analyze data from a setting with a pure notch of known size: a fiscal stimulus coupon program in China that gave consumers a fixed subsidy amount for spending in excess of a specified amount, first studied by Ding et al. (2025). They estimate the marginal propensity to consume (MPC) in this setting using the notch estimator in Kleven and Waseem (2013). Our estimator is able to fit the observed bunching distributions well; in this setting the source of lumpiness has a particularly clear interpretation, as consumers must add or remove discrete items in order to adjust their spending around the subsidy threshold. This allows us to extract an additional economically meaningful parameter, namely, the average distance between consumption size opportunities.

Third, we analyze data on bunching around a compliance-cost notch of *unknown* size: value-added tax (VAT) registration thresholds. This application illustrates a strength of our method: the ability to recover an unobserved money-metric notch value from observed bunching behavior. For this application we use novel administrative tax data from South Africa, which (like many developing countries) relies more heavily on VAT and corporate income relative to income taxes. In addition to estimating the elasticity of taxable turnover due to VAT tax rate changes around the threshold, this allows us to estimate the compliance cost of registering for VAT. This compliance cost is a key parameter in the optimal setting of taxable turnover thresholds (see, e.g., Keen and Mintz (2004)).

Finally, we study a setting in which taxpayers appear to treat a statutory tax kink *as if* it were a notch, suggesting potential unobserved real or psychological costs corresponding to the tax bracket threshold, for example, due to confusion between average and marginal tax rates. Here too, we employ novel South African administrative tax data where we observe parallel “notch-like” bunching behavior at tax kinks among both individuals and small businesses. This finding is consistent with independent work by Gelber et al. (2025) documenting notch-like “left-bunching” in incomes around a kink produced by the U.S. Social Security earnings test. Our method allows us to estimate the extent to which behavior departs from what would be expected under a pure kink. Our analysis also uncovers an interesting source of heterogeneity: we find that small businesses with paid tax practitioners exhibit less bunching diffusion than other firms, and also less notch-like bunching behavior, consistent with practitioners helping

these businesses to understand and take advantage of tax incentives by fine-tuning their taxable incomes.

Relative to existing literature, the main contribution of our new model and estimator is to the theoretical and empirical work on bunching in the presence of frictions, discussed above. Our empirical findings also connect to two other areas of the public finance literature. First, our finding of notch-like behavior at statutory kinks contributes to the literature on behavioral frictions and misperceptions about the tax code. Rees-Jones (2018) uses bunching behavior around the threshold at which taxpayers face a net refund or balance due in order to quantify their degree of loss aversion. Rees-Jones and Taubinsky (2020) experimentally study misperceptions of the income tax code, finding that a large number of respondents “iron,” misinterpreting an average tax rate as the relevant marginal rate. Using exogenous variation in worker knowledge about a notch in the Norwegian income tax system, Kostøl and Myhre (2021) estimate that at least 30 percent of estimated optimization frictions are due to workers’ imperfect knowledge about the tax system. Outside the domain of taxes, Ito (2014) present evidence that consumers respond (at the margin) to average rather than marginal electricity prices.

Second, some of the empirical applications that we study add to the literature quantifying behavioral responses to taxation in developing economies. Particularly relevant is the subset of papers estimating the elasticity of corporate taxable income (e.g., Devereux, Liu and Loretz, 2014), which is an important parameter in developing economies, given their greater relative reliance on the corporate income tax base (Gordon and Li, 2009). For examples, see Best et al. (2015) in Pakistan, Bachas and Soto (2021) in Costa Rica, and Pillay (2021), Kemp (2019), Boonzaaier et al. (2019) and Lediga, Riedel and Strohmaier (2019) in our setting of South Africa.

The rest of the paper proceeds as follows. In Section 2, we show how sparsity-based frictions modify the frictionless model of bunching at tax kinks and notches, and we establish microfoundations connecting the model to canonical economic models from the labor and search literatures. Section 3 derives the bunching densities implied by the uniform sparsity model, establishes identification of its parameters, and describes our maximum likelihood estimator. Section 4 presents our four empirical applications. Section 5 concludes. Computational code for the uniform sparsity model is available at <https://github.com/bblockwood/DiffuseBunching>.

2 Model

2.1 Baseline bunching model with frictionless choice

Our starting point is a generalization of the canonical frictionless bunching estimator presented in Saez (2010) and illustrated in Figure 1. Taxpayers with heterogeneous productivities—for example, individuals earning labor income or firms earning profits—choose incomes z under an income tax $T(z)$. The top panel of Figure 1 plots after-tax income as a function of pre-tax income; for consistency with the bunching literature, we will refer to these as “consumption” (c) and “earnings” (z), respectively.

Different tax schedules cause taxpayers to select different incomes, giving rise to different income distributions. In the example plotted in Figure 1, the linear tax $T_0(z)$ results in the cumulative distribution function (CDF) of incomes labeled $H_0(z)$, while the linear tax $T_1(z)$ results in the CDF $H_1(z)$. Corresponding income densities $h_0(z)$ and $h_1(z)$ are shown in the bottom panel.

The horizontal distance between H_0 and H_1 quantifies the income response Δz to a reform from T_0 to T_1 at each quantile n of the distribution. Formally,

$$\Delta z(n) = H_1^{-1}(n) - H_0^{-1}(n). \quad (1)$$

This response remains well defined even if taxpayers reorder under the reform, for example due to heterogeneous elasticities or income adjustment frictions of the kind considered in the next section. Indeed, in the presence of such heterogeneity or adjustment frictions, the response Δz is generally the statistic of primary interest for policy makers, because it quantifies the fiscal effects from tax-induced behavioral distortions. $\Delta z(n)$ can be expressed in elasticity form as:³

$$e(n) = \frac{d \ln z}{d \ln(1 - T')} \approx \frac{\ln\left(\frac{z + \Delta z(n)}{z}\right)}{\ln\left(\frac{1 - T'_1}{1 - T'_0}\right)}. \quad (3)$$

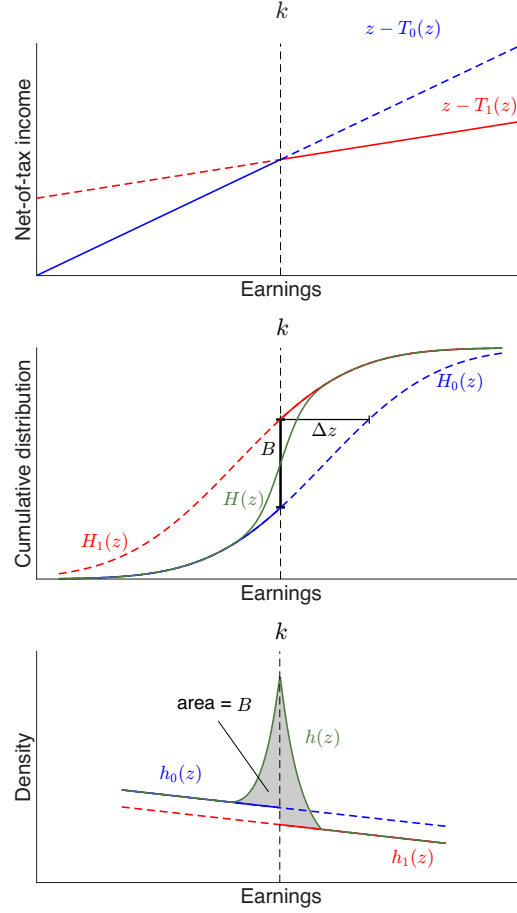
This being a static model, the CDFs H_0 and H_1 should be interpreted as steady-state distributions under different tax schedules, so that $e(n)$ represents the long-run response to a

³When Δz is small relative to z , $e(n)$ is approximately the compensated elasticity. Formally, it is a weighted average of the compensated and uncompensated elasticities of taxable income $e_c(n)$ and $e_u(n)$:

$$e(n) = \left(\frac{k}{z_0(n)}\right) e_c(n) + \left(1 - \frac{k}{z_0(n)}\right) e_u(n). \quad (2)$$

where $z_0(n) := H_0^{-1}(n)$ denotes the n th quantile of the income distribution under T_0 and k is the income at which T_0 and T_1 intersect. See Kleven (2016), footnote 5, where this is derived using the Slutsky equation.

Figure 1: Income Distributions and Densities Under a Kinked Tax Schedule



The top panel shows a taxpayer's budget constraints under two linear income tax functions, $T_0(z)$ and $T_1(z)$. These can construct a “kinked” tax schedule (with resulting budget constraint plotted by the solid line) consisting of $T_0(z)$ and $T_1(z)$ below and above income k , respectively. The middle panel plots the income CDFs $H_0(z)$ and $H_1(z)$ that arise under each linear tax function. Absent frictions, the income CDF under the kinked tax schedule is $H_0(z)$ below k and $H_1(z)$ above k , with a jump B at the threshold k . With frictions, the transition from $H_0(z)$ to $H_1(z)$ around the kink at k is gradual, as plotted by $H(z)$. The bottom panel plots the income densities $h_0(z)$, $h_1(z)$, and $h(z)$, corresponding to the CDFs $H_0(z)$, $H_1(z)$, and $H(z)$, respectively.

reform from T_0 to T_1 .⁴

The insight in Saez (2010) is that the observed income distribution under a piecewise-linear income tax,

$$T(z) := \begin{cases} T_0(z) & \text{if } z \leq k \\ T_1(z) & \text{if } z > k, \end{cases} \quad (4)$$

can provide information about the income response Δz to a reform from T_0 to T_1 . This logic

⁴In adjustment cost models, such as Gelber, Jones and Sacks (2020), agents either pay an adjustment cost and fully reoptimize or else do not respond at all. Such models produce predictions about bunching dynamics, e.g., bunching increases over time as more agents adjust. Our static model complements such models, predicting diffusion even in the static (steady state) equilibrium.

is illustrated in Figure 2. Assume that each income choice is the solution to a taxpayer's frictionless optimization problem, so that their selected income represents a point of tangency between their budget constraint and an upward-sloping indifference curve arising from a concave, three-times differentiable utility function $u(c, z)$ which trades off the utility of consumption against the disutility of exerting effort to earn income, as illustrated in the case of a linear tax in panel (a). Then, taxpayers who choose incomes below k under the kinked tax schedule T (see panel (b)) face the same local budget constraint as they would under the linear tax T_0 , while taxpayers earning above k under T face the same local budget constraint as they would under the linear tax T_1 . As a result, theory predicts that the observed income distribution under the kinked tax T will coincide with $H_0(z)$ at incomes below k and with $H_1(z)$ at incomes above k , with a discontinuous vertical jump at k of magnitude

$$B := H_1(k) - H_0(k), \quad (5)$$

corresponding to an atom of mass in the observed income density at k . Appendix A illustrates this logic in the presence of a tax notch, which also produces an atom of mass at the threshold k as well as a “hole” (density equal to zero) at incomes just above the bracket threshold which are utility-dominated by income k .

The vertical distance B between these CDFs is related to the horizontal distance Δz by the following equation:

$$H_0(k) + B = H_0(k + \Delta z). \quad (6)$$

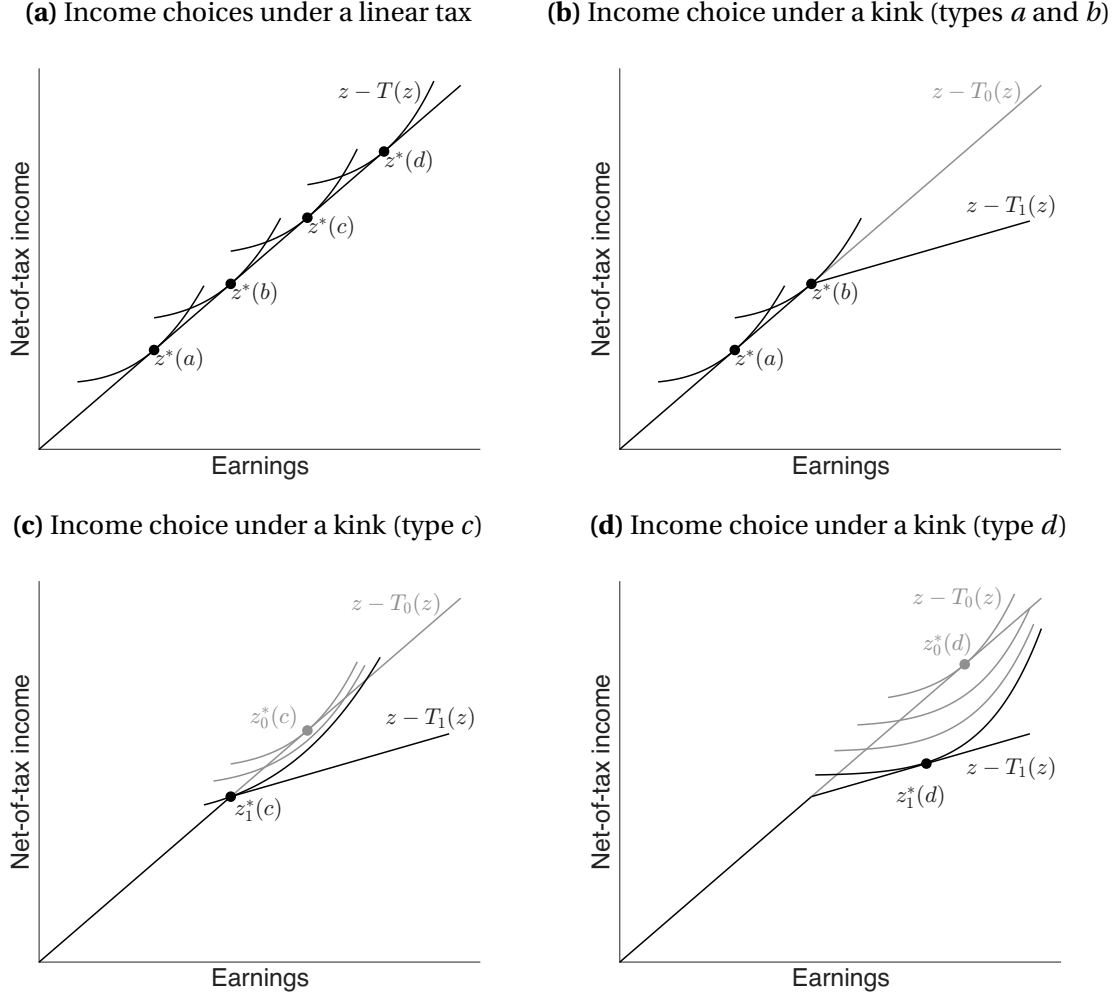
Strictly speaking, this condition identifies the income response only at a specific quantile of the income distribution—the quantile that earns k under the linear tax T_1 —but applications of the bunching estimator approach often assume that the elasticity is constant across incomes, either globally or in a region around the kink k .

Employing equation (6), we can use a Taylor expansion of $H_0(z)$ around $z = k$ to write B in terms of Δz and the local density $h_0(k)$ and its derivatives:

$$B = h_0(k)\Delta z + \frac{h'_0(k)}{2}\Delta z^2 + \frac{h''_0(k)}{3!}\Delta z^3 + \dots \quad (7)$$

If the density $h_0(z)$ is locally linear near k , so that higher-order terms are negligible, the

Figure 2: Frictionless bunching model with a progressive tax kink



This figure illustrates income choices around a tax kink under the conventional frictionless model. Panel (a) illustrates the optimal choice of income, z^* , for four selected types of taxpayers under a linear income tax. Panels (b), (c), and (d) illustrate the optimal choice for each type under a kinked income tax, where the tax changes from the $T_0(z)$ to $T_1(z)$ at the threshold k . Incomes z_0^* and z_1^* denote the optimal choice under the linear tax $T_0(z)$ and $T_1(z)$, respectively.

income response Δz can be written as an implicit function of B ,

$$\Delta z = \frac{B}{h_0(k) + \frac{h'_0(k)}{2} \Delta z} \approx \frac{B}{\left(\frac{h_0(k) + h_0(k + \Delta z)}{2} \right)}, \quad (8)$$

where the second equality uses $h_0(k + \Delta z) \approx h_0(k) + h'_0(k) \Delta z$ for small Δz . Substituting equation (8) into equation (3) yields the bunching estimator derived in Saez (2010).⁵

⁵To produce equation (5) in Saez (2010), rearrange equation (8) above to be $\Delta z = k \left[\left(\frac{1 - T'_1}{1 - T'_0} \right)^e - 1 \right]$ and substitute it into equation (3), noting that $h_1(k) = h_0(k + \Delta z) \left(\frac{1 - T'_0}{1 - T'_1} \right)^e$.

This frictionless model predicts an atom of excess bunching mass in the observed income density. In practice, observed bunching mass is typically diffuse, like the green density line $h(z)$ in Figure 1 (see, for example, Saez (2010)). Such diffuse bunching is usually interpreted as evidence of income-adjustment frictions. In their presence, the vertical distance B between the latent CDFs still quantifies the key empirical parameter of interest—the steady-state income response Δz —according to equation (7), but it can no longer be estimated by measuring a vertical discontinuity in the income CDF at k .

To estimate B in this setting, Saez (2010) proposes quantifying the excess mass in the observed density $h(z)$ around k , which is identical to B when integrated relative to the counterfactual densities $h_0(z)$ and $h_1(z)$:⁶

$$B \equiv \int_{-\infty}^k (h(z) - h_0(z)) dz + \int_k^{\infty} (h(z) - h_1(z)) dz. \quad (9)$$

This relationship is illustrated by the shaded region of excess mass in the bottom panel of Figure 1, which is equal to B .

This strategy requires knowledge of the counterfactual densities h_0 and h_1 , one or both of which is often unknown, and so much of the empirical bunching literature proposes ways to estimate the excess mass B from the observed density $h(z)$ alone. Chetty et al. (2011) influentially propose estimating B by integrating over the observed density $h(z)$ relative to an estimated counterfactual density obtained by fitting a flexible polynomial to $h(z)$ outside of a visually specified bunching window. Kleven and Waseem (2013) extend this approach to handle notches, wherein the tax *level* (and not just the marginal tax rate) jumps discontinuously at a threshold k . While these approaches implicitly accommodate adjustment frictions by allowing for diffuse bunching, they do not model the underlying optimization process.

As we show in the next section, a more explicit treatment of the diffusion process can improve the estimation of the bunching mass B in the presence of adjustment frictions by better distinguishing between the diffuse bunching that is “expected” around a kink and the underlying counterfactual densities. Moreover, by explicitly modeling these adjustment frictions and then fitting the observed shape of the diffuse mass around a kink, we can recover additional information about income-adjustment frictions, costs, and tax misperceptions that is discarded when the integrated bunching mass is reduced to a scalar measure of excess mass.

⁶This identity follows from the fact that $B = H_1(k) - H_0(k) = [H(k) - H_0(k)] + [H_1(k) - H(k)]$, where the bracketed terms are equal to the two integrals in equation (9), respectively.

2.2 A model of sparsity-based frictions

We introduce a simple modification to the frictionless model of Saez (2010): rather than selecting incomes from a continuum, taxpayers choose their preferred income from a sparse set of opportunities.

Formally, each taxpayer realizes a random *income opportunity set* from an *income opportunity process* and selects their utility-maximizing income from this set. We assume that income opportunities are drawn independently, so that the income opportunity process is a Poisson point process, with a taxpayer-type-specific intensity function $\lambda_n(x)$, so that the expected number of income opportunities available to a type- n taxpayer in a given interval $[\underline{z}, \bar{z}]$ is $\int_{\underline{z}}^{\bar{z}} \lambda_n(x) dx$. The intensity function $\lambda_n(x)$ is assumed to be continuously differentiable and strictly positive around the taxpayer's frictionless target income $z^*(n)$. We further assume that opportunities arrive with positive intensity across the income line, so that each taxpayer's opportunity set is almost surely nonempty—equivalently, the expected number of opportunities $\Lambda_n = \int \lambda_n(x) dx$ is infinite—as holds under the homogeneous-intensity benchmark introduced below (Assumption 1). This is what allows the type-conditional density below to be written without an explicit normalization.

Unlike in the frictionless model, here taxpayers of the same type (productivity, elasticity, etc.) will choose different incomes due to their differing opportunity sets. The *type-conditional density* of incomes among taxpayers of type n is characterized by the following proposition.

Proposition 1. *The type-conditional density of incomes at $\tilde{z}$ among taxpayers of type n with utility $u(c, z|n)$ under income tax $T(z)$ is given by⁷*

$$g(\tilde{z}|n) = \underbrace{\lambda_n(\tilde{z})}_{\text{intensity at } \tilde{z}} \times \underbrace{\exp \left[- \int_{z \in \Theta(\tilde{z}|n)} \lambda_n(z) dz \right]}_{\text{Pr(no opportunities in } \Theta(\tilde{z}|n))}, \quad (10)$$

where

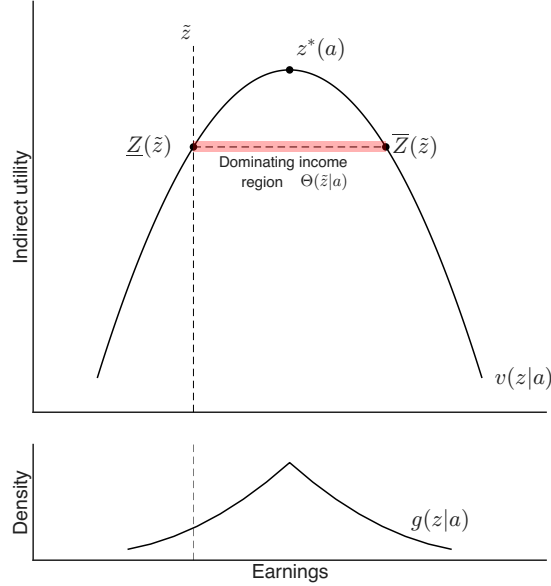
$$\Theta(\tilde{z}|n) := \left\{ z \mid u(z - T(z), z|n) \geq u(\tilde{z} - T(\tilde{z}), \tilde{z}|n) \right\} \quad (11)$$

denotes the upper contour set of incomes that utility-dominate $\tilde{z}$ for a taxpayer of type n .

Proof. Let $\mathcal{Z}$ be an income opportunity set realized from a point process with intensity $\lambda_n(\cdot)$. Consider a small income interval $[\tilde{z}, \tilde{z} + d\tilde{z}]$. The probability that $\mathcal{Z}$ contains an opportunity in

⁷Equation (10) is the density of chosen income among type- n taxpayers whose opportunity set is nonempty. Integrating it over $\tilde{z}$ gives $1 - e^{-\Lambda_n}$, the probability of drawing at least one opportunity, where $\Lambda_n = \int \lambda_n(z) dz$ is the expected number of opportunities; the density among taxpayers with a nonempty opportunity set is obtained by dividing by this factor. Under our maintained assumption that the opportunity set is almost surely nonempty ($\Lambda_n = \infty$), this normalization equals one and (10) is itself a proper density, so we omit it throughout. The finite-intensity case, in which the normalization $1/(1 - e^{-\Lambda_n})$ is material, is treated explicitly in the proof of Corollary 2 (Appendix B).

Figure 3: Type-conditional income density under a linear tax (type a)



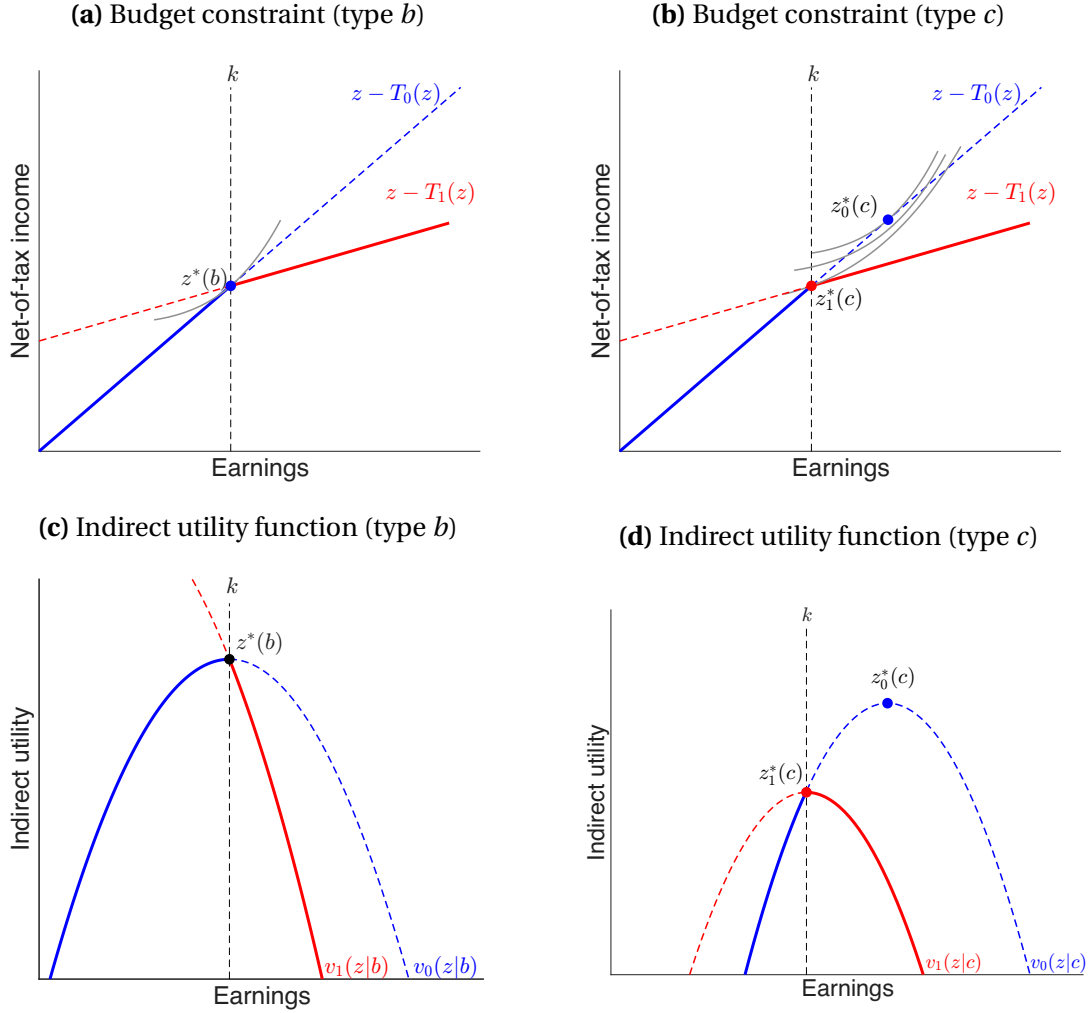
This figure illustrates the calculation of the type-conditional income density in the uniform sparsity model among a -type agents at a particular income level $\tilde{z}$, under a locally linear income tax. The top plot displays the taxpayer's indirect utility function over incomes. An agent who has $\tilde{z}$ in their income opportunity set will select this income iff they do not have some other income opportunity in the shaded “dominating region.” The type-conditional density $g(\tilde{z}|a)$ is equal to this conditional probability multiplied by the probability of drawing $\tilde{z}$.

this interval is $\lambda_n(\tilde{z}) d\tilde{z} + o(d\tilde{z})$. The probability that $\mathcal{Z}$ contains no opportunities in $\Theta(\tilde{z}|n)$ is $\exp\left[-\int_{z \in \Theta(\tilde{z}|n)} \lambda_n(z) dz\right]$. Independence of disjoint sets for a Poisson process lets us multiply these terms; dividing by $d\tilde{z}$ and taking the limit as $d\tilde{z} \rightarrow 0$ yields (10). $\square$

The logic behind this result is illustrated in Figure 3, which plots the indirect utility function $v(z|a)$ over income for a taxpayer of type a (see Figure 2) who faces a locally linear income tax. The type-conditional income density at income $\tilde{z}$ is equal to the intensity of having an opportunity at $\tilde{z}$ multiplied by the probability that no other opportunities fall in the shaded region of incomes $\Theta(\tilde{z}|n)$ that provide higher utility than $\tilde{z}$. The type-conditional density reaches its maximum at the taxpayer's frictionless target income $z^*(n)$, at which point $\Theta(z|n)$ becomes a singleton containing only $z^*(n)$ and the second term on the right-hand side of equation (10) is equal to 1.

Figures 4 and 5 illustrate how this logic extends to a tax kink. Figures 4(a) and 4(b) plot the budget constraints and indifference curves for types b and c from Figure 2 under the kinked tax schedule $T(z)$. Figures 4(c) and 4(d) plot the resulting indirect utility functions for each type, which are constructed by retaining the relevant segments of the indirect utility functions that would arise under each of the linear tax schedules $T_0(z)$ and $T_1(z)$. Figure 5 illustrates implications for the type-conditional income densities $g(z|b)$ and $g(z|c)$. The formula for the

Figure 4: Utility from income choices around a tax kink

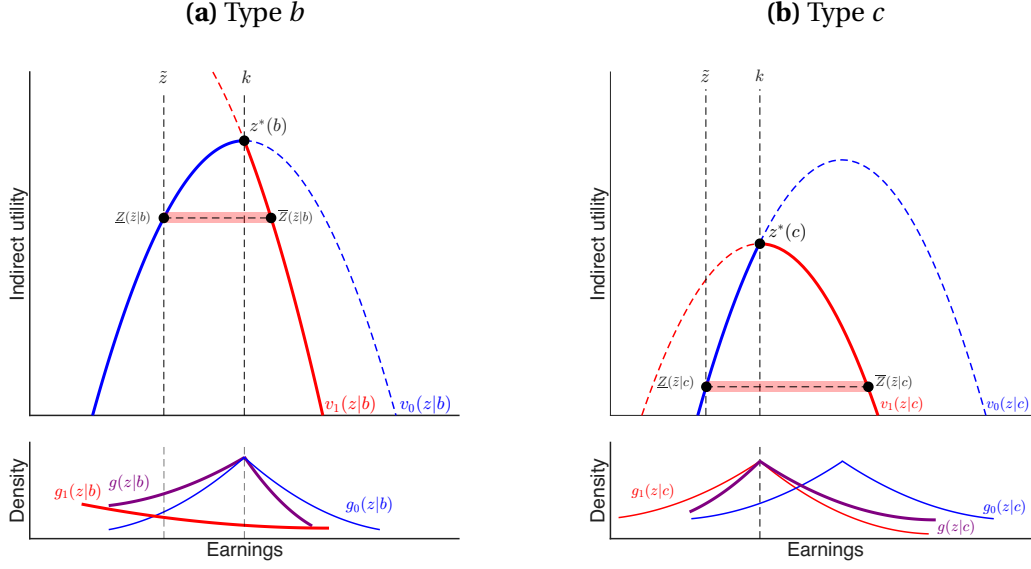


Panels (a) and (c) illustrate the construction of the indirect utility function around a progressive tax kink for the marginal non-buncher (type b). Panel (a) shows the taxpayer's budget constraint, plotted as a solid line, where $T_0(z)$ and $T_1(z)$ are the linear income taxes below and above the bracket threshold k , respectively. Panel (c) plots the indirect utility functions $v_0(z|b)$ and $v_1(z|b)$, which would be obtained if the linear tax functions $T_0(z)$ or $T_1(z)$ applied across all incomes. Type b 's indirect utility function under the kinked tax schedule, plotted as a solid line, is given by $v_0(z|b)$ below k and $v_1(z|b)$ above k . Panels (b) and (d) show analogous illustrations for the marginal buncher (type c). This taxpayer's optimal frictionless income choices under the linear taxes $T_0(z)$ and $T_1(z)$ are denoted $z_0^*(c)$ and $z_1^*(c)$.

type-conditional density in Proposition 1 carries through: the distortionary effect of the tax kink operates entirely through its impact on the dominating income regions $\Theta(z|n)$, which are again plotted as shaded regions in Figure 5. Because the kink point k maximizes indirect utility for both types b and c (and all types in between), the type-conditional income density reaches its maximum at k for those types.

Appendix A illustrates this logic in the context of a notch. It highlights an important implication of this model: some taxpayers choose an income opportunity in the “dominated

Figure 5: Type-conditional income density around a kink



This figure illustrates how the indirect utility functions from Figure 4 are used to compute the type-conditional income densities. The panels show the calculation for the marginal non-buncher (panel (a)) and the marginal buncher (panel (b)). Each panel illustrates the calculation of the type-conditional income density $g(z|n)$ at a (different) income $\bar{z}$. We first identify the range of incomes that utility-dominate $\bar{z}$ for each taxpayer, corresponding to the horizontal dashed line, and we proceed as in Figure 3. The type-conditional densities are plotted in purple. For reference, the type-conditional density under the counterfactual linear taxes $T_0(z)$ and $T_1(z)$ are plotted in blue and in red, respectively.

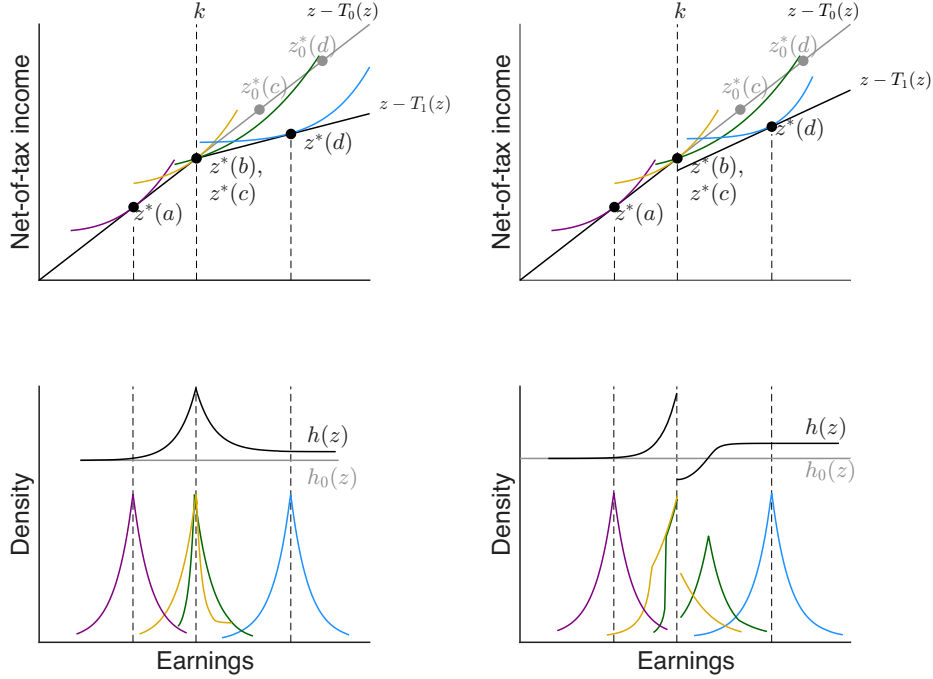
region” just above the notch because all of their other income opportunities happen to be more distant and yield lower utility. This implication contrasts with the frictionless model, in which the density is zero above the notch because those incomes are strictly dominated by choosing to earn the threshold income k .

Because types are unobservable, type-conditional densities also are not directly observable. Instead, we observe the income density $h(z)$ which is the sum of the type-conditional densities across all types. The observed income density $h(z)$ is the aggregation of the type-conditional densities at a given income z across all taxpayer types:

$$h(z) = \int_n g(z|n) f(n) dn. \quad (12)$$

Because each type-conditional density integrates to one—every taxpayer almost surely draws an opportunity—the aggregate $h(z)$ is itself a proper density. If the type density is smooth and the tax schedule is linear, then the type-conditional density effectively acts as a smoothing filter through which the type density is passed, producing an observed density that is also smooth. In the presence of a kink, the observed income density around the kink arises as the aggregation of the many type-conditional densities like those in Figure 5, each of which has

Figure 6: Aggregating type-conditional densities into an observable income density



The top left panel of this figure shows the optimal frictionless income choice for agents of types a , b , c , and d in the presence of a kink at k , with each type's maximal indifference curve plotted in a different color. The panel below it illustrates the type-conditional income densities in corresponding colors, plotted on the same horizontal axis. Summing across the type-conditional densities of these and the continuum of intervening types produces the observed income density, $h(z)$, which exhibits diffuse bunching around the bracket threshold. The counterfactual income density $h_0(z)$, which would be observed under the linear tax function $T_0(z)$, is plotted in gray for reference. The top right panel shows the analogous construction for a tax notch, with the resulting observed income density plotted below.

their own maximum at the kink but exhibits diffusion around the kink, producing diffusion in the aggregated income density, as illustrated in Figure 6. Similarly, in the presence of a notch, the aggregation of the many type-conditional densities like those in Figure 5 produces a distinctive pattern: diffuse excess mass below the notch, a local maximum precisely at the notch, and a discontinuous drop in density to the lower (yet still positive) density above the notch. We explore the quantitative behavior of these patterns in our estimation section below.

2.3 Microfoundations for sparsity-based frictions

In this section we relate the sparsity-based frictions model to a range of existing economic models of search, job choice, and adjustment frictions. We do not claim full structural equivalence between any of these microfoundation models and the sparsity-based frictions model, because any particular microfoundation may have richer primitives and imply additional observables beyond the income margin. Instead, we ask a narrower question:

when does a given microfounded model induce the same observed *income distributions* as the sparsity-based frictions model? To formalize this question, we begin with a definition of *income-distribution equivalence*.

Definition 1 (Income-distribution equivalence). *A model is income-distribution equivalent to the sparsity-based frictions model if, for any tax function T , there exists a utility function $u(c, z \mid n)$, a type distribution $f(n)$, and an income opportunity process with type-specific intensity function $\lambda_n(z)$ such that the model induces the same observed income density*

$$h(z) = \int_n g(z \mid n) f(n) dn$$

as the sparsity-based frictions model for all incomes z in the support of interest.

This definition allows us to distinguish between three cases. First, a model might be *exactly* income-distribution equivalent to the sparsity-based frictions model: although it may be richer in other dimensions, it induces exactly the same distribution of chosen incomes under a given tax function for an appropriate choice of utility function, type distribution, and income opportunity process. Second, a model may not be exactly income-distribution equivalent, but it may nevertheless be *asymptotically* income-distribution equivalent, meaning that it induces an income distribution that converges to one produced by a sparsity-based frictions model under a rare-events limit. We present microfoundations that fall into each of the above two categories. Finally, we present a number of models that are not income-distribution equivalent to the sparsity-based frictions model (either exactly or asymptotically). We produce simulated income distributions in each case, demonstrating which features of bunching patterns are consistent with the sparsity-based frictions model equivalents and which are not.

2.3.1 Microfoundation with exact income-distribution equivalence

Poisson-arrival search with recall: A natural motivation comes from search models in which opportunities arrive stochastically and workers choose from among the offers they receive. In the Mortensen and Pissarides (1994) tradition, job-specific shocks arrive as a Poisson jump process, with each new shock drawn from a common distribution. In directed-search models with multiple applications, workers may receive more than one offer and then choose among them. For example, in Albrecht, Gautier and Vroman (2006), each worker submits a fixed finite number of applications, vacancies may make offers to selected applicants, and workers who receive multiple offers choose among them. A search environment in which opportunities arrive as independent Poisson events over a fixed decision horizon, after which the worker chooses the utility-maximizing realized option, is income-distribution equivalent to the sparsity-based frictions model under the following conditions.

Corollary 1 (Search models with Poisson offer arrival). *Consider a worker of type n in a search environment over a fixed decision horizon. Suppose that:*

1. *job meetings, applications, or offer opportunities arrive as a Poisson process with rate α over a decision horizon $\mathcal{T}_n$, and each arrival generates an earnings opportunity drawn independently from density v_n ;*
2. *each realized opportunity implies an earnings level z ;*
3. *conditional on type n , the worker's utility from any realized opportunity depends only on after-tax earnings and pre-tax earnings, so that utility can be written as*

$$u(z - T(z), z \mid n);$$

4. *at the end of the decision horizon, the worker chooses the utility-maximizing realized opportunity.*

Then the worker's realized earnings opportunities form a Poisson opportunity process on earnings space with intensity function

$$\lambda_n(z) = \alpha |\mathcal{T}_n| v_n(z),$$

where $|\mathcal{T}_n|$ denotes the length of the decision horizon. The worker's chosen earnings are therefore distributed exactly as in Proposition 1. Aggregating across workers, the search environment is income-distribution equivalent to the sparsity-based frictions model.

Proof sketch. Over the decision horizon $\mathcal{T}_n$, the worker receives a random set of realized opportunities generated by a Poisson process with arrival rate α , where each arrival independently generates an earnings draw from density v_n . The induced set of earnings opportunities is therefore a Poisson opportunity process on earnings space with intensity function $\lambda_n(z) = \alpha |\mathcal{T}_n| v_n(z)$. Conditional on type n , the worker then chooses the utility maximizing element of that realized earnings-opportunity set. Proposition 1 therefore applies directly. $\square$

2.3.2 Microfoundations with asymptotic income-distribution equivalence

Nonsequential product search: In product search models, consumers sample product and/or price offers, learning the price p and the attributes of each sampled product (Burdett and Judd, 1983; Wolinsky, 1986). We consider the case of nonsequential search, in which consumers sample some number of offers M and then choose the utility-maximizing opportunity from their sampled set, similar to the setting in Wildenbeest (2011). To extend such models to

settings with job offers, we allow price to be negative so that a job can be interpreted as a product with pre-tax compensation $z = -p$. One special case of this model is considered in Saez (1999), the working paper that preceded Saez (2010), in which taxpayers draw a menu of labor opportunities from a uniform distribution centered around their frictionless target.

At finite M , this environment is not exactly the sparsity-based frictions model, because the number of sampled opportunities is fixed rather than Poisson. A natural comparison is therefore to a *Poissonized* sparsity-based frictions version of this model whose opportunity intensity equals the expected number of nonsequential draws times the primitive offer density. The next corollary shows that, for each type, the exact- M nonsequential search density converges in total variation to the corresponding Poissonized model's density as $M \rightarrow \infty$. Consequently, the observable earnings density also converges.

Corollary 2 (Nonsequential search). *Suppose each type- n taxpayer draws M independent earnings opportunities from a continuous density v_n and selects the one maximizing $u_n(z) := u(z - T(z), z \mid n)$, where $u_n(Z)$ is continuously distributed under $Z \sim v_n$ so that ties occur with probability zero. Compare this to the sparsity-based frictions model with the same utility and opportunity intensity $\lambda_{n,M}(z) = Mv_n(z)$. Then for every type n the nonsequential-search type-conditional density converges in total variation to its sparsity-based counterpart as $M \rightarrow \infty$, and consequently, for any fixed population type density $f(n)$, so does the observable income density. The nonsequential search model is therefore asymptotically income-distribution equivalent to the sparsity-based frictions model.*

Proof. Appendix B derives the closed forms of the two type-conditional densities, bounds their total-variation distance, and shows it vanishes as $M \rightarrow \infty$. $\square$

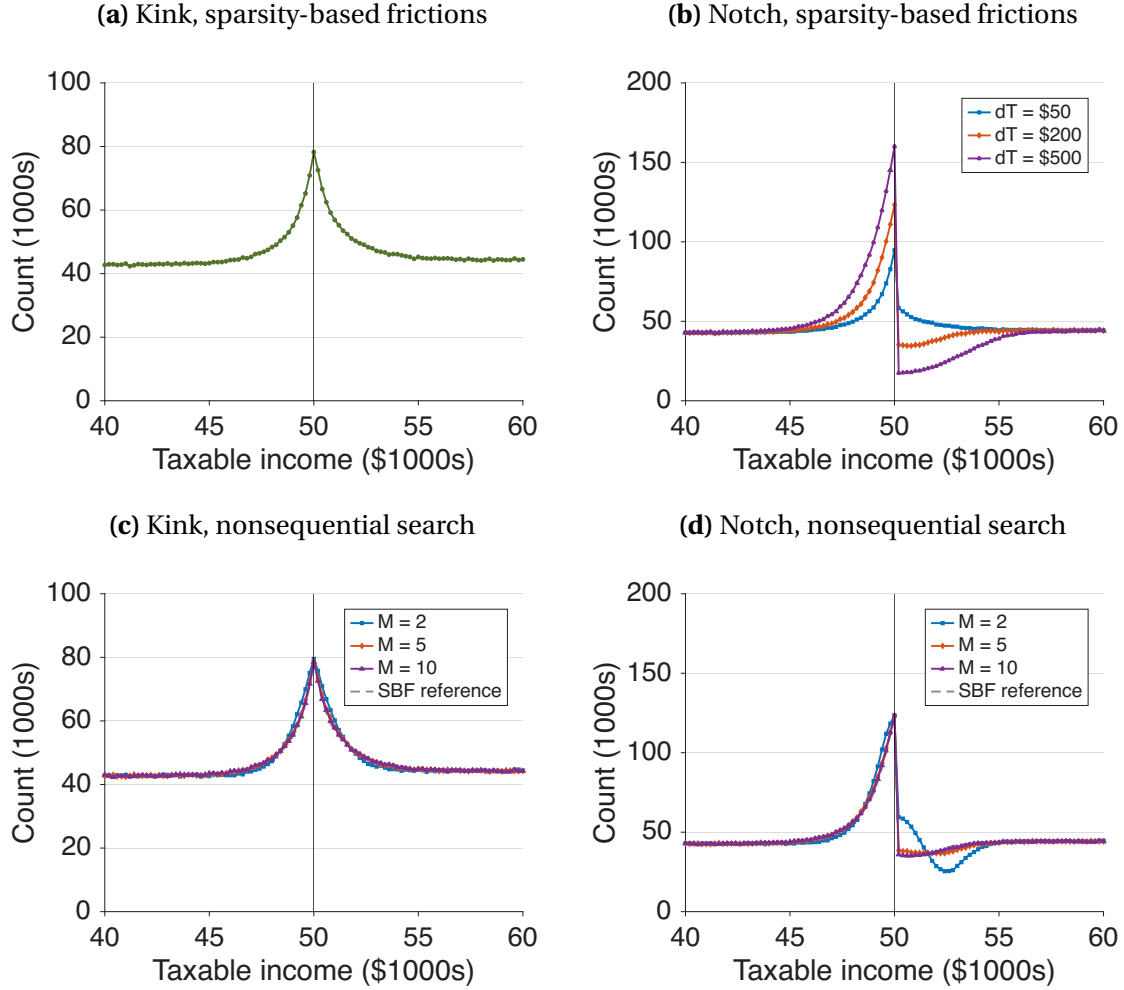
2.3.3 Simulations

Figure 7 presents simulated bunching patterns arising from the sparsity-based frictions model. These simulations assume a quasilinear, isoelastic utility function as in Saez (2010), in which types differ only in their income-generating ability (productivity) n :

$$u(c, z \mid n) = c - \frac{n}{1 + 1/e} \left(\frac{z}{n} \right)^{1+1/e}.$$

We set the elasticity of taxable income to $e = 0.3$ in our baseline specification, and the tax is piecewise linear with a marginal tax rate that increases from 0.1 to 0.2 at the threshold $k = \$50,000$. The intensity of income opportunities is assumed to be constant, $\lambda_n(z) = \bar{\lambda}$, with $1/\bar{\lambda} = \$2,000$ in our baseline so that income opportunities are \$2,000 apart in expectation. The underlying type density $f(n)$ is assumed to be locally constant.

Figure 7: Bunching patterns under sparsity-based frictions and nonsequential search



This figure plots simulated income densities around a bracket threshold under sparsity-based frictions (SBF) and under nonsequential search with normal draws. In all panels, the marginal tax rate rises from $t_0 = 0.1$ to $t_1 = 0.2$ at the threshold $k = \$50,000$, the elasticity of taxable income is $e = 0.3$, and the type density $f(n)$ is locally constant. Each simulation uses a population of $N = 2 \times 10^7$ taxpayers and displays \$200 income bins. Panel (a) plots the SBF density around a kink, with opportunity intensity $\bar{\lambda} = 1/\$2,000$. Panel (b) plots SBF densities around a notch for three notch sizes $dT \in \{\$50, \$200, \$500\}$. Panel (c) plots densities at the kink under a nonsequential search model in which each taxpayer draws M income opportunities from a normal distribution centered at their target income, with variance chosen so that the density of draws at the target equals $\bar{\lambda}$, for $M \in \{2, 5, 10\}$. Panel (d) is the analogue of panel (c) around a \$200 notch.

Panel 7(a) displays the observed income distribution in our baseline specification. Simulated incomes produce a distinctive “tent-shaped” pattern of bunching around the tax kink, generated by the accumulation of the diffuse type-conditional income densities of taxpayers whose frictionless target income is $z^*(n) = k$, as in Figure 6. As we will show in Section 3.1, this shape is approximated by a double exponential (Laplace) distribution.

Panel 7(b) displays bunching patterns around a notch. In these simulations, the tax *level* increases by an amount ΔT when a taxpayer’s income surpasses k . We plot simulations with

$\Delta T = \$50$, $\Delta T = \$200$, and $\Delta T = \$500$ to illustrate the effects of different notch sizes on the observed distribution. This panel illustrates the distinctive implications of the sparsity-based frictions model for bunching behavior around notches: on the desirable side of the notch—i.e., in this case, at incomes below the notch—there is excess mass with a distinctive exponential shape similar to the bunching pattern on both sides of the kink in panel (a). The bunching mass reaches its peak exactly at the notch threshold, then drops discontinuously to the density on the undesirable side of the notch. Unlike in the frictionless model, however, the observed density remains positive even in the “dominated region” where income-consumption bundles are dominated by the threshold income value k , reflecting that in the sparsity-based frictions model some agents’ best available income opportunity happens to be in that region.

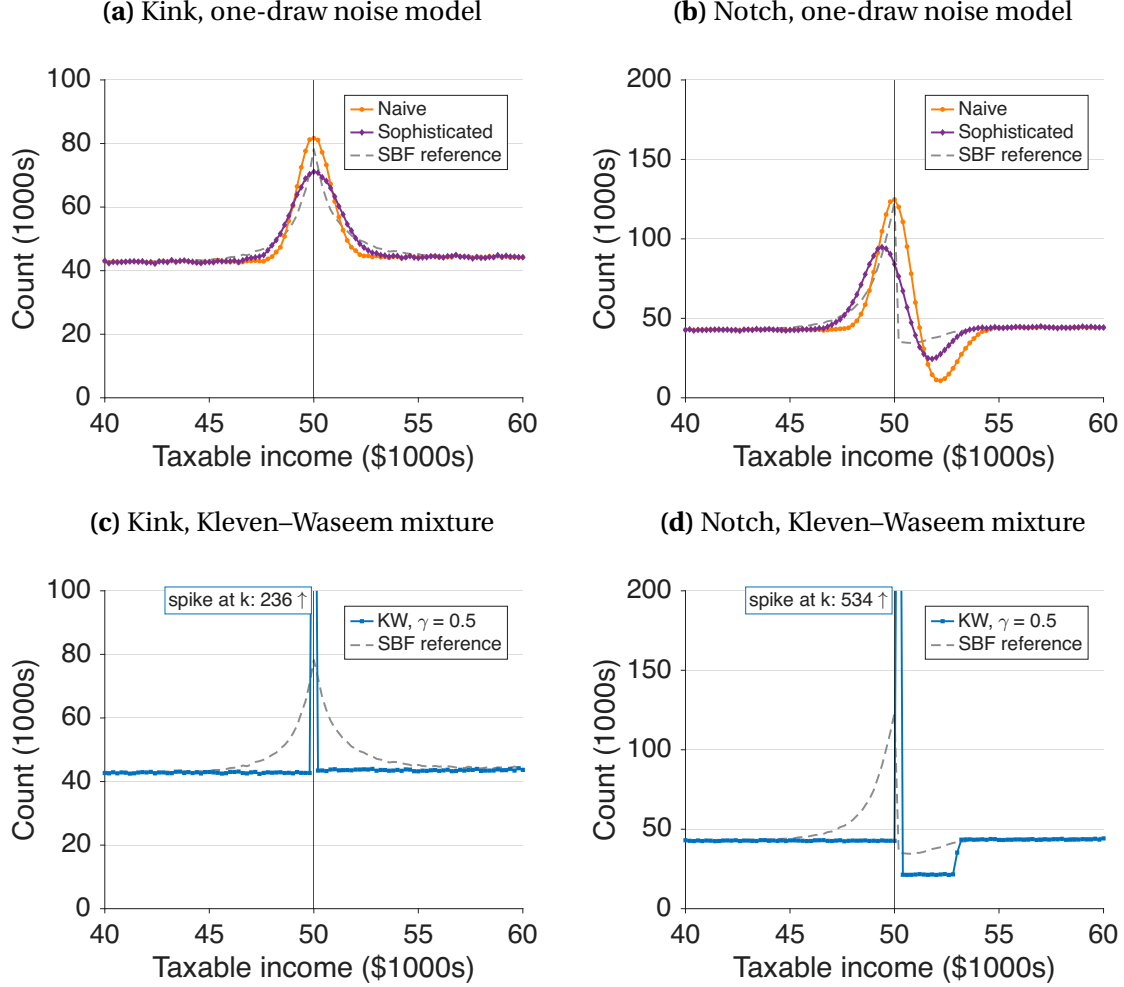
Panel 7(c) illustrates the convergence behavior of the nonsequential search microfoundation described above in Section 2.3.2. Here we assume that each type draws M incomes from a normal distribution centered around their target income $z^*(n)$, with the variance of the normal distribution set so that the resulting opportunity intensity around $z^*(n)$ is the same as in the sparsity-based frictions simulations in panel (a). From Corollary 2 we know that this density converges to the one displayed in panel (a) as $M \rightarrow \infty$. Here we illustrate that the convergence happens very rapidly. We plot simulations with $M = 2$, $M = 5$, and $M = 10$, demonstrating that even for these relatively low numbers of income opportunities the income distribution of the nonsequential search model is well approximated by the sparsity-based frictions model. Panel 7(d) displays analogous simulations of the nonsequential search model in the presence of a \$200 notch, showing that the same rapid convergence to the sparsity-based frictions benchmark—here to the $\Delta T = \$200$ curve from panel 7(b)—carries over to the notch setting.

2.3.4 Models without income-distribution equivalence

Some natural models of frictional bunching which have been used in the bunching literature are not income-distribution equivalent to the sparsity-based frictions model, even asymptotically. Here we consider two prominent examples: a one-draw noise model in which agents target an income but realize a noisy version, and a model in the spirit of Kleven and Waseem (2013) in which a fraction of agents are entirely unresponsive to the kink or notch. Figure 8 displays simulated bunching patterns for both models, with simulation parameters chosen identically to those underlying Figure 7 so that the qualitative differences in bunching shapes are attributable to the model rather than to calibration choices.

One-draw with noise model: In this model, originally proposed by Saez (1999) and also studied by Gelber et al. (2025), each taxpayer of type n chooses a target income $\hat{z}(n)$ and realizes income $z = \hat{z}(n) + \varepsilon$, where ε is iid noise drawn from $\mathcal{N}(0, \sigma^2)$. We consider two

Figure 8: Bunching patterns under non-equivalent friction models



This figure plots simulated income densities around a bracket threshold under two friction models that are not income-distribution equivalent to the sparsity-based frictions (SBF) model. Tax, preference, and type-density parameters, as well as the simulated population of $N = 2 \times 10^7$ taxpayers, are identical to those in Figure 7. Each panel displays the corresponding SBF density from Figure 7 for reference. Panels (a) and (b) plot the one-draw noise model, in which each agent of type n aims for their frictionless target income $z^*(n)$ and instead realizes a noisy version $z^*(n) + \varepsilon$ with ε iid and drawn from $\mathcal{N}(0, \sigma^2)$ with $\sigma = \mu / \sqrt{2\pi} \approx \798 , so that the noise density at the target equals $\bar{\lambda}$. The sophisticated variant maximizes expected utility over the noise distribution. Panel (b) uses a \$200 notch. Panels (c) and (d) plot the Kleven–Waseem model, in which a fraction $\gamma = 0.5$ of agents are entirely unresponsive to the kink/notch (choosing income under the linear schedule T_0) and the remaining $1 - \gamma$ are fully frictionless responders. Because the responsive (frictionless) bunchers in this model produce a point mass at k that dwarfs the smooth density components, the spike at k in panels (c) and (d) is clipped on the y-axis and labeled with its bin count.

variants. A *naive* agent ignores the noise and targets the frictionless optimum $\hat{z}(n) = z^*(n)$ given the actual tax schedule; a *sophisticated* agent anticipates the noise distribution and chooses

$$\hat{z}(n) = \arg \max_{\hat{z}} \mathbb{E}_{\varepsilon} [u(\hat{z} + \varepsilon - T(\hat{z} + \varepsilon), \hat{z} + \varepsilon | n)].$$

Figure 8 panels (a) and (b) plot simulated densities for both variants under a kink and a \$200 notch, with $\sigma = \mu/\sqrt{2\pi}$ chosen so that the noise density at the target equals the sparsity-based frictions intensity $\bar{\lambda}$ used in Figure 7.

The resulting densities differ in informative ways from the sparsity-based frictions density, which is also plotted in panels (a) and (b) for reference. In the case of a kink (panel a), the observed bunching around k inherits the qualitative shape of the noise distribution, in this case producing a bunching pattern that is approximately normal and differentiable at k .⁸ This contrasts with the sharp-peaked tent-shaped (double exponential) bunching pattern often observed in practice which—as we show in the next section—is a general feature of the sparsity-based frictions model. In a context where the income noise happened to have a Laplace distribution centered at zero, the two models might be indiscernible around a kink.

The difference between the models is more pronounced in the case of a notch (panel b), because the observed income distribution is revealed to be *smooth and continuous* across the notch threshold k , rather than exhibiting the discontinuous drop at k that is evident in the sparsity-based frictions model. The intuition for this difference is straightforward: when income targets are disturbed by a noise term with a continuously differentiable distribution, discontinuities in the target income distribution are smoothed out. Thus an empirical discontinuity at k is a key signature suggesting bunching patterns may be better represented by the sparsity-based frictions model than by the “one draw with noise” model.

Kleven/Waseem fixed-fraction unresponsive model: Following Kleven and Waseem (2013), we also consider a two-type mixture in which a fraction γ of agents are entirely unresponsive to the kink or notch—they choose income as if facing the linear schedule T_0 with marginal rate t_0 everywhere—while the remaining $1 - \gamma$ are frictionless responders to the actual schedule T . Panels 8(c) and 8(d) plot simulated densities for $\gamma = 0.5$. In the case of a kink, this model produces bunching mass qualitatively similar to the frictionless specification, with a point mass located at k . In the case of a notch, this model rationalizes positive density in the dominated region of incomes above the notch, which contains only the γ fraction of unresponsive types

⁸In the naive model one-draw noise model, the bunching mass exactly inherits the distribution of noise (here normally distributed) as would-be bunchers choose k as their target and realize $k + \varepsilon$. In the sophisticated version, the spread of the bunching mass is wider, because taxpayer types near the bottom (or top) of the frictionless bunchers choose to reduce their expected losses from noise realization by reducing (or raising) their targeted income relative to k .

whose frictionless target happens to land there. This model does not predict diffusion in the bunching mass, instead generating a sharp point mass at k from the responsive bunchers, plus a smooth γ -weighted counterfactual density across the entire income window from the unresponsive agents. Thus, empirical diffusion in the bunching mass to the left of a notch is a signature suggestive of the sparsity-based frictions model relative to the fixed-fraction unresponsive model.

Together, these results highlight that although the sparsity-based frictions model does not approximate all natural models of microfoundations for income frictions, it does produce distinctive empirical patterns that are qualitatively different from other models, suggesting that the pattern of bunching can be used as diagnostic choice for model selection. In Section 4 we consider several empirical applications that appear to exhibit the signatures of the sparsity-based frictions model. First, however, we turn to the identification and estimation of the sparsity-based frictions model's parameters.

3 Identification and Estimation

The previous section showed how to generate income densities from known parameters under the sparsity-based frictions model. Although that tells us whether the sparsity-based frictions model is qualitatively appropriate for a given setting, it does not show how to recover its parameters from the income distribution alone. We turn to that problem now.

We first show a useful result, namely, that under a linear tax, type-conditional densities are locally Laplace around each type's target income. This local approximation is globally exact if the opportunity intensity $\lambda_n(z)$ is uniform and indirect utility is quadratic, and we can quantify the approximation error away from this benchmark, showing that it is quantitatively negligible for realistic applications like the ones presented in our empirical section. If, moreover, the local intensity and the local elasticity of taxable income are homogeneous across types, then the observed income distribution takes a tractable analytic form. We call this case the “uniform sparsity model,” which serves as our benchmark specification for estimation. We then prove two additional results about bunching at kinks and notches under the uniform sparsity model. First, given a known counterfactual income distribution under a linear tax, the (uniform) opportunity intensity and the elasticity of taxable income are separately identified. Second, if the counterfactual density is not known, it can be estimated under an identifying assumption that it is sufficiently regular across the region of the kink or notch. Finally, we lay out our procedure for recovering model parameters and standard errors using maximum likelihood methods.

3.1 Laplace behavior of the type-conditional density

3.1.1 Local Laplace behavior

The following lemma formalizes the way in which the type-conditional income density converges to a Laplace distribution locally around a type's frictionless target income $z^*(n)$, which we sometimes write simply as z^* when only one type is involved to simplify notation.

Lemma 1 (Local Laplace form of the type-conditional density). *Let $v(z | n) := u(z - T(z), z | n)$, and suppose T is locally linear near an interior maximizer z^* of $v(\cdot | n)$ with $v''(z^* | n) < 0$. Then, as $z \rightarrow z^*$,*

$$g(z | n) = \lambda_n(z) \exp(-2\lambda_n(z^*)|z - z^*| + O((z - z^*)^2)), \quad (13)$$

so to first order $g(z | n) \approx \lambda_n(z^) \exp(-2\lambda_n(z^*)|z - z^*|)$: a Laplace form centered at z^* with scale $(2\lambda_n(z^*))^{-1}$.⁹*

Proof sketch. A second-order Taylor expansion of $v(\cdot | n)$ around z^* shows that the dominating set is an interval of length $2|z - z^*| + O((z - z^*)^2)$; applying Proposition 1 and continuity of λ_n gives the result. Details are in Appendix C. $\square$

This local approximation has a striking implication: near a taxpayer's frictionless target income, the type-conditional density takes a familiar parametric form determined entirely by the local opportunity intensity at that target. In particular, the shape does not locally depend on the elasticity of the local indirect utility function. The intuition for this result can be seen from Figure 3: whatever the curvature of the indirect utility function, the dominating region $\Theta(z | n)$ is an interval with z on one end and the utility-equivalent $\bar{z}$ (in the notation of the above proof) on the other, centered around the target income z^* . In a local region around the target z^* , where indirect utility is approximately quadratic, this interval has approximately width $2|z - z^*|$, regardless of the curvature and, hence, elasticity.

3.1.2 Conditions for global Laplace behavior

The above lemma motivates the natural benchmark case in which the Laplace form for type-conditional density holds globally, rather than only as a local limit. This benchmark case holds under two assumptions:

Assumption 1 (Homogeneous intensity). *The type-specific opportunity intensity is homogeneous, $\lambda_n(z) \equiv \bar{\lambda}_n > 0$.*

⁹Here $O((z - z^*)^2)$ is standard “big- O ” notation for a remainder term whose magnitude is bounded by a constant multiple of $(z - z^*)^2$ as $z \rightarrow z^*$. Because it shrinks faster than the linear term $|z - z^*|$, it is negligible near z^* and drops out of the first-order (Laplace) approximation.

Assumption 2 (Quadratic utility). *For each type n with target income $z^*(n)$, the indirect utility function is quadratic, $v(z | n) = v(z^*(n) | n) - \frac{\xi_n}{2}(z - z^*(n))^2$, where $\xi_n > 0$.*

Under Assumptions 1 and 2, the local Laplace approximation from Lemma 1 becomes exact globally, as shown by the following corollary.

Corollary 3 (Exact Laplace form under the benchmark conditions). *Under Assumptions 1 and 2, the type-conditional density is*

$$g(z | n) = \bar{\lambda}_n \exp(-2\bar{\lambda}_n |z - z^*(n)|) \quad \text{for all } z. \quad (14)$$

That is, $g(\cdot | n)$ is globally Laplace with center $z^(n)$ and scale parameter $(2\bar{\lambda}_n)^{-1}$.*

Proofsketch. Quadratic utility makes the dominating set exactly the symmetric interval of length $2|z - z^*(n)|$, so with constant intensity Proposition 1 specializes to (14). Details are in Appendix C. $\square$

The curvature parameter ξ_n controls the income response to a change in tax rates: raising the marginal rate by Δt lowers type n 's target income by $|\Delta z_n| = \Delta t / \xi_n$.¹⁰ This level response is inversely proportional to ξ_n , as is the local ETI: $e_n = \frac{dz^*(n)}{d(1-t)} \frac{1-t}{z^*(n)} = \frac{1-t}{\xi_n z^*(n)}$, and we will sometimes therefore refer to ξ_n as (inversely) quantifying the local ETI.

3.1.3 Quantifying Laplace approximation error

Although Corollary 3 delivers an exact Laplace form under benchmark Assumptions 1 and 2, in practice, the opportunity intensity may vary across incomes and the indirect utility function may depart from quadratic. In Appendix D we quantify the resulting approximation error: Proposition 4 shows that when the intensity λ_n is uniformly close to $\lambda_n(z^*)$ and indirect utility is uniformly close to quadratic on an interval around z^* , the type-conditional density and its Laplace approximation are both sandwiched between two explicit Laplace envelopes, with the distance between the envelopes controlled by the sizes of the two deviations. The error is quantitatively negligible for empirically realistic calibrations: evaluating the bound for the quasilinear isoelastic utility used in the simulations of Section 2.3, on an interval containing over 99.7% of the type-conditional density, the maximum absolute error is just 0.43% of the peak density. Going forward, we therefore use the Laplace approximation as the foundation of our remaining empirical results, although in practice the results that follow can easily be numerically computed for other specific utility functions and opportunity intensities.

¹⁰Without loss of generality we can normalize indirect utility by the local marginal utility of consumption at the target $z^*(n)$, so that ξ_n is directly interpretable in terms of a money-metric response.

3.2 The uniform sparsity model and resulting bunching densities

The Laplace result of the previous subsection holds within types. Under Assumptions 1 and 2, each type's conditional density is exactly Laplace, but with a scale parameter $(2\bar{\lambda}_n)^{-1}$ that may vary across types and a center $z^*(n)$ whose location depends on the type's elasticity. To obtain a tractable expression for the observed income density $h(z)$ —the object we actually observe empirically—we impose two further restrictions that make these features common across types.

Assumption 3 (Cross-type homogeneity). *The opportunity intensity and the income response to tax changes are common across types: $\bar{\lambda}_n \equiv \bar{\lambda} > 0$ and $\xi_n \equiv \xi > 0$ for all n .*

Assumption 3 fixes a single opportunity intensity $\bar{\lambda}$, which we summarize by the *lumpiness parameter*

$$\mu := \frac{1}{\bar{\lambda}}, \quad (15)$$

the expected spacing between adjacent income opportunities, together with a single curvature ξ of the quadratic indirect utility of Assumption 2, analogous to the assumption of a homogeneous (local) ETI in conventional bunching estimation methods.

The remainder of this subsection characterizes, in terms of the primitives (μ, ξ) and the distribution of target incomes, the income densities that this model generates: the observed density under a linear tax (immediately below) and the bunching density around a kink; the analogous characterization around a notch is derived in Appendix F.

3.2.1 Density under a linear tax

We first characterize the observed income density in terms of the model primitives above in the interior of a linear tax, i.e., sufficiently far from any kinks, notches, or corner solutions that they do not distort the observed income density. Under the uniform sparsity model, Corollary 3 and Assumption 3 imply that every type-conditional density is Laplace with the same scale parameter $\mu/2 = (2\bar{\lambda})^{-1}$, differing across types only in its center $z^*(n)$. Aggregating these densities over types therefore reduces to a single smoothing operation.

To express this aggregation compactly, assume that target income $z^*(n)$ varies monotonically with type n , so that target incomes can serve as an index of types. We denote the density of target incomes by $h^*(z)$, with distribution function defined by $H^*(z^*(n)) = F(n)$. Under a linear tax this target-income density is exactly the frictionless counterfactual density that the same population would generate in the absence of sparsity-based frictions.

Corollary 4 (Observed density as a Laplace smoothing of target incomes). *Under the uniform sparsity model (Assumptions 1, 2, and 3), and assuming $z^*(n)$ is strictly monotone, the observed*

income density under a linear tax is

$$h(z) = \int_s \bar{\lambda} \exp(-2\bar{\lambda}|z - s|) h^*(s) ds, \quad \bar{\lambda} = \frac{1}{\mu}, \quad (16)$$

where $h^*(s)$ denotes the density of target incomes. That is, the observed income density is the convolution of the target-income density with a Laplace kernel of scale $\mu/2$.

Proof sketch. Substitute (14) into (12) and change variables from n to $s = z^*(n)$. Details are in Appendix C. $\square$

Equation (16) shows that, under a linear tax, the observed density depends on the model primitives only through the lumpiness parameter μ , which sets the bandwidth $\mu/2$ of the Laplace smoothing kernel; the curvature ξ (and thus the elasticity of taxable income) does not enter. Intuitively, sparsity diffuses each type around its target, but under a linear tax that target is the type's frictionless choice and is unaffected by e .

Under this model, if the target income density $h^*(z)$ is smooth then the convolution in (16) implies that the relationship between the target income density $h^*(z)$ and the observed income density $h(z)$ takes a simple form, characterized by the following corollary.

Corollary 5 (Local recovery of the target-income density). *Suppose the conditions of Corollary 4 hold. Then at any income z at which the target-income density h^* is continuous—equivalently, any z in the interior of a linear-tax region, where the observed density h is twice continuously differentiable—the observed and target-income densities satisfy*

$$h^*(z) = h(z) - \frac{\mu^2}{4} h''(z). \quad (17)$$

Proof sketch. Split the convolution (16) at $s = z$ and differentiate twice under the integral; the resulting second-order identity rearranges to (17). Details are in Appendix C. $\square$

In words, the target income density is equal to the observed income density minus a smoothing factor proportional to the local curvature of the observed density and the squared lumpiness parameter. Intuitively, if the target income density is strongly curved around some income z , e.g., because that income is a local density maximum, then the density smoothing due to diffusion will reduce the observed density at z relative to the frictionless target income density.

3.2.2 Bunching density around a kink

We now characterize the bunching density under the uniform sparsity model around a tax kink, where the marginal rate rises from t_0 below tax bracket threshold k to $t_1 = t_0 + \Delta t$.

Begin with the target incomes. Under the kinked schedule the frictionless target incomes follow the standard bunching pattern of Section 2 (Figure 1): types whose optimum under the lower rate t_0 lies below k are unaffected by the kink and continue to target that income; types whose optimum under the higher rate t_1 lies above k target their (lower) t_1 -optimum; and an intermediate range of types—those with a t_0 -optimum in $(k, k + \Delta z)$, where Δz is the income response defined in (1)—target the kink itself. Writing h_0^* and h_1^* for the densities of target incomes under the linear schedules T_0 and T_1 , the target-income distribution under the kink therefore consists of the density h_0^* at incomes below k , the density h_1^* at incomes above k , and an atom located exactly at k whose mass is the bunching mass $B = H_1(k) - H_0(k)$ of (5), equivalently $B = H_0(k + \Delta z) - H_0(k)$ by (6).

Smoothing this target distribution with the common Laplace kernel yields a remarkably simple characterization of the observed density on either side of the kink: it is the relevant linear-tax counterfactual plus a single exponential term quantifying the additive bunching mass, expressible directly in terms of structural primitives.

Proposition 2 (Bunching density around a kink). *Under the uniform sparsity model, let h_0 and h_1 be the counterfactual income densities under the linear taxes T_0 and T_1 , let B be the bunching mass, and suppose h_0 and h_1 are linear across the bunching window $[k, k + \Delta t/\xi]$. Then the observed income density near the kink is*

$$h(z) = \begin{cases} h_0(z) + \frac{\Delta t}{2\xi} \left(\frac{h_0(k) + h_1(k)}{\mu} + h_0'(k) \right) e^{-2(k-z)/\mu}, & z \leq k, \\ h_1(z) + \frac{\Delta t}{2\xi} \left(\frac{h_0(k) + h_1(k)}{\mu} - h_0'(k) \right) e^{-2(z-k)/\mu}, & z \geq k, \end{cases} \quad (18)$$

the local counterfactual plus exponential excess bunching mass decaying away from k at the Laplace rate $2/\mu$. The density peaks at the kink k , where

$$h(k) = \frac{h_0(k) + h_1(k)}{2} \left(1 + \frac{\Delta t}{\mu\xi} \right). \quad (19)$$

If the counterfactuals are not linear across the window then these expressions contain an additional term proportional to their curvature; the general formulas and the proof are in Appendix E.

Proof sketch. Under the kinked schedule the distribution of target incomes is the counterfactual target density h_0^* below k , the density h_1^* above k , and a point mass of size B at k —the bunchers, whose frictionless targets are displaced to the threshold. The observed density is the Laplace smoothing of this distribution (Corollary 4), which acts on the three pieces separately. Away from k , smoothing the smooth parts returns the respective counterfactual densities h_0 and h_1 ; the point mass at k is spread by the kernel into a two-sided exponential decaying at the Laplace

rate $2/\mu$ —the diffuse bunching. The observed density is therefore the local counterfactual plus this exponential, as in (18), with the amplitude on each side fixed by two conditions: continuity of h at k , and the downward slope kink $h'(k^-) - h'(k^+) = 4B/\mu^2$ that the point mass induces. When the counterfactual is linear across the bunching window the smoothed parts coincide with h_0 and h_1 and their slopes agree, collapsing the amplitudes and the peak to the stated closed forms; Appendix E treats curved counterfactuals, which add a term proportional to the counterfactual's curvature. $\square$

Equation (18) has a transparent structure. Absent a kink ($B = 0$ and $h_0 = h_1$) it returns the counterfactual h_0 . With a kink, each side carries an exponential bunching mass addition of scale $\mu/2$, so lumpier opportunities (larger μ) spread a given amount of excess mass over a wider income range—exactly as in the simulations of Appendix Figure A4(b). The bunching mass height at k is governed by B/μ , the diffuse analog of the frictionless bunching atom: the mass B , spread over income opportunities that are μ apart in expectation, raises the average counterfactual density $\frac{1}{2}(h_0(k) + h_1(k))$ at the peak by the proportional factor $\Delta t/(\mu\xi)$. The two amplitudes differ only through the counterfactual step $\frac{1}{2}(h_1(k) - h_0(k))$: because the T_1 counterfactual is the T_0 counterfactual translated down by $\Delta z = \Delta t/\xi$, the bunching mass is taller on the side toward which the counterfactual declines, with the amplitudes adjusting to keep h continuous, and the two exponential tails together integrate to the bunching mass B (exactly, when the counterfactual is locally linear). The two structural parameters enter through different features. The integrated excess mass $B \approx \frac{\Delta t}{2\xi}(h_0(k) + h_1(k))$ depends on ξ but not μ , so it identifies the curvature ξ ; given ξ , the peak height then identifies the lumpiness μ through the term B/μ . We develop this separation formally in Section 3.3.

The observed income density around a *notch* can be characterized with the same tools, although dominating income regions need no longer be compact, so type-conditional densities lose their Laplace form. Appendix F derives the analogous characterization (see Proposition 5): the observed density is an aggregation of deformed Laplace kernels, with closed-form expressions for the peak at the threshold and for the discontinuous drop to the strictly positive density in the dominated region just above it—reproducing the signatures of notch bunching visible in Figure 7(b). The notch also sharpens identification: with the notch size ΔT known, the width of the dominated region, $\Delta z^D = \sqrt{2\Delta T/\xi}$, pins down ξ through a channel distinct from the diffuse excess mass, while μ governs both the width of the bunching mass below k and the size of the drop.

3.3 Model identification

We now turn to identification of the model's two structural parameters: the elasticity of taxable income (via curvature ξ) and the lumpiness parameter μ . The closed-form bunching densities of Section 3.2 make the identification logic transparent, because they express the two most salient features of the observed density around a threshold—the total excess mass and the height of the peak—directly in terms of ξ , μ , and the counterfactual density.

Identification proceeds differently depending on what is observed. When a counterfactual density is available (e.g., from a pre-reform period or a comparison group) the two parameters are separately point-identified with no further restrictions, as we show next (Section 3.3.1). When no counterfactual is observed, identification instead rests on a smoothness assumption on the underlying ability density, which we state explicitly and operationalize in estimation (Section 3.3.2).

3.3.1 Separate identification of elasticity and lumpiness under known counterfactual

Suppose we observe the counterfactual density h_0 that would prevail under the linear tax T_0 —for instance, from a pre-reform period or a comparison group. The closed forms of Section 3.2 then separately identify the elasticity e and μ through two distinct features of the observed density.

The elasticity is identified by the excess mass. The bunchers are the types whose T_0 -target falls in $(k, k + \Delta z)$, so the mass displaced by the threshold is $B = H_0(k + \Delta z) - H_0(k) = \int_k^{k+\Delta z} h_0^*(s) ds$ (equation (5)), where $h_0^* = h_0 - \frac{\mu^2}{4} h_0''$ is recovered from the observed counterfactual h_0 by Corollary 5; by Proposition 2 this is $B \approx \frac{\Delta t}{2\xi} (h_0(k) + h_0(k + \Delta z))$. The displaced mass is in turn read off the observed density: above the bunching window the observed density coincides with the upper-tax counterfactual $h_1(z) = h_0(z + \Delta z)$, so the leftward shift of the upper tail relative to the known h_0 reveals the income response $\Delta z = \Delta t / \xi$. Because the displaced mass is strictly monotone in Δz and depends on the data only through the observed h_0 , this relation pins down the curvature ξ —and hence the local elasticity $e = (1 - t) / (\xi z^*)$ of Section 3.1.2, with no need for μ . In contrast, the lumpiness governs how the excess mass is *spread around the threshold*, not how much of it there is. At a notch, the elasticity is identified even more directly, since the dominated-region width $\Delta z^D = \sqrt{2\Delta T / \xi}$ from equation (29) recovers ξ from a single observable feature of the density.

The lumpiness is identified by the peak, given the elasticity. With ξ in hand—and hence Δz and the implied upper-tax counterfactual $h_1(z) = h_0(z + \Delta z)$ —the observed density at the threshold, $h(k) = B / \mu + \frac{1}{2} (h_0(k) + h_1(k)) + \frac{\mu}{4} (h_1'(k) - h_0'(k))$ from (27), leaves μ as the only unknown, since the excess mass B and the counterfactual values $h_1(k) = h_0(k + \Delta z)$ and

$h'_1(k) = h'_0(k + \Delta z)$ are all determined by h_0 and ξ . The peak height therefore determines μ : a smaller μ concentrates the fixed mass B into a taller, narrower spike at k .

Given that these two features (excess mass and peak) of the distribution pin down e and μ , the remainder of the bunching profile is then overidentifying. Proposition 2 (Proposition 5 at a notch) predicts the entire density on both sides of the threshold given e , μ , and h_0 , so the shape of the distribution (over and above the excess mass and peak) carries redundant information about the parameters that can be used to test the model. We next informally discuss identification when the counterfactual is not known, followed by more formal identification arguments.

3.3.2 Model identification with unknown counterfactual

When no counterfactual is observed, as in many settings with cross-sectional data under a single tax schedule, then recovering the structural parameters requires a restriction on the underlying ability density: Blomquist et al. (2021) show that if the type distribution is unrestricted, the excess mass at a kink or notch does not identify the elasticity, because a given mass could be consistent with either a high elasticity and few taxpayers near the threshold or a low elasticity and many such taxpayers. A common resolution is to adopt the identifying assumption that the ability density is smooth across the threshold, implemented by fitting a polynomial of a given order to the counterfactual income density (Chetty et al., 2011). We adapt this implementation by instead fitting a polynomial to the *type* density—rather than to the observed income density—as this is in fact consistent with the identifying assumption. Several features make our setting less exposed to the Blomquist et al. (2021) critique than the standard estimator. First, diffuse bunching reveals the *shape* of the excess mass, not only its total, supplying identifying restrictions beyond the single bunching moment. Second, our model delivers a sharp identifying restriction that requires no knowledge of the counterfactual: a smooth counterfactual density cannot produce a discontinuity in the slope of the observed density; yet under the uniform sparsity model the observed density has exactly such a discontinuity at the bunching peak at k . By Proposition 2 this kink is generated entirely by the bunching mass diffused through the Laplace kernel, and the counterfactual slopes $h'_0(k)$ and $h'_1(k)$ cancel exactly (Appendix E), leaving:

$$h'(k^-) - h'(k^+) = \frac{4B}{\mu^2}. \quad (20)$$

The one-sided slopes at k can be estimated directly from the data, so equation (20) identifies the combination B/μ^2 with no reference to the level or shape of the counterfactual density.

The two structural parameters load on distinct features of the bunching distribution. The

excess mass decays away from the threshold k at the Laplace rate $2/\mu$ —a shape no smooth counterfactual shares—so the exponential *rate* of the bunching mass separately identifies μ , and for a given μ , the slope discontinuity recovers the mass $B = \frac{\mu^2}{4} [h'(k^-) - h'(k^+)]$, whose conversion into an elasticity requires only the counterfactual density's *level* at k , not its shape through the bunching region. This is why the estimator of Section 3.4 can use data arbitrarily close to the threshold rather than excising a bunching window: our model recognizes the kink in the density as the signature of bunching. A notch produces, in addition, a *level* discontinuity in the observed density (Proposition 5)—another feature no smooth counterfactual can generate—which carries identifying information in the same spirit. The following proposition formalizes the separation.

Proposition 3 (Separate identification under local smoothness). *Suppose the counterfactual target density h_0^* is locally a polynomial of degree Q near k (more generally, real-analytic and smooth on a scale longer than the diffusion length $\mu/2$), and that bunching is positive. Then, under the uniform sparsity model:*

- (i) (Lumpiness from the decay rate.) *On either side of k , μ is the unique scale $\nu > 0$ for which $h - \frac{\nu^2}{4} h''$ is again locally a polynomial of degree Q ; equivalently, $h^{(n+2)}(z) = \frac{4}{\mu^2} h^{(n)}(z)$ for every $n > Q$. This determines μ with no reference to ξ or to the counterfactual's coefficients.*
- (ii) (Elasticity from the mass, given μ .) *Given μ , the deconvolution $h_0^* = h - \frac{\mu^2}{4} h''$ recovers the counterfactual on each side, and the bunching mass $B = \frac{\mu^2}{4} [h'(k^-) - h'(k^+)]$ identifies ξ —hence e —as the unique root of $B = \int_k^{k+\Delta t/\xi} h_0^*(s) ds$.*

The argument uses only local smoothness of the counterfactual, never its shape through the bunching region.

Proof. See Appendix G. □

Part (i) is the operative step: of the two ingredients of the observed density, only the diffuse bunching is non-smooth on the diffusion scale, decaying at the exponential rate $2/\mu$. The map $h \mapsto h - \frac{\nu^2}{4} h''$ leaves any smooth counterfactual a polynomial but cancels this exponential only when its bandwidth ν equals the true scale μ , so μ is read off the decay rate of the excess mass—its curvature alone—with no reference to ξ or the counterfactual. (This decay rate is distinct from the slope discontinuity at k , which part (ii) uses to recover the mass B .) The argument fails only in the knife-edge case where the counterfactual itself curves on the scale μ , making its curvature indistinguishable from bunching.

3.4 Model estimation

This subsection describes the maximum-likelihood estimator we use throughout the empirical applications in Section 4. The estimator jointly recovers the elasticity e , the lumpiness parameter μ , the polynomial coefficients θ governing the underlying ability density, and (optionally) the notch size ΔT , which we can allow to be an estimated parameter by treating it as a revealed feature of taxpayer behavior. The numerical details of this estimation method are described in the Numerical Implementation companion document distributed with our replication code (<https://github.com/bblockwood/DiffuseBunching>).

When the counterfactual income density is unobserved, we infer the underlying ability density $f(n)$ by imposing the smoothness assumption discussed in Section 3.3.2, namely, that the ability density is smooth around the bracket threshold k .¹¹ We operationalize this identification strategy by assuming that the ability density follows a polynomial of order Q , i.e.,

$$f(n|\theta) = \theta_0 + \theta_1 n + \theta_2 n^2 + \dots = \sum_{q=0}^Q \theta_q n^q \quad (21)$$

for a vector $\theta = \{\theta_0, \theta_1, \dots, \theta_Q\}$.

We then estimate the parameters of the model— e , μ , θ , and (if desired) ΔT —using maximum likelihood. Letting i index the observations in the data, with X_i denoting each observation's income, our starting point for the likelihood function is

$$L(e, \mu, \Delta T, \theta) = \prod_i h(X_i | e, \mu, \Delta T, \theta). \quad (22)$$

While this method may appear to resemble the widely used Chetty et al. (2011) approach of fitting a flexible polynomial to the observed income distribution outside of a selected bunching window, it is substantively different in two important ways. First, by structurally accounting for the distortion pattern produced by the bracket threshold, there is no need to select (visually or otherwise) an excluded bunching window around the threshold when computing the best-fit values of θ . Instead, even data near the bracket threshold helps to identify θ , because the model treats the diffuse bunching there as signal rather than contamination—the basis for the shape-based identification of Section 3.3.2. This logic points to the greater robustness of this estimation method to choices about the polynomial degree Q relative to standard bunching estimators, where additional flexibility may attempt to fit excess mass that spills outside the

¹¹This implies that the counterfactual income density $h_0(z)$ is also smooth. See Moore (2022) for a discussion of what can be identified by bunching estimators without estimating the elasticity directly, and Bertanha, McCallum and Seegert (2023) for the potential to exploit notches for elasticity identification with weaker assumptions on the density.

excluded bunching window. We confirm this reasoning in simulations in Appendix H.3.

Second, this approach assumes that the smooth polynomial structure is a feature of the underlying *ability distribution*, $f(n)$, rather than of the observed income distribution outside the bunching window. As illustrated in Figure 1(c), the frictionless model predicts a discontinuity in the income density around the bracket threshold due to the jump in types and the condensed mapping from types to income under higher marginal tax rates. By estimating the polynomial coefficients on the type distribution directly, this approach does not impose smoothness across the bracket threshold.

We compute standard errors for our maximum likelihood point estimates using the usual maximum-likelihood asymptotic variance, the inverse observed-information matrix. In Appendix H we assess this estimator on repeated rounds of simulated data with known parameters, and compare it to the conventional bunching estimator. Our method recovers the true elasticity with accurate confidence intervals at every simulated value of the lumpiness parameter μ , whereas the conventional estimator is biased downward—by about 50 percent when bunching is substantially diffuse—with confidence intervals that lie entirely below the true elasticity (Appendix Figure A7). The bias arises because diffuse bunching mass spills beyond any finite bunching window, pulling the estimated counterfactual density up into the bunching region; it is compounded by the integration constraint imposed when fitting that counterfactual. Appendix H.2 details these issues.

4 Empirical Applications

We apply our estimation method to four empirical bunching contexts. First, we consider the canonical example of bunching at a tax kink: the earned-income tax credit (EITC) threshold first studied by Saez (2010). We then consider three additional examples meeting two criteria: 1) the tax policy design (i.e. kink or notch, presence of compliance costs, etc.) is widely prevalent around the world; and 2) our method can glean economically-important insights from moments of the data discarded by conventional approaches.

We first study the canonical case of diffuse bunching at a kink, originally studied in Saez (2010) and later in Mortenson and Whitten (2020). We then apply our method to the case of a known notch value, namely, the consumption coupon fiscal stimulus program studied in Ding et al. (2025). Our method is especially informative here: the estimated lumpiness parameter captures the discreteness of adding or removing items from the consumption basket. The estimated elasticity in this application then provides an estimate of the size of the consumption response to fiscal stimulus coupons. Third, we study a notch of unknown size: the much-studied VAT registration threshold for small businesses. Here, bunching at the

threshold lets us estimate the otherwise-unobserved compliance cost of becoming VAT-liable. Finally, we study a statutory kink whose bunching signature is notch-like, perhaps because taxpayers misperceive the kink as a notch, or face an unobserved cost.¹² The asymmetry in the bunching mass then identifies the size of the “as-if” notch.

4.1 Application 1: Diffuse Bunching at a Canonical Kink

The canonical example of bunching at a kink is seen at the EITC threshold in the United States studied in Saez (2010) and more recently in Mortenson and Whitten (2020). We focus on the first kink, where the EITC amount “plateaus” (i.e. stops increasing). At this point in the schedule, substantial diffuse bunching is evident in the distribution, illustrated in the income histograms in Figure 9.¹³

We use two datasets to apply our estimator in this context: the first, comprising replication data from Mortenson and Whitten (2020), has finer granularity as it covers a larger number of years. The second, comprising replication data from Saez (2010), also records self-employed workers separately. In both datasets we focus on the sub-sample of workers with one child.

Figure 9(a) plots the histogram of incomes among all workers from Mortenson and Whitten (2020).¹⁴ It also displays the best fit of the uniform sparsity model bunching estimator, which estimates an elasticity of 0.21, substantially higher than 0.12, the elasticity estimated by the conventional Chetty et al. (2011) method. We also estimate a μ value (the average distance between earning opportunities) of \$600 relative to the kink-point of approximately \$8,600.

Figure 9(b) presents the original Saez (2010) bunching data at the first kink, along with our model’s best fit. We estimate an elasticity of 0.54, again substantially higher than the original Saez (2010) estimate of 0.21, and also higher than the estimate of 0.28 using the Chetty et al. (2011) method with a polynomial approximation for the counterfactual density.¹⁵ The estimated μ in this sample is \$1,700.

Figure 9(c) applies the uniform sparsity estimator to the income histogram for self-employed workers from Saez (2010). Our estimated elasticity is 1.60, again higher than the original Saez (2010) elasticity of 1.10, and also higher than the Chetty et al. (2011) estimate of 1.11. Estimated μ is again \$1,700, with a tighter confidence interval than in Figure 9(b).

Together, these estimates show that the uniform sparsity model consistently estimates EITC elasticities that are substantially higher than conventional estimation approaches.

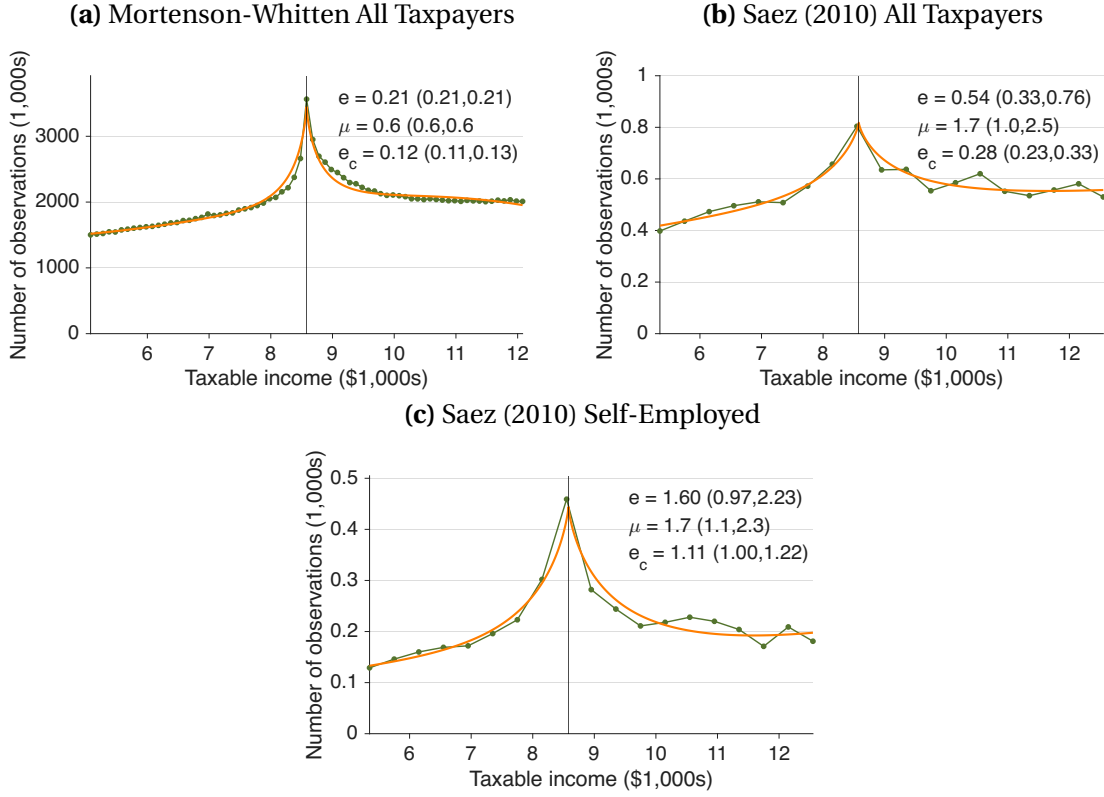
¹²Joel Slemrod calls such misperceptions “botches.”

¹³See Saez (1999) Table 1 for details on the kink-sizes and income brackets. Here we follow Saez (2010) in centering the data around the first kink to make all years comparable.

¹⁴We thank the authors for access to the data displayed in the histograms in their paper.

¹⁵The Saez (2010) elasticity results come from Table 2, panel (A) in that paper.

Figure 9: Bunching at the EITC kink



Green points plot the income histograms of single parents with one child in the United States around the first EITC kink. Panel (a) uses the microdata from Mortenson and Whitten (2016). Panel (b) uses the replication data from all taxpayers in Saez (2010). Panel (c) uses the self-employed sample from Saez (2010). Orange lines plot the best-fit income density under our uniform sparsity model. Each panel reports our parameter estimates e (elasticity of taxable income) and μ (average distance between income opportunities in \$1000s), as well as the elasticity e_c estimated using the conventional method. Numbers in parentheses indicate the 95 percent confidence interval on parameter estimates generated by the MLE method.

These differences are consistent with the differences in elasticity estimates delivered by the simulation results that compare the two methods in Appendix H.

4.2 Application 2: Bunching at a Statutory Notch of Known Size

This application demonstrates, in a very different setting from the previous example, our method's ability to estimate the elasticity from a notch of known size. Ding et al. (2025) study a subsidy program in China that paid consumers bonus coupons for making purchases over a specified order size.¹⁶ This policy structure, which was designed as a fiscal stimulus program, produces a notch in the mapping between costs and the value of goods purchased; the cost falls discontinuously when this value exceeds a specified threshold. Ding et al. (2025) focus

¹⁶We thank the authors for their openness to our use of the histograms data in their paper; no identifiable data is necessary for the production of these estimates.

their estimation on the marginal propensity to consume out of the subsidy amount, but we note that this setting is also well-suited to estimating other behavioral parameters, including the spending elasticity (the analog of the elasticity of taxable income in this setting) and the lumpiness parameter. The latter has a particularly clear interpretation in this setting, as consumers adjust their order size by adding or removing discrete items from their cart.

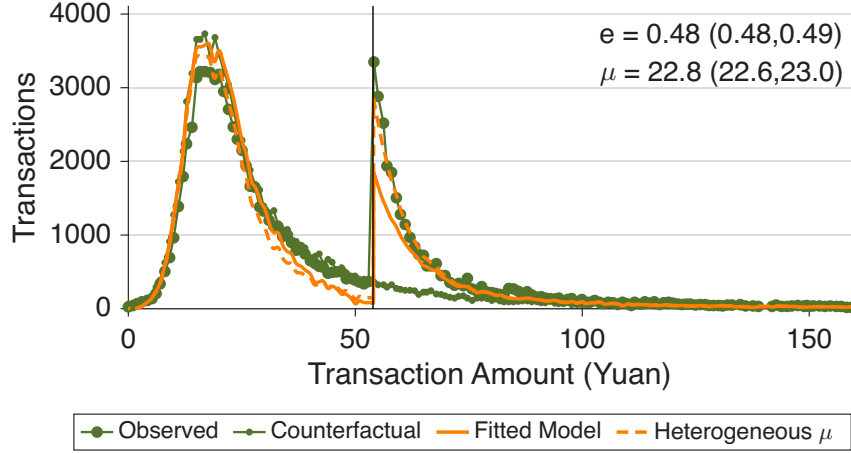
This application also illustrates the generality of the estimation method in three ways. First, the running variable is consumer expenditures rather than earnings (our focus in the discussions thus far). Second, the beneficial side of the notch is to the right of the threshold, rather than to the left (more commonly the case with earnings tax kinks and notches). And third, because the policy was a temporary treatment, the data also contain an observable counterfactual distribution of purchases that obtains in the absence of the notch subsidy, so there is no need to estimate the counterfactual in this setting.

Figure 10 displays the histogram of purchase sizes among consumers treated with the notch subsidy. It also displays (with smaller markers) the counterfactual distribution of purchases observed in the absence of the subsidy. Consistent with the model of sparsity-based frictions, the bunching mass at the notch exhibits pronounced asymmetry, with diffuse mass on the desirable side of the notch. That diffusion, combined with the lumpiness inherent in consumers' choices over discrete menu items, suggests to us that this is a setting in which sparsity-based frictions are likely to be a more realistic model of underlying frictions than the Kleven and Waseem (2013) notch-bunching model of heterogeneous responsiveness.

The solid orange line in Figure 10 displays the best fit of the uniform sparsity bunching estimator. When we apply the method, we use the counterfactual density provided by the data, rather than an estimated polynomial. Like the observed density, the best-fit model-predicted density exhibits diffuse asymmetric bunching to the right of the notch. The notch value (ΔT) in this setting is known (18 yuan), and we impose it as a parameter of the tax function. We estimate a lumpiness parameter equalling $\mu = 22.8$, which represents the cost in yuan of the average item that is marginal to being added or dropped from a consumer's basket, and a spending elasticity of 0.48. Because these parameters are distinct from the marginal propensity to consume, they can help us understand how consumers might respond to other types of subsidies, such as a flat subsidy on all purchases rather than a discontinuous notch.

Heterogeneity in lumpiness. Although the model-predicted (solid-line) density in Figure 10 captures key qualitative features of the observed income histogram—most notably the diffuse bunching to the right of the notch—we also note that it exhibits a smaller “peak” at the notch, and a slower rate of decay (moving rightward) from that peak, than is evident in the data. We suspect that this mismatch is driven by our imposition of a homogeneous lumpiness

Figure 10: Bunching at a Known Notch Value: Threshold Coupons



This figure plots bunching in the distribution of online order amounts around the 54 yuan coupon threshold studied in Ding et al. (2025). The coupon gives 18 yuan off a purchase of 54 or greater. Series are labeled as follows. “Observed”: the observed distribution of order amounts when consumers had access to the coupons (period t). “Counterfactual”: the distribution of order amounts among the same sample of consumers during the pre-period before coupon access. “Fitted model”: the best-fit model-generated distribution from our uniform sparsity model. “Heterogeneous μ ”: the best-fit model-generated distribution from the uniform sparsity model when we allow for two values of the μ parameter.

parameter μ . If that parameter is instead heterogeneous across consumers—for example, if some consumers are choosing between small marginal items in their order baskets while others are choosing between more expensive items—then the observed density would be a mixture of densities, some with a narrow sharp peak above the notch, and others with a lower peak and more diffusion. The mixture would produce a taller peak with a faster rate of decay. We therefore re-estimate the model allowing for heterogeneous lumpiness μ . For illustrative purposes, we constrain this heterogeneity to add just one degree of freedom, finding the best-fit combination of two μ values while retaining homogeneity in the spending elasticity. As shown by the dashed line in Figure 10, allowing for this heterogeneous lumpiness indeed produces a better fit to the high peak and fast rate of decay observed in the bunching mass.

4.3 Application 3: Bunching at Regulatory Notch of Unknown Size

Our third application makes use of the uniform sparsity model’s ability to estimate a notch value of unknown size from observed bunching behavior. We focus on a widely prevalent feature of tax policies, namely, value-added tax (VAT) registration thresholds which require that firms with revenues above a specified threshold must register for, collect, and remit VAT, imposing costs on them. Keen and Mintz (2004) note that these VAT compliance costs are an important

determinant of optimal VAT thresholds, and Liu and Lockwood (2015), among others, show bunching at VAT registration thresholds, consistent with VAT registration imposing compliance costs on affected firms. By estimating the notch value at the VAT registration threshold, our method allows us to infer the money-metric size of compliance costs distinct from the change in statutory tax incentives at that threshold.

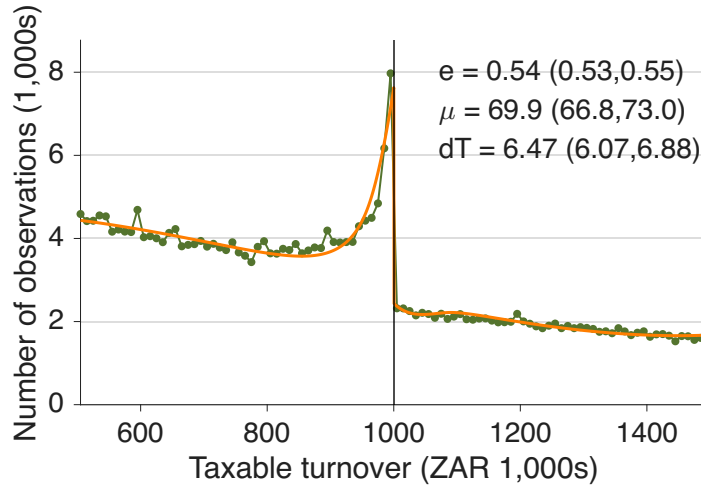
We study the VAT registration threshold in a novel setting, South Africa, using extensive administrative tax data spanning 2018-2022. Firms are required to register for the Value-Added Tax (VAT) with the South African Revenue Service (SARS) if their total taxable turnover exceeds R1 million in any consecutive 12-month period. This registration is mandatory, and once registered, the business must charge VAT—currently at a standard rate of 15%—on most goods and services it supplies. Firms with taxable turnover below this threshold may also register voluntarily if their turnover exceeds R50,000 over the past 12 months. Registered businesses can claim input tax credits on VAT paid for business-related purchases, thereby offsetting their VAT liability. Registration is done online through the SARS eFiling system or at a SARS branch.

To approximate the change in the tax level associated with crossing the VAT registration threshold (to distinguish that change from compliance costs) we compute the marginal income tax rate for firms below and above the notch. We approximate the marginal income tax rate by taking the slope of a smoothed binscatter of average tax liability across revenues for firms below and above the notch. We compute this marginal tax rate below the notch only for the subset of firms that do not *voluntarily* register for VAT, in order to identify the change in tax level that arises due to the change in VAT registration status.¹⁷ Voluntary registration is allowed in South Africa, and we find that 49% of firms below the notch choose to register for VAT voluntarily.

Figure 11 shows the distribution of firms in R10,000 bins of taxable turnover around the VAT registration thresholds. There is a clear pattern of bunching in firm turnover to the left of the VAT registration threshold. Our model is able to fit the raw bunching data well, including the diffuse bunching to the left and the positive density to the right of the notch—both patterns that are apparent in previous VAT registration applications. We estimate an elasticity of taxable turnover to VAT tax rates of $e = 0.54$. Our estimate of the lumpiness parameter μ equals R69900, representing the average distance between turnover opportunities being 6.99% of taxable turnover (for firms with taxable turnover of R1 million). Finally, we estimate an additional discrete notch value beyond the change in tax rates of $\Delta T = \text{R}6470$ in taxable turnover. This implies that firms behave as though VAT registration imposes an additional cost equivalent

¹⁷This issue is studied extensively in Liu et al. (2021), who find that approximately half of British firms just below a VAT notch voluntarily register. They find that firms are more likely to voluntarily register for the VAT when they benefit more from input credits than they are hurt by the VAT tax on output (i.e. firms that do more business-to-business sales relative to business-to-consumer sales, and when they face lower product market competition, because more of the VAT tax can be passed on to their customers).

Figure 11: Bunching at a VAT registration threshold



Green points plot the histograms of taxable turnover of all businesses in South Africa around a VAT registration threshold, after which VAT registration becomes compulsory. Orange lines plot the best-fit income density under the uniform sparsity model. The figure reports parameter estimates e (elasticity of taxable turnover), μ (average distance between taxable turnover opportunities in R1000s), and dT (the estimated “as-if” discrete change in tax liability at the bracket threshold, in R1000s). Numbers in parentheses indicate the 95 percent confidence interval on parameter estimates generated by the MLE method.

to approximately 0.647% of the R1 million turnover threshold.¹⁸ This estimate is potentially helpful for evaluating optimal VAT turnover thresholds.

4.4 Application 4: Notch-like Bunching Around a Kink

For our final application, we consider a variation on standard bunching-at-a-kink behavior, in which the observed pattern of bunching around a statutory kink resembles observed behavior around a tax notch. This application is motivated by our observation, also in South African administrative data, that (both personal and small business) taxpayers appear to exhibit strong notch-like patterns of asymmetric bunching around tax thresholds that are statutory kinks.¹⁹

Figure 12 shows this pattern of asymmetric bunching for two distinct populations. In panel (a), we plot the histogram of taxable incomes among small businesses around their

¹⁸Because the VAT threshold and our outcome variable are both defined in turnover, this turnover-equivalent estimate is our preferred interpretation of the result. For comparison, translating this value into a profit-equivalent amount requires an assumption about the relevant profit margin. Using the average taxable profit margin of South African firms over 2018–2022 (6.18%), gives a taxable income-equivalent value of R400.

¹⁹Like many developing countries, South Africa relies more heavily on corporate income taxes than most developed economies. In 2017, corporate taxes accounted for 16.2 percent of total tax revenue in South Africa, considerably higher than the OECD average of 9.7 percent, but in line with the average shares for Africa (18.5 percent) and Latin America (15.4 percent). Data from the OECD accessed at: https://stats.oecd.org/Index.aspx?DataSetCode=CTS_REV. The South African corporate tax system is tiered, with a progressive, kinked tax schedule for “Small Business Corporations” (SBCs), and a flat 28 percent tax applying to other (larger) resident corporations. Businesses can optionally register as an SBC if they meet a set of requirements, the most pertinent being that their annual revenue must be below R20 million (about \$1.4 million US).

first corporate income tax kink, at which the marginal tax rate rises from 0 to 7 percent.²⁰ The bunching pattern exhibits pronounced asymmetric diffuse bunching to the left of the threshold, with no excess mass—and even a small apparent depression—to the right of the threshold. Panel (b) displays a similar pattern for self-employed individuals around the first kink in the personal income tax schedule, where the income tax rate rises from 0% to 18%.²¹ Although the asymmetry remains pronounced in panel (b), the density to the right of the threshold does not exhibit a depression, but rather a muted excess mass relative to smooth interpolation from the far left to the far right. The sparsity-based frictions model can fit either of these patterns around a notch, depending on the combination of the elasticity and lumpiness parameters, as illustrated by Appendix Figure A4(d).

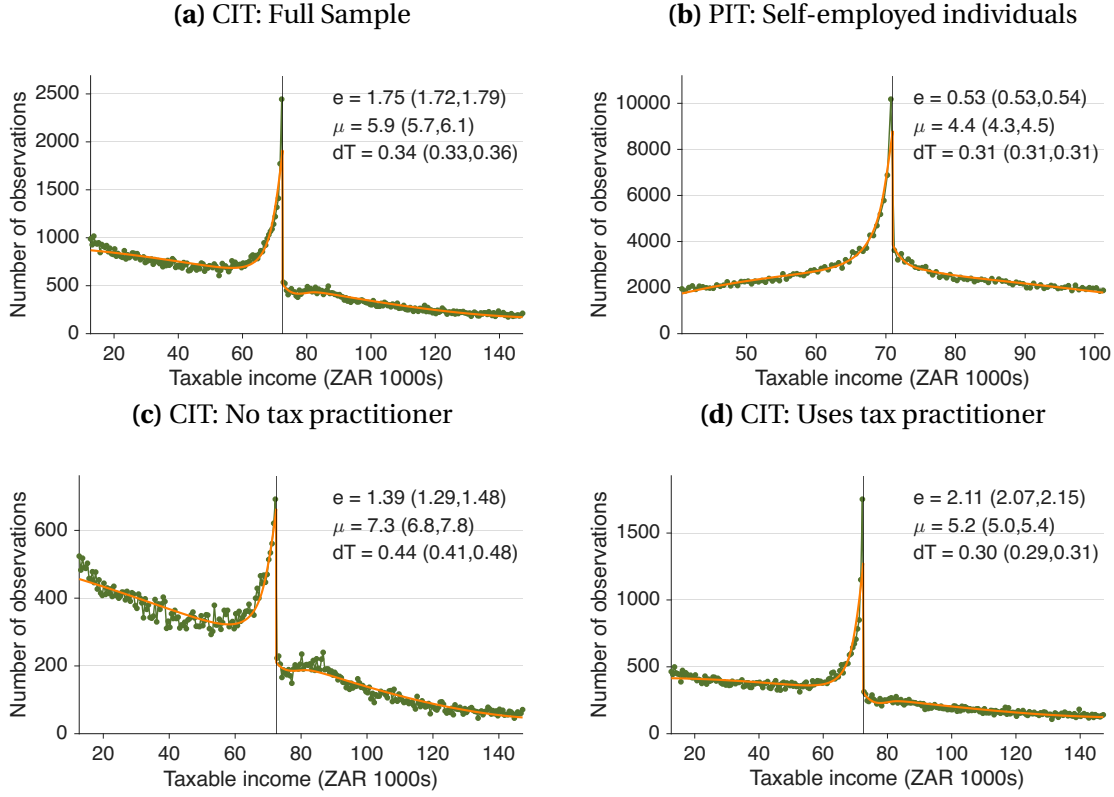
Applying the sparsity-based frictions estimator to the density in Figure 12(a), we estimate a notch value of R340, or about \$24 US. This estimate is statistically significant, indicating that the model strongly rejects pure kink behavior. That said, the as-if notch value is not large, consistent with a modest perceived cost of earning above the threshold. Figure 12(a) focuses on the lowest of three prominent tax kinks in the small business income tax schedule. Appendix Figure A10 shows bunching patterns in the taxable incomes of small businesses around the middle kink (at R360,000) and the upper kink (at R550,000). For reasons explored below, we partition these populations by whether they use a tax practitioner. Although much more muted than at the lower kink, asymmetry remains present at these higher tax kinks as well, with estimated notch values significantly above zero in all cases except Figure A10(d), suggesting that notch-like behavior at a kink is not only observed in the very smallest businesses.

A potential mechanism underlying notch-like bunching around a tax kink is taxpayer confusion about average vs. marginal tax rates. A large literature suggests agents may confuse marginal versus average changes in incentives (Ito (2014), Rees-Jones and Taubinsky (2020)). If taxpayers earning just below the threshold mistakenly believe that a small increase in earnings would cause their average (rather than their marginal) tax rate to rise, that would produce a notch-like pattern of bunching of the sort observed in Figure 12. Recent evidence from the U.S. Social Security Earnings Test shows that some taxpayers appear to perceive statutory kinks as notches, which generates left-bunching consistent with this mechanism (Gelber et al. (2025)).

²⁰SBCs account for 38 percent of all formally registered companies, although due to their smaller size, they account for less than 20 percent of total corporate tax revenues. For the purposes of this paper, we focus on the population of SBCs from 2014 to 2018 because they face a piecewise-linear kinked tax schedule ideal for bunching estimation.

²¹The lowest kink point in the schedule is a result of the standard deduction—all South Africans receive a tax deduction equal to the amount of taxable income at the lowest threshold (R70,000 or \$4,860 per annum in 2017). This means that any income below this threshold is taxed at a 0% marginal tax rate, whereafter the marginal tax rate jumps to 18%, in a similar fashion to that of the first SBC kink. We find little evidence of bunching at kinks for wage earning South African workers.

Figure 12: Asymmetric bunching at a kink: South African Evidence



Panel (a) displays the distribution of corporate taxable incomes around the first kink in the corporate income tax (CIT) Small Business tax schedule, where the marginal tax rate rises from 0% to 7%. Green points represent the empirical histograms of taxpayer incomes; orange lines plot the best-fit income density under the uniform sparsity model. Panel (b) displays the corresponding distributions of personal incomes for self-employed taxpayers at the first kink in the personal income tax schedule, where marginal tax rates change from 0% to 18%. Panels (c) and (d) display the CIT distributions at the first kink among firms without and with tax practitioners, respectively. Each figure also reports parameter estimates e (elasticity of taxable income), μ (average distance between income opportunities in R1000s), and dT (the estimated “as-if” discrete change in tax liability at the bracket threshold, in R1000s). Numbers in parentheses indicate the 95 percent confidence interval on parameter estimates generated by the MLE method.

Heterogeneity in “Botching”

Here we explore differences in notch-like bunching at a kink across a dimension of taxpayer heterogeneity that is observable in our tax data and may be related to taxpayers’ ability to understand and take advantage of tax incentives, namely, the use of a hired tax preparer. Figures 12(c) and 12(d) partition small businesses by their tax practitioner usage. At the lowest kink, there is a clear hole in the distribution for firms that do not use a tax practitioner (panel (c))—consistent with firms interpreting the lowest kink as a notch; this hole is not clearly evident at the lowest kink for firms that use a tax practitioner (panel (d)). These results uncover how firms using tax practitioners appear less prone to “botching” (i.e., they have less “notch-like” behavior around a kink); at the lowest kink firms without tax practitioners act as if

the average tax liability increases by approximately R440 at the first kink, while firms with tax practitioners act as if the average tax liability increases by R300 (a 32% reduction).

Figure A10 shows plots like those in Figure 12(c) and (d) but for the middle and upper tax kinks. The panels partition businesses on their tax practitioner usage, and each reports estimated elasticities. The raw histograms exhibit more pronounced bunching among firms that use tax practitioners. We do not observe a consistent pattern in the relative income elasticity between businesses that do and do not use tax practitioners: although the elasticity is higher among firms that use tax practitioners at the lowest kink, the reverse is true at the middle kink. At the upper kink, the elasticities in each group are statistically indistinguishable. Yet, there is a clear and consistent difference between the lumpiness parameters of the different groups of firms: adjustment frictions appear to be smaller for firms who use tax practitioners at every kink.²² This is consistent with such firms fine-tuning their incomes more precisely in response to tax incentives, or paying closer attention to the set of possible actions—real economic activity or reporting behavior—which can be used to target incomes more precisely to a desired level.

5 Conclusion

This paper develops a new bunching-based elasticity estimator using a novel approach to modeling income adjustment frictions. We adopt a general model of frictions in which taxpayers choose between discrete income opportunities, capturing a range of microfoundations. We show that multiple different income adjustment and search processes are well approximated by a parsimonious limiting case, in which opportunities are drawn from a Poisson process governed by a single “lumpiness” parameter. This new parameter quantifies the expected distance between adjacent opportunities. The model predicts key patterns observed in empirical bunching settings, such as diffuse bunching around kinks and positive mass above notches.

We demonstrate the versatility and applicability of this new approach by using it to study a set of four canonical and novel settings. The new method—consistent with simulations that we conduct—uncovers estimated elasticities in the classic US EITC kink that are nearly twice as large as in previous work. The method is also well able to capture the bunching patterns observed in a recently studied fiscal stimulus coupon program in China featuring a pure notch of known size, additionally extracting an economically-meaningful parameter, namely, the

²²At the middle kink we estimate $\mu = 28.6$ with a 95% CI of [19.5,37.8] for firms without a tax practitioner, as compared to $\mu = 9.5$ with a 95% CI of [7.0,12.1] with one. At the upper kink our μ estimate of 27.2 is noisier, with a 95% CI [-6.3,60.6], but directionally higher than the corresponding $\mu = 6.4$ with a tax practitioner (95% CI of [4.6,8.2]).

average distance between consumption opportunities.

In two other applications on recently-harnessed administrative data from South Africa, the method yields useful new economic insights. When we apply the method to observed bunching around VAT registration thresholds, we provide a new money-metric estimate of the compliance and hassle factors of VAT registration for small businesses. And when we apply this method to administrative tax data on small firms and individuals around kinks in both the corporate and personal income tax schedules in South Africa, we find evidence that both firms and individuals treat the lowest tax kink like a notch. We estimate substantially lower income adjustment frictions among firms with paid tax preparers, consistent with finer income targeting among that group. By quantifying the extent of adjustment frictions, our approach allows us to recover the “as-if” notch value when taxpayers treat a kink like a notch.

Our proposed bunching estimator can be applied to estimate behavioral responses in other settings with kinked budget sets, including with non-income tax instruments, nonlinear pricing schedules, or non-monetary payoffs. More generally, the model of uniform sparsity is a useful and parsimonious description of behavior that can be extended to a wider range of settings.

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Online Appendix

Diffuse Bunching with Frictions: Theory and Estimation

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A Details of Bunching in the Presence of a Tax Notch

Figures A1–A3 illustrate details of the bunching model around kinks and notches in the frictionless model and with sparsity-based frictions.

Figure A1 illustrates bunching patterns that arise from a kink (panels (a) and (b)) or a notch (panels (c) and (d)) in a model without income frictions. The left panels (a) and (c) illustrate the choices of individual types of taxpayers, each of whom is depicted by a discrete dot. The right panels translate this discrete choice behavior into income densities with a continuum of types. In the presence of a kink, the frictionless model predicts an atom of mass at the bracket threshold with smooth densities on either side, generally with a discontinuity at the threshold due to the leftward shift and income compression of taxpayers who face the higher marginal tax rate above the threshold. In the presence of a notch, the model again predicts an atom of mass at the threshold and a smooth density to the left, but with an absence of mass (density equal to zero) in a dominated region of incomes just above the bracket threshold.

Figures A2 and A3 are analogous to Figures 4 and 5 in the paper, but in the presence of a notch rather than a kink. The notch in the budget constraint produces a discontinuity in the indirect utility functions plotted in Figures A2(c) and A2(d). For illustrative purposes, type c is selected to be the type that is just indifferent between two levels of income, k and their optimal income choice under $T_1(z)$. (This is distinct from Figure 6 in the paper, where type c strictly prefers k .) As illustrated in Figure A3, the logic of Proposition 1 again carries through—the type-conditional density scales with the probability of drawing an income opportunity inside the shaded region of dominating incomes—although the notch sometimes produces a set of dominating income $\Theta(z|n)$ consisting of disjoint intervals, as shown in Figure A3(b). Under sparsity-based frictions, type-conditional densities exhibit positive density even in the so-called “dominated region” to the right of the notch because for some taxpayers an income opportunity in that region may be preferable to all of their other opportunities.

B Proof of Corollary 2

This section provides the deferred proof of Corollary 2 in Section 2.3.

Proof. Fix a type n and write $u_n(z) = u(z - T(z), z \mid n)$, and let $G_n(t) := \int_{\{y: u_n(y) \leq t\}} v_n(y) dy$ denote the CDF of the utility delivered by a single draw $Z \sim v_n$.

Let $Z_1, \dots, Z_M$ be the M iid draws from v_n . Because $u_n(Z)$ has a continuous distribution, ties occur with probability zero. The chosen opportunity—the utility-maximizing draw—lies in $[z, z + dz]$ exactly when one of the M draws falls in that interval, with probability

$$M v_n(z) dz + o(dz),$$

and the other $M - 1$ draws all yield weakly lower utility than $u_n(z)$, with probability

$$[G_n(u_n(z))]^{M-1}.$$

Multiplying these gives

$$g_M^{NS}(z \mid n) = M v_n(z) [G_n(u_n(z))]^{M-1}.$$

For the sparsity-based frictions model with intensity $\lambda_{n,M}(z) = M v_n(z)$, Proposition 1 implies that the unnormalized chosen-opportunity density (see the discussion following Proposition 1) is

$$M v_n(z) \exp\left(-\int_{\{y: u_n(y) \geq u_n(z)\}} M v_n(y) dy\right).$$

Again using continuity of the utility distribution to ignore ties,

$$\int_{\{y: u_n(y) \geq u_n(z)\}} v_n(y) dy = 1 - G_n(u_n(z)).$$

Therefore the corresponding subdensity is

$$M v_n(z) \exp(-M[1 - G_n(u_n(z))]).$$

Its total mass is

$$1 - e^{-M},$$

because the SBF opportunity process draws a $\text{Poisson}(M)$ number of opportunities and the event of drawing none has probability e^{-M} . Conditioning on the event that at least one

opportunity is drawn therefore yields

$$g_M^{SBF}(z | n) = \frac{M v_n(z) \exp(-M[1 - G_n(u_n(z))])}{1 - e^{-M}}.$$

Now define

$$\Delta_M(n) := \int |g_M^{NS}(z | n) - g_M^{SBF}(z | n)| dz$$

and note that if $Z \sim v_n$ this can be written as

$$\Delta_M(n) = \int M \phi(G_n(u_n(z))) v_n(z) dz = M \mathbb{E}[\phi(G_n(u_n(Z)))]$$

for ϕ appropriately defined. Using the probability integral transform,

$$S := G_n(u_n(Z)) \sim \text{Unif}[0, 1].$$

Hence we can rewrite $\Delta_M(n)$ as

$$\Delta_M(n) = M \mathbb{E}[\phi(S)] = \int_0^1 \left| M s^{M-1} - \frac{M e^{-M(1-s)}}{1 - e^{-M}} \right| ds.$$

In particular, $\Delta_M(n)$ does not depend on n , and we thus write it as Δ_M hereafter.

Since $\frac{1}{1 - e^{-M}} = 1 + \frac{e^{-M}}{1 - e^{-M}}$, the triangle inequality gives

$$\Delta_M \leq \int_0^1 M |s^{M-1} - e^{-M(1-s)}| ds + \frac{e^{-M}}{1 - e^{-M}} \int_0^1 M e^{-M(1-s)} ds = \int_0^1 M |s^{M-1} - e^{-M(1-s)}| ds + e^{-M},$$

where we have used $\int_0^1 M e^{-M(1-s)} ds = 1 - e^{-M}$. Substituting $r = M(1 - s)$ in the remaining integral,

$$\Delta_M \leq \int_0^M |(1 - r/M)^{M-1} - e^{-r}| dr + e^{-M}.$$

For each fixed $r \geq 0$, $(1 - r/M)^{M-1} \rightarrow e^{-r}$, so the integrand (extended by zero for $r > M$) converges pointwise to zero. For $M \geq 2$ and $r \in [0, M]$, the inequality $1 - x \leq e^{-x}$ gives

$$(1 - r/M)^{M-1} \leq e^{-r(M-1)/M} \leq e^{-r/2},$$

so the integrand is dominated by $2e^{-r/2}$, which is integrable on $[0, \infty)$. Dominated convergence yields $\Delta_M \rightarrow 0$, and thus we have

$$\int |h_M^{NS}(z) - h_M^{SBF}(z)| dz \leq \int \Delta_M(n) f(n) dn \rightarrow 0,$$

proving the corollary. □

C Proofs of Lemmas and Corollaries in Section 3

This section collects the deferred proofs of Lemma 1 and Corollaries 3, 4, and 5.

Proof of Lemma 1. By Proposition 1, $g(z | n) = \lambda_n(z) \exp[-\int_{\Theta(z|n)} \lambda_n(x) dx]$. Writing $\delta = z - z^*$, the Taylor expansion $v(z | n) = v(z^* | n) + \frac{1}{2} v''(z^* | n) \delta^2 + O(\delta^3)$ implies that the utility-equivalent income on the opposite side of z^* is $\bar{z} = z^* - \delta + O(\delta^2)$, so the dominating set is the interval between z and $\bar{z}$, of length $|\Theta(z | n)| = 2|\delta| + O(\delta^2)$. Since λ_n is continuously differentiable, $\int_{\Theta(z|n)} \lambda_n(x) dx = 2\lambda_n(z^*)|z - z^*| + O((z - z^*)^2)$, giving (13); replacing $\lambda_n(z)$ by $\lambda_n(z^*)$ to first order gives the Laplace approximation. □

Proof of Corollary 3. Under Assumptions 1 and 2, the equal-utility income on the opposite side of $z^*(n)$ satisfies

$$|\bar{z} - z^*(n)| = |z - z^*(n)|, \quad (23)$$

so the dominating set $\Theta(z | n)$ is the interval between z and $\bar{z}$, with length $|\Theta(z | n)| = 2|z - z^*(n)|$. Because $\lambda_n(z) \equiv \bar{\lambda}_n$, Proposition 1 implies that equation (10) specializes to equation (14). □

Proof of Corollary 4. Substituting the type-conditional density in (14), evaluated at the common intensity $\bar{\lambda}_n \equiv \bar{\lambda}$, into the expression for the observed income density in equation (12) gives

$$h(z) = \int_n \bar{\lambda} \exp(-2\bar{\lambda}|z - z^*(n)|) f(n) dn.$$

Since $z^*(n)$ is strictly monotone, we change variables from n to $s = z^*(n)$; the resulting density of target incomes is $h^*(s)$, yielding equation (16). □

Proof of Corollary 5. Write $a := 2\bar{\lambda} = 2/\mu$ and split (16) at $s = z$ as $h(z) = \bar{\lambda} [I_1(z) + I_2(z)]$, where $I_1(z) := \int_{-\infty}^z e^{-a(z-s)} h^*(s) ds$ and $I_2(z) := \int_z^{\infty} e^{-a(s-z)} h^*(s) ds$. Differentiating under the integral (the boundary terms at $s = z$ cancel) gives $I_1' = h^* - aI_1$ and $I_2' = -h^* + aI_2$, so $h' = \bar{\lambda} a (I_2 - I_1)$. Differentiating once more, $h'' = \bar{\lambda} a [(-h^* + aI_2) - (h^* - aI_1)] = a^2 h - 2\bar{\lambda} a h^*$. Since $2\bar{\lambda} a = (2\bar{\lambda})^2 = a^2$, this rearranges to (17). □

D Quantifying Laplace Approximation Error

This section states and proves the bound on the error of the Laplace approximation to the type-conditional density, referenced in Section 3.1. Provided the opportunity intensity λ_n is uniformly close to $\lambda_n^* := \lambda_n(z^*)$ on an interval I around z^* and indirect utility is uniformly close

to quadratic on I , the type-conditional density $g(\cdot | n)$ and the Laplace reference $L(\cdot | n)$ are both sandwiched between two Laplace envelopes, the distance between which can be quantified in terms of the deviations.

Proposition 4 (Laplace approximation error). *Fix a type n and an interval $I = [z_{Lo}, z_{Hi}]$ containing the interior maximizer z^* of $v(\cdot | n)$; let $\lambda_n^* := \lambda_n(z^*)$, $\xi_n := -v''(z^* | n) > 0$, $v_Q(z | n) := v(z^* | n) - \frac{\xi_n}{2}(z - z^*)^2$, and $L(z | n) := \lambda_n^* \exp(-2\lambda_n^*|z - z^*|)$. Suppose $\Theta(z | n) \subset I$ for every $z \in I$ and that, for some $A_n \in [0, \xi_n/2)$ and $B_n \in [0, 1)$,*

$$|v(z | n) - v_Q(z | n)| \leq A_n(z - z^*)^2 \quad \text{and} \quad |\lambda_n(z) - \lambda_n^*| \leq B_n \lambda_n^* \quad \text{for all } z \in I. \quad (24)$$

With $\rho_n := \sqrt{(\xi_n + 2A_n)/(\xi_n - 2A_n)}$, define the Laplace envelopes

$$\begin{aligned} \bar{L}_n(z) &:= (1 + B_n) \lambda_n^* \exp(-(1 - B_n)(1 + \rho_n^{-1})\lambda_n^*|z - z^*|), \\ \underline{L}_n(z) &:= (1 - B_n) \lambda_n^* \exp(-(1 + B_n)(1 + \rho_n)\lambda_n^*|z - z^*|). \end{aligned}$$

Then both $g(\cdot | n)$ and $L(\cdot | n)$ lie in $[\underline{L}_n, \bar{L}_n]$ on I , so the absolute approximation error satisfies

$$|g(z | n) - L(z | n)| \leq \max(\bar{L}_n(z) - L(z | n), L(z | n) - \underline{L}_n(z)). \quad (25)$$

When $A_n = B_n = 0$ the envelopes coincide with $L(\cdot | n)$ and the bound is zero, recovering Corollary 3.

The two error sources enter (25) separately: A_n governs how much $\Theta(z | n)$ deviates from a symmetric interval of length $2|z - z^*|$ and feeds through ρ_n , while B_n governs how much the integral of λ_n over Θ deviates from $\lambda_n^*|\Theta|$ and enters the envelope prefactors. Each gap in (25) equals $\lambda_n^*B_n$ at $z = z^*$ and decays exponentially as $|z - z^*| \rightarrow \infty$, so the bound attains a finite supremum at an intermediate distance from z^* .

To gauge the quantitative size of this absolute error for an empirically plausible calibration, we evaluate the proposition for the quasilinear isoelastic utility used in the simulations of Section 2.3, with target $z^*(n) = \$50,000$, elasticity $e = 0.3$, and uniform opportunity intensity $\bar{\lambda}_n = 1/\$2,000$ (so $B_n = 0$). On the analysis interval $I = [z^* - 3\mu, z^* + 3\mu]$, which contains over 99.7% of the type-conditional density, the smallest A_n for which (24) holds is approximately 5.4% of ξ_n , giving $\rho_n \approx 1.115$. The resulting bound from (25) is at most 2.1% of the peak density anywhere on I , while the actual maximum absolute error is just 0.43% of the peak density—both small enough that the Laplace approximation is essentially indistinguishable from the exact density for our estimation purposes.

Proof of Proposition 4. By Proposition 1, $g(z | n) = \lambda_n(z) \exp[-\int_{\Theta(z|n)} \lambda_n(x) dx]$. Set $\delta := z - z^*$

and let $\bar{z} = z^* - \bar{\delta}$ ($\bar{\delta} \geq 0$, on the opposite side of z^*) satisfy $v(\bar{z} | n) = v(z | n)$. The quadratic bound in (24) gives $-(\frac{\xi_n}{2} + A_n)(w - z^*)^2 \leq v(w | n) - v(z^* | n) \leq -(\frac{\xi_n}{2} - A_n)(w - z^*)^2$ for $w \in I$; applying it at $w = z$ and $w = \bar{z}$ yields $\rho_n^{-1}|\delta| \leq \bar{\delta} \leq \rho_n|\delta|$, so $|\Theta(z | n)| \in [(1 + \rho_n^{-1})|\delta|, (1 + \rho_n)|\delta|]$. With the intensity bound, $\int_{\Theta(z | n)} \lambda_n(x) dx \in [(1 - B_n)(1 + \rho_n^{-1})\lambda_n^*|\delta|, (1 + B_n)(1 + \rho_n)\lambda_n^*|\delta|]$, and multiplying by $\lambda_n(z) \in [(1 - B_n)\lambda_n^*, (1 + B_n)\lambda_n^*]$ gives $\underline{L}_n(z) \leq g(z | n) \leq \bar{L}_n(z)$. The same envelopes contain $L(\cdot | n)$ —e.g., the lower bound reduces to $(1 - B_n) \leq \exp([(1 + B_n)(1 + \rho_n) - 2]\lambda_n^*|z - z^*|)$, true since the right-hand side is at least 1—yielding (25). When $A_n = B_n = 0$, $\rho_n = 1$ and both envelopes equal $L(\cdot | n)$. $\square$

E Proof of Proposition 2

This section provides the deferred proof of Proposition 2 in Section 3.2, characterizing the observed income density in a neighborhood of a kink. Throughout, h_0^* and h_1^* denote the target-income densities under the linear schedules T_0 and T_1 , h_0 and h_1 the corresponding counterfactual observed densities (their Laplace smoothings, defined for all incomes), B the bunching mass, $\bar{\lambda} = 1/\mu$, and $\Delta z = \Delta t/\xi$ the constant level response.

The density profile and exact amplitudes

Under the kinked schedule the frictionless target incomes follow the standard bunching pattern: their density is the lower-tax counterfactual target density h_0^* below k and the upper-tax target density h_1^* above k , together with a point mass of B bunchers exactly at k . By Corollary 4 the observed density h is the Laplace smoothing of this distribution—its convolution with the kernel $\bar{\lambda}e^{-2\bar{\lambda}|x|}$, $\bar{\lambda} = 1/\mu$ —which we evaluate on each side of the kink.

Away from k the target density is smooth, and there Corollary 5 links the observed and target densities through $h^* = h - \frac{\mu^2}{4}h''$. Below k the target density is h_0^* , and the lower-tax counterfactual h_0 already satisfies $h_0^* = h_0 - \frac{\mu^2}{4}h_0''$; subtracting this from the same identity for h cancels the common right-hand side, so the difference $g := h - h_0$ satisfies $g - \frac{\mu^2}{4}g'' = 0$, i.e. $g = \frac{\mu^2}{4}g''$, whose solutions are the exponentials $e^{\pm 2z/\mu}$. Only the decaying one can appear, since the kink leaves incomes far below it undisturbed ($h \rightarrow h_0$ as $z \rightarrow -\infty$); writing its still-unknown coefficient as C_- ,

$$h(z) = h_0(z) + C_- e^{-2(k-z)/\mu}, \quad z \leq k,$$

and symmetrically $h(z) = h_1(z) + C_+ e^{-2(z-k)/\mu}$ for $z \geq k$, with a second unknown coefficient C_+ . The constants C_- and C_+ are the heights of the diffuse bunching mass just to the left and right of the threshold; this is the piecewise form (18), and two conditions at k determine them.

First, smoothing by a continuous kernel leaves h continuous, so the two branches agree at $z = k$:

$$h_0(k) + C_- = h_1(k) + C_+.$$

Second, smoothing the (bounded) density part of the target distribution gives a curve with no slope break at k , so the only slope discontinuity in h comes from the smoothed point mass: its contribution $B\bar{\lambda}e^{-2\bar{\lambda}|z-k|}$ has a slope that drops by $4B/\mu^2$ as z crosses k . Hence $h'(k^-) - h'(k^+) = 4B/\mu^2$, and substituting the derivatives of (18) yields

$$h'_1(k) - h'_0(k) - \frac{2}{\mu}(C_+ + C_-) = -\frac{4B}{\mu^2}.$$

Solving these two linear equations gives the exact amplitudes

$$C_{\pm} = \frac{B}{\mu} + \frac{\mu}{4}(h'_1(k) - h'_0(k)) \mp \frac{1}{2}(h_1(k) - h_0(k)). \quad (26)$$

Evaluating (18) at $z = k$ from below, $h(k) = h_0(k) + C_-$, gives the exact peak

$$h(k) = \frac{B}{\mu} + \frac{h_0(k) + h_1(k)}{2} + \frac{\mu}{4}(h'_1(k) - h'_0(k)). \quad (27)$$

Equations (26)–(27) are the general (nonlinear) counterparts of the linear-window expressions in Proposition 2.

The structural approximation and its error

It remains to re-express the amplitudes in the structural parameters (μ, ξ) and to bound the error. Under the constant level response $\Delta z = \Delta t/\xi$ the bunchers are exactly the types with t_0 -target income in $(k, k + \Delta z)$, so $B = \int_k^{k+\Delta z} h_0^*(s) ds$. Because the response is a uniform translation, the target densities satisfy $h_1^*(w) = h_0^*(w + \Delta z)$; in particular $h_0^*(k + \Delta z) = h_1^*(k)$, so the endpoints of the integrand over the bunching window are $h_0^*(k)$ and $h_1^*(k)$, and the trapezoidal rule with remainder gives

$$B = \frac{\Delta z}{2}(h_0^*(k) + h_1^*(k)) - \frac{\Delta z^3}{12}h_0^{*''}(\zeta), \quad \zeta \in (k, k + \Delta z).$$

Applying the exact deconvolution identity $h_j^*(k) = h_j(k) - \frac{\mu^2}{4}h_j''(k)$,

$$\frac{B}{\mu} = \frac{\Delta z}{2\mu}(h_0(k) + h_1(k)) - \underbrace{\frac{\mu\Delta z}{8}(h_0''(k) + h_1''(k)) - \frac{\Delta z^3}{12\mu}h_0^{*''}(\zeta)}_{=:R}. \quad (28)$$

The same translation gives the remaining ingredients of $C_{\pm}$ up to higher order: $h_1(k) - h_0(k) = h_0(k + \Delta z) - h_0(k) = \Delta z h'_0(k) + O(\Delta z^2)$ and $h'_1(k) - h'_0(k) = \Delta z h''_0(k) + O(\Delta z^2)$. Substituting into the exact amplitudes (26) and collecting,

$$C_{\pm} = \frac{\Delta z}{2\mu} (h_0(k) + h_1(k)) \mp \frac{\Delta z}{2} h'_0(k) - R + \frac{\mu \Delta z}{4} h''_0(k) + O(\Delta z^2),$$

and using $\Delta z = \Delta t / \xi$ the leading terms reproduce the linear-window amplitudes of Proposition 2 (equation (18)). Likewise, the first term of (28) combines with $\frac{1}{2}(h_0(k) + h_1(k))$ in (27) to give the linear-window peak (19).

Finally we bound the error. Every dropped term is a second derivative of a counterfactual density near k multiplied by a factor of either $\mu \Delta z$ or $\Delta z^3 / \mu$; in particular the error is a density (the second derivative carries units income^{-3} and the factor income^2), not a raw length scale, and it vanishes with the counterfactual curvature. When diffusion is at least as wide as the bunching window, $\mu \geq \Delta z$, we have $\Delta z^3 / \mu \leq \mu \Delta z$, so both prefactors are $O(\mu \Delta z) = O(\mu \Delta t / \xi)$ and the leading error term is $-\frac{\mu \Delta z}{8}(h''_0(k) + h''_1(k))$, inherited from R . All such terms vanish when h_0 and h_1 are linear across the bunching window $[k, k + \Delta t / \xi]$, so the structural approximations are then exact. $\square$

F Bunching Density Around a Notch

This section characterizes the observed income density around a threshold carrying a notch, complementing the kink characterization of Proposition 2 in Section 3.2. In the presence of a notch, dominating income regions need not be compact, as illustrated in Figure A3(b). As a result, type-conditional densities lose their Laplace form and the observed density no longer exhibits two-sided exponential decay. Nevertheless, the density can be quantified as shown in the following proposition.

Proposition 5 (Bunching density around a kink-and-notch). *Consider a threshold k carrying both a kink and a notch: $T(z) = T_0(z)$ for $z \leq k$ and $T_1(z)$ for $z > k$, with marginal rates $t_0, t_1 = t_0 + \Delta t$ and $T_1(k) = T_0(k) + \Delta T$. Let h_0, h_1 be the linear-tax counterfactual densities, $h_j^* = h_j - \frac{\mu^2}{4} h_j''$ the target densities, $\Delta z = \Delta t / \xi$, $\bar{\lambda} = 1 / \mu$, and $B = H_1(k + \Delta z^D) - H_0(k)$ the bunching mass. No frictionless taxpayer locates in the dominated region $(k, k + \Delta z^D)$, whose width follows from the marginal buncher's indifference between k and its interior optimum, $\frac{\xi}{2}(\Delta z^D)^2 = \Delta T$:*

$$\Delta z^D = \sqrt{2\Delta T / \xi} \quad (\text{independent of } \Delta t). \quad (29)$$

Indexing each type by its T_0 -target s , the observed density is

$$h(z) = \bar{\lambda} \int_{-\infty}^{\infty} \exp[-\bar{\lambda} \Theta(z, s)] h_0^*(s) ds, \quad (30)$$

where $\Theta(z, s)$ is the length of the type's dominating set (derived in the proof below): the undeformed Laplace kernel $\Theta = 2|z - s|$ for targets $s \leq k$, and a deformed kernel pulling relocated targets above the dominated region back toward k . The density peaks at the kink,

$$h(k) = \frac{B}{\mu} + \left[\frac{1}{2} h_0(k) - \frac{\mu}{4} h_0'(k) \right] + \int_{k+\Delta z^D}^{\infty} \bar{\lambda} e^{-2\bar{\lambda} \sqrt{(s-k)^2 - (\Delta z^D)^2}} h_1^*(s) ds, \quad (31)$$

and is continuous except at k , where it drops to the strictly lower, still-positive right-limit

$$h(k^+) = \bar{\lambda} \int_{-\infty}^{\infty} \exp\left(-\bar{\lambda} \left[\sqrt{(s-k)^2 + (\Delta z^D)^2} - (s-k) + 2(s-k-\Delta z)_+ \right] \right) h_0^*(s) ds, \quad (32)$$

with $(x)_+ := \max(0, x)$. As $\Delta T \rightarrow 0$ the dominated region closes and the drop vanishes, returning the characterization to the kink density of Proposition 2; as $\Delta t \rightarrow 0$ it reduces to a pure notch.

Equations (31)–(32) reproduce the three signatures of notch bunching visible in Figure 7(b): diffuse excess mass below k that peaks at the threshold, a discontinuous drop at k , and strictly positive density in the dominated region where the frictionless model predicts none. The notch also sharpens identification: with ΔT known, the dominated-region width (29) pins down ξ through a channel distinct from the diffuse excess mass, while μ governs both the width of the bunching mass below k and the size of the drop.

The remainder of this section proves the proposition. Throughout, $\bar{\lambda} = 1/\mu$; h_0, h_1 are the counterfactual densities and $h_j^* = h_j - \frac{\mu^2}{4} h_j''$ the target densities under T_0, T_1 (Corollary 5), related by the kink translation $h_1^*(s) = h_0^*(s + \Delta z)$ with $\Delta z = \Delta t/\xi$; and we shift coordinates so that $k = 0$, indexing a type by its T_0 -target σ (density h_0^*). Set $D := \Delta z^D$.

Indirect utility, the marginal buncher, and the dominating set

The schedule is $T(z) = T_0(z)$ for $z \leq k$ and $T(z) = T_1(z)$ for $z > k$, where T_0, T_1 are the linear taxes of rates t_0, t_1 with $T_1(k) = T_0(k) + \Delta T$. Under the uniform sparsity model with quadratic indirect utility of curvature ξ , the type with T_0 -target σ has money-metric indirect utility equal to the rate- t_0 value on the lower branch ($x \leq 0$) and the rate- t_1 value on the upper branch ($x > 0$), downward parabolas with vertices at the T_0 - and T_1 -targets σ and $\sigma - \Delta z$:

$$v(x) = \begin{cases} M_0 - \frac{\xi}{2}(x - \sigma)^2, & x \leq 0, \\ M_1 - \frac{\xi}{2}(x - (\sigma - \Delta z))^2, & x > 0, \end{cases}$$

where M_0, M_1 are the branch peak values. Write $\phi_0(0) := M_0 - \frac{\xi}{2}\sigma^2$ and $\phi_1(0) := M_1 - \frac{\xi}{2}(\sigma - \Delta z)^2$ for the one-sided values at the threshold. Because $T_1(k) = T_0(k) + \Delta T$, consumption at 0 is ΔT lower on the upper branch, so $\phi_1(0) = \phi_0(0) - \Delta T$ (money-metric) and v falls by ΔT as income crosses 0 from below.

The marginal buncher is the highest-target type still choosing 0, indifferent between 0 (value $\phi_0(0)$) and its upper-branch optimum at the vertex $\sigma - \Delta z$ (value M_1). Indifference $\phi_0(0) = M_1$, together with $\phi_1(0) = M_1 - \frac{\xi}{2}(\sigma - \Delta z)^2 = \phi_0(0) - \Delta T$, gives $\frac{\xi}{2}(\sigma - \Delta z)^2 = \Delta T$, i.e. the upper vertex sits at $\sigma - \Delta z = D = \sqrt{2\Delta T/\xi}$, independent of Δt . Hence the dominated region is $(0, D)$, establishing (29), with bunchers occupying $\sigma \in (0, D + \Delta z)$ and mass $B = H_1(k + D) - H_0(k)$.

For income δ at value $\ell := v(\delta)$, the dominating set is the super-level set $\{x : v(x) \geq \ell\}$, the union of one interval per branch: half-width $\rho_0 = \sqrt{2(M_0 - \ell)/\xi}$ about σ on the lower branch, and $\rho_1 = \sqrt{2(M_1 - \ell)/\xi}$ about $\sigma - \Delta z$ on the upper branch (empty when the radicand is negative). Clipping the lower interval to $\{x \leq 0\}$ and the upper to $\{x > 0\}$ —disjoint sets—gives the dominating-set length $\Theta(z, s) = L_0 + L_1$ with

$$L_0 = |[\sigma - \rho_0, \sigma + \rho_0] \cap (-\infty, 0]|, \quad L_1 = |[(\sigma - \Delta z) - \rho_1, (\sigma - \Delta z) + \rho_1] \cap (0, \infty)|. \quad (33)$$

Aggregating $\bar{\lambda} e^{-\bar{\lambda}\Theta(z,s)}$ over targets with density h_0^* gives (30). The full profile is evaluated numerically; the peak and the drop, derived next, are in closed form.

Peak

At $\delta = 0^-$, $\ell = \phi_0(0)$, so $\rho_0 = |\sigma|$ and (since $M_1 - \phi_0(0) = \frac{\xi}{2}(\sigma - \Delta z)^2 - \Delta T$) $\rho_1 = \sqrt{(\sigma - \Delta z)^2 - D^2}$. Evaluating (33):

- *Below* ($\sigma \leq 0$): $L_0 = 2|\sigma|$; the upper interval, centered at $\sigma - \Delta z \leq 0$ with radius $\rho_1 < |\sigma - \Delta z|$, lies entirely below 0, so $L_1 = 0$ and $\Theta = 2|\sigma| = 2|z - s|$ (undeformed Laplace, against h_0^*).
- *Bunchers* ($0 < \sigma < D + \Delta z$): $\sigma > 0$ forces $L_0 = 0$, and the upper vertex $\sigma - \Delta z < D$ makes ρ_1 imaginary, so $L_1 = 0$ and $\Theta = 0$.
- *Relocated* ($\sigma > D + \Delta z$, i.e. upper vertex $\sigma - \Delta z > D$): $L_0 = 0$ and the upper interval lies entirely above 0, so $\Theta = 2\rho_1 = 2\sqrt{(\sigma - \Delta z)^2 - D^2}$.

Hence $h(0^-) = \bar{\lambda} \int_{-\infty}^0 e^{-2\bar{\lambda}|\sigma|} h_0^* d\sigma + \bar{\lambda} \int_0^{D+\Delta z} h_0^* d\sigma + \bar{\lambda} \int_{D+\Delta z}^{\infty} e^{-2\bar{\lambda}\sqrt{(\sigma-\Delta z)^2-D^2}} h_0^* d\sigma$. The middle term is $\bar{\lambda}B = B/\mu$. The first is the partial Laplace smoothing from below: with kernel $w(s) = \bar{\lambda}e^{2\bar{\lambda}s}$ satisfying $w' = 2\bar{\lambda}w$, substituting $h_0^* = h_0 - \frac{\mu^2}{4}h_0''$ and integrating by parts twice (boundary terms at $-\infty$ vanish) gives $\frac{1}{2}h_0(0) - \frac{\mu}{4}h_0'(0)$, exactly as in the proof of Proposition 2. In the third, substitute $s = \sigma - \Delta z$ and use $h_0^*(s + \Delta z) = h_1^*(s)$ to obtain $\bar{\lambda} \int_D^{\infty} e^{-2\bar{\lambda}\sqrt{s^2-D^2}} h_1^*(s) ds$. Restoring k ,

these three terms are exactly (31). Since h rises toward 0 from below and (next) drops at 0, $h(0^-)$ is the peak.

Discontinuous drop

At $\delta = 0^+$, $\ell = \phi_1(0) = \phi_0(0) - \Delta T$, so $\rho_0 = \sqrt{\sigma^2 + D^2}$ and (since $M_1 - \ell = M_1 - \phi_1(0) = \frac{\xi}{2}(\sigma - \Delta z)^2$) $\rho_1 = |\sigma - \Delta z|$. Because $\sigma + \rho_0 > 0 > \sigma - \rho_0$, the lower interval clips to give $L_0 = \rho_0 - \sigma = \sqrt{\sigma^2 + D^2} - \sigma$; the upper interval, centered at $\sigma - \Delta z$ with radius $|\sigma - \Delta z|$, contributes $L_1 = 2(\sigma - \Delta z)_+$. Thus

$$\Theta(0^+, \sigma) = \sqrt{\sigma^2 + D^2} - \sigma + 2(\sigma - \Delta z)_+,$$

which, restoring k and aggregating, is (32).

The type-conditional density $\bar{\lambda}e^{-\bar{\lambda}\Theta(\cdot, s)}$ is continuous in z wherever $\nu(\cdot)$ is, hence everywhere except at 0, the sole discontinuity of ν . Comparing limits target by target, $\Theta(0^+, \sigma) - \Theta(0^-, \sigma)$ equals $\sqrt{\sigma^2 + D^2} + \sigma \geq 0$ for $\sigma \leq 0$; equals $\sqrt{\sigma^2 + D^2} - \sigma + 2(\sigma - \Delta z)_+ > 0$ for bunchers; and for relocated σ equals $[\sqrt{\sigma^2 + D^2} - \sigma] + 2[(\sigma - \Delta z) - \sqrt{(\sigma - \Delta z)^2 - D^2}] > 0$, a sum of two strictly positive terms. Hence $\Theta(0^+, \sigma) > \Theta(0^-, \sigma)$ for every σ whenever $D > 0$, so the integrand in (30) is strictly smaller at 0^+ and $h(0^+) < h(0^-) = h(k)$: the density drops discontinuously at the notch. Positivity of h throughout $(0, D)$ is immediate, since the integrand in (30) is strictly positive.

Nesting

As $\Delta T \rightarrow 0$, $D \rightarrow 0$: the dominated region closes, the deformed kernel $\sqrt{(\sigma - \Delta z)^2 - D^2}$ becomes $|\sigma - \Delta z|$ so the relocated term reverts to the undeformed Laplace contribution $\frac{1}{2}h_1(0) + \frac{\mu}{4}h'_1(0)$ of Proposition 2, and $\Theta(0^+, \sigma) \rightarrow \Theta(0^-, \sigma)$ so the drop vanishes—recovering the general kink peak (27). As $\Delta t \rightarrow 0$, $\Delta z \rightarrow 0$ and $h_1 = h_0$, giving the pure notch: the upper and lower vertices coincide, $\rho_1 = \sqrt{\rho_0^2 - D^2}$ throughout, and (31)–(32) reduce to integrals of $\sqrt{\sigma^2 - D^2}$ and $\sqrt{\sigma^2 + D^2} + |\sigma|$ against the single counterfactual h_0^* . $\square$

G Separate identification with an unknown counterfactual: proof

This appendix proves Proposition 3 of Section 3.3.2. We use the kink characterization of Proposition 2: on the left of k , $h(z) = h_0(z) + C_- e^{-2(k-z)/\mu}$ with $C_- \neq 0$ under positive bunching, and analogously on the right.

Proof of Proposition 3. (i) The operator $1 - \frac{\mu^2}{4}\partial^2$ maps degree- Q polynomials bijectively onto degree- Q polynomials, so h_0^* of degree Q implies the counterfactual $h_0 = h_0^* + \frac{\mu^2}{4}h_0^{*''} + \dots$ is itself a degree- Q polynomial. Hence on the left $h = P_Q + C_-e^{-2(k-z)/\mu}$ with $\deg P_Q = Q$. For a trial scale ν ,

$$h - \frac{\nu^2}{4}h'' = (P_Q - \frac{\nu^2}{4}P_Q'') + C_-(1 - \nu^2/\mu^2)e^{-2(k-z)/\mu},$$

whose first term has degree Q ; the whole expression is a degree- Q polynomial iff the exponential coefficient vanishes, i.e. iff $\nu = \mu$ (using $C_- \neq 0$). Differentiating n times, $h^{(n)} = P_Q^{(n)} + C_-(2/\mu)^n e^{-2(k-z)/\mu}$, so for $n > Q$ we have $P_Q^{(n)} = 0$ and $h^{(n+2)} = \frac{4}{\mu^2}h^{(n)}$. If instead h_0^* is real-analytic and smooth on a scale longer than $\mu/2$, its derivatives grow more slowly than $(2/\mu)^n$, the bunching mass dominates, and $h^{(n+2)}/h^{(n)} \rightarrow 4/\mu^2$. None of these statements involves ξ or the coefficients of P_Q .

(ii) Given μ , the bunching mass is annihilated by $1 - \frac{\mu^2}{4}\partial^2$, so on the left $h - \frac{\mu^2}{4}h'' = h_0 - \frac{\mu^2}{4}h_0'' = h_0^*$ (Corollary 5); deconvolution thus recovers h_0^* on each side, and by continuity h_0^* up to k . The slope discontinuity (20) gives $B = \frac{\mu^2}{4}[h'(k^-) - h'(k^+)]$. The bunchers are exactly the types with T_0 -target in $(k, k + \Delta z)$, $\Delta z = \Delta t/\xi$, so $B = \int_k^{k+\Delta t/\xi} h_0^*(s) ds$; the right-hand side is continuous and strictly decreasing in ξ (the upper limit falls and the integrand is positive), so the equation has a unique root ξ , which with $e = (1 - t)/(\xi z^*)$ identifies the elasticity. $\square$

H Simulations

Using simulated data with known underlying parameters, we can assess the performance of our proposed estimation method, as compared to the conventional approach, in the presence of sparsity-based frictions. Given the large literature estimating elasticities from kinks, as opposed to notches, we focus primarily on the conventional “kink-based” bunching estimators as in Saez (2010) and Chetty et al. (2011).

We specify a simulated tax kink using the same parameters as in Figures 7 and A4: the marginal tax rate rises from $t_0 = 0.1$ to $t_1 = 0.2$ at the threshold $k = 300,000$. We simulate income densities assuming a baseline elasticity of $e_0 = 0.3$ and a lumpiness parameter of $\mu_0 = 10,000$, where the “0” subscript denotes the true parameters of the data-generating process, as distinct from model estimates of the parameters, which are denoted $\hat{e}$ and $\hat{\mu}$. We construct these simulations using linear underlying ability density, $f(n|\theta) = \theta_0 + \theta_1 n$, with $\theta_0 = 1000$ and $\theta_1 = -50$. Each simulation uses a taxpayer population of 100,000, which produces an amount of sampling noise similar to our empirical distributions in Figure A10. (The simulations in Figures 7 and A4 used a much higher population size of 2 million to illustrate the shape of the bunching mass with less sampling noise.)

We simulate these income distributions in two steps. First, we draw ability values (n_i) from the known ability density $f(n|\theta)$ in the vicinity of the tax bracket threshold.²³ For each ability draw, we then simulate a set of income opportunities drawn from a Poisson process, from which we choose, for each agent, the highest-utility option.²⁴

H.1 Performance of our estimator and the conventional bunching estimator

To assess the performance of our estimation method, we simulate many rounds of data from the same data-generating process with sparsity-based frictions, and in each case, we apply our estimation procedure to jointly estimate $\hat{e}$ and $\hat{\mu}$. We are interested in whether the distribution of these estimates is centered around the parameters of the data-generating process e_0 and μ_0 , and how often the estimated confidence intervals contain the true value.

One example round of simulated data is displayed in Figure A5(a). The green dots plot the simulated income histogram. The estimated parameters $\hat{e}$ and $\hat{\mu}$ resulting from our maximum likelihood estimation are reported in the upper corner, along with the 95 percent confidence interval for each estimate. The orange line plots the model-predicted income density under these estimated parameter values.

Before comparing to the conventional estimate, we analyze multiple simulation draws from the uniform sparsity model to understand (1) standard error performance and (2) simulation based evidence for separate identification of the elasticity and friction parameters.

Figure A6(a) plots the joint distribution of estimates ($\hat{e}, \hat{\mu}$) across 1000 simulation rounds, using the same data generating process underlying Figure A5(a). The histogram of each marginal distribution is displayed outside of each axis. A number of notable features emerge. First, for both $\hat{e}$ and $\hat{\mu}$, the distribution of estimates is centered around the true parameter value. Averaging across simulation rounds, the average value of $\hat{e}$ is 0.307 and the average value of $\hat{\mu}$ is 10.1 (measured in 1000s), close to the true values of $e_0 = 0.3$ and $\mu_0 = 10$.

²³Specifically, we draw 100,000 values of n_i between a set of bounds $\underline{n}$ and $\bar{n}$, with the probability of drawing any value n proportional to $f(n|\theta)$. To choose the lower bound $\underline{n}$, we note that due to frictions, the set of agents who earn a given z will include types whose target incomes are well below and well above z . Therefore, to simulate the income density near the bounds of an income range $[\underline{z}, \bar{z}]$, we must draw from an ability density with target incomes well outside that range. We choose $\underline{n}$ and $\bar{n}$ such that $z^*(\underline{n}) = \underline{z} - 100,000$ and $z^*(\bar{n}) = \bar{z} + 100,000$.

²⁴To simulate income opportunity sets, we exploit the fact that differences between adjacent elements in a Poisson process are iid draws from an exponential distribution with mean μ . Thus, we can construct a random income opportunity set spanning an arbitrarily wide range around a type's preferred income $z^*(n)$ by joining a random set of above-target opportunities, $\{z^*(n) + \varepsilon_a, z^*(n) + \varepsilon_a + \varepsilon_b, z^*(n) + \varepsilon_a + \varepsilon_b + \varepsilon_c, \dots\}$, with a random set of below-target opportunities $\{z^*(n) - \varepsilon_i, z^*(n) - \varepsilon_i - \varepsilon_j, z^*(n) - \varepsilon_i - \varepsilon_j - \varepsilon_k, \dots\}$, where the ε values are iid draws from an exponential distribution with mean μ . In the context of a kink, where indirect utility functions are concave, only a single element must be drawn in each set, since more distant draws are guaranteed to yield lower utility. For a notch, with non-concave indirect utility functions, a larger number of opportunities is drawn, such that each agent's range of income opportunities spans across the local maxima in their indirect utility functions.

Second, the spread of both distributions provides an indication of sampling error. In each round of simulated data, the maximum likelihood estimation procedure also provides a standard error estimate, and so a key question is whether this estimate gives an accurate picture of the degree of precision in the estimate. To explore this, we can compare the standard deviation of the distribution of $\hat{e}$ estimates, which is 0.026, to the average *estimate* of the standard error, which is 0.026, indicating that the maximum likelihood estimate of the standard error provides a good sense of the true degree of sampling uncertainty. In the case of μ , the standard deviation of the distribution of $\hat{\mu}$ is 1.121, and the average value of the estimated standard error is 1.099.

A third notable feature of Figure A6(a) is the upward slope in the cluster of joint estimates. This indicates that when $\hat{e}$ is overestimated due to sampling bias, it is likely that $\hat{\mu}$ is overestimated as well. To explore this phenomenon, Figure A6(b) plots model-generated income densities for five combinations of (e, μ) . The thick solid line plots the baseline density with $e = 0.3$ and $\mu = 10$. The other four lines correspond to the (e, μ) pairs corresponding to the four square-shaped points in Figure A6(a).

In Figure A6(b), the densities corresponding to the points to the northwest and southeast of the baseline are easy to visually distinguish from the baseline, exhibiting substantially lower and higher densities at the kink, respectively. A higher elasticity e increases the density at the kink point by raising the total amount of bunching mass. A *lower* value of the lumpiness parameter also increases the density at the kink point, by concentrating the excess mass more tightly around the kink (Figure A6(b)). Thus, the parameter combinations to the southeast of the baseline in Figure A6(a) correspond to densities with substantially higher density around the kink point, like the tallest density displayed in Figure A6(b). The reverse is true for parameter combinations to the northwest of the baseline values, where the levels of both parameters (low e and high μ) reinforce each other to push down the density at the kink. In contrast, parameter combinations to the northeast and southwest of the baseline have opposing effects on the density at the kink. They are still distinct, indicating that the model is identified, but their difference is more subtle, involving the density at intermediate points in between the kink point and the bounds of the income window. The pattern of points in Figure A6(a) corresponds to this visual impression: in the presence of sampling error, it is easier to distinguish—in a statistical sense—between data-generating processes with parameter pairs on the northwest-southeast axis than those on the northeast-southwest axis in Figure A6(b).²⁵

In sum, these points paint a clear picture of the performance of the maximum likelihood estimator when the model is correctly specified. Estimates of the elasticity and the lumpiness

²⁵Put differently, the estimator that we propose would find it easier to distinguish between data generated from “low e , high μ ” and “low μ , high e ” combinations than between “low μ , low e ” and “high μ , high e ” combinations.

parameter appear consistent in that they are distributed around the true parameters of the data-generating process, and standard errors estimated by maximum likelihood are very close to the standard deviation of the distribution of estimates. They also highlight an important aspect of this model: estimation error in e and μ are likely to have the same sign. This result has important implications for the comparison of this model to the conventional elasticity estimator based on bunching mass.

H.2 Specifying the Conventional Bunching Estimator

We now apply the conventional bunching estimator to the same simulation data. We apply the conventional bunching estimator based on Saez (2010) to estimate the income elasticity of the simulated datasets underlying Figure A5. We use as our baseline the implementation described in Chetty et al. (2011), which builds on Saez (2010) by estimating a counterfactual using a smoothed polynomial regression. We also later discuss comparisons to the methods used in Saez (2010) and Mortenson and Whitten (2016), the working paper that preceded Mortenson and Whitten (2020).

This estimation procedure involves two steps, first estimating a counterfactual income density based on the income density excluding data points near the kink, and then using the counterfactual density to estimate the excess mass from which the elasticity is recovered. To estimate the counterfactual density, we fit a polynomial of a specified degree to the observed income density, excluding the data in a specified window around the kink, using the following specification:

$$C_j = \sum_{i=0}^Q \beta_i^0 \cdot (Z_j)^i + \sum_{i=R_l}^{R_u} \gamma_i^0 \cdot \mathbf{1}[Z_j = i] + \epsilon_j^0. \quad (34)$$

Here, Q denotes the order of the polynomial, and R_l and R_u denote the lower and upper bounds of the “bunching window” near the kink, which is excluded from the polynomial estimation.²⁶ When estimating the polynomial regression, we follow Chetty et al. (2011) and impose an “integration constraint” such that the total count of observations across the empirical distribution equals the integral of observations under the counterfactual density across the plotted region.²⁷

²⁶The convention in Chetty et al. (2011) is to set a symmetric bunching window, such that $R_l = -R_u$. We allow for the possibility of an asymmetric bunching window, following the approach in Bosch, Dekker and Strohmaier (2020), which we detail below.

²⁷Kleven (2016) notes that imposing an integration constraint may bias the elasticity estimate: “This approach may introduce bias, especially in relatively flat distributions in which interior responses do not affect bin counts (except at the very top of the distribution away from the threshold being analyzed). It would be feasible to implement a conceptually more satisfying approach that does not have this potential bias, but for the reasons stated above, it will matter very little in most applications.” As we discuss below, our results confirm that the integration constraint introduces bias, and that the introduced bias may be substantial.

The second step is to compute the excess mass of incomes around the kink relative to this counterfactual density. Using equation (34), we compute the counterfactual mass in each bin within the bunching window, $\hat{C}_j^0$. Subtracting this predicted mass from the observed density yields the estimated excess number of individuals who report incomes near the kink relative to this counterfactual distribution:

$$\hat{B} = \sum_{i=R_l}^{R_u} (C_i - \hat{C}_i^0) = \sum_{i=R_l}^{R_u} \hat{\gamma}_i^0. \quad (35)$$

We then map this excess mass estimate to an estimated elasticity using the approximation from Chetty et al. (2011):

$$\hat{e} \approx \frac{\hat{B}}{z^* \cdot h_0(z^*) \cdot \log\left(\frac{1-t_0}{1-t_1}\right)}. \quad (36)$$

Standard errors for $\hat{e}$ are estimated using a bootstrap procedure. We resample with replacement from the underlying distribution of firms 1000 times, re-estimating the elasticity each time, and defining the standard error as the standard deviation of the distribution of $\hat{e}$ estimates.

This conventional estimation method relies on three parameter inputs: the lower and upper bounds of the bunching window (R_l and R_u) and the order of the polynomial (Q). These are often left to the discretion of the researcher to be chosen via “visual inspection.” We instead follow the algorithmic approach proposed in Bosch, Dekker and Strohmaier (2020), which allows the polynomial order and the bunching region to be informed by the data itself.²⁸

Figure A5(b) presents the results. The bunching window is bounded by dashed lines, and the orange line displays the fitted counterfactual density outside that window.²⁹ The estimated elasticity and bootstrap-based 95 percent confidence interval is reported in the corner.

²⁸This approach proceeds in five steps: (1) Estimate equation (34) with no bunching window—so that the polynomial estimation excludes only the bins adjacent to the kink—for a range of polynomial orders, retaining the specification that minimizes the Bayesian Information Criterion (BIC). (2) Define the lower bound of the bunching window as the leftmost set of two adjacent bins below the threshold where the actual count in each bin exceeds the 95 percent confidence interval of the predicted bin counts from equation (34), and define the upper bound using an analogous procedure to the right of the kink. (3) Repeat steps (1) and (2), widening the bunching window by one bin above and below the kink each time. Each such iteration produces a candidate set of bounds for a bunching window. (4) From the resulting distributions of candidate bounds, choose the modal lower bound and upper bound to define the preferred bunching window. (5) Using this preferred bunching window, re-estimate the final counterfactual regression with the preferred polynomial order as in Step (1).

²⁹Strictly speaking, this line represents the counterfactual *frequency*, equal to the counterfactual density scaled up by the bin width of the empirical histogram in order to render the plots visually comparable.

H.3 Comparison of uniform sparsity to conventional method elasticities

Comparing the elasticity estimates from the two methods, we note that the conventional bunching estimator in Figure A5(b) underestimates the true elasticity of the data-generating process by 25 percent. It also provides a misleading sense of precision: the 95 percent confidence interval does not contain the true elasticity. In contrast, the sparsity-based friction estimator in panel (a) is close to the true value of $e_0 = 0.3$, which is spanned by the 95 percent confidence interval.

To compare the relative performance of these estimators more generally, we apply them to 1000 different rounds of simulated data. Figure A7(a) plots the histogram of elasticity estimates from the conventional bunching estimator and from our proposed estimation method. Consistent with the results from the single simulation round, the distribution of elasticity estimates from the conventional bunching estimator lies substantially below the true elasticity e_0 . The average of elasticity estimates under the conventional approach is 0.243, and the bootstrap-based 95 percent confidence intervals contain the true e_0 in less than 10 percent of the cases. In contrast, the distribution of elasticity estimates from our proposed estimation method is centered around e_0 , with an average value of $\hat{e}$ across these simulation rounds of 0.307. The estimated confidence intervals from our approach also provide an accurate sense of precision: across the 1000 estimation rounds, the 95 percent confidence intervals contained the true e_0 in 95.3 percent of cases.

The downward bias in the conventional bunching estimator appears to be driven by frictions, as illustrated by Figure A7(b). To construct this figure, we reproduce distributions like those in Figure A5(a) using several different values of the lumpiness parameter μ_0 . Figure A7(b) plots the mean and the 95 percent quantile interval of each distribution at each value of μ . When the lumpiness parameter is small—approaching the continuous-income-choice model—the mean estimate of $\hat{e}$ under the conventional approach is close to the true value of $e_0 = 0.3$. However, as μ_0 rises, the conventional estimator exhibits substantial bias, underestimating the true parameter by more than 50 percent at the highest plotted value of μ_0 . These estimates also provide a misleading sense of precision: the 95 percent quantile intervals remain about the same size as μ_0 rises, and their upper bound falls far below e_0 . In contrast, under our method, the distribution of $\hat{e}$ remains centered around e_0 as frictions increase. The 95 percent quantile interval grows with μ_0 , reflecting the increasing imprecision in the elasticity estimate as lumpiness increases. This imprecision accurately reflects the greater difficulty of discerning diffuse bunching mass from underlying features of the smooth ability density when frictions are substantial.

Why do frictions cause the conventional bunching estimator to be biased downward? We highlight two contributing factors. The first arises because diffusion in the bunching

mass makes it difficult to distinguish excess mass from patterns in the counterfactual income density. In the uniform sparsity model of frictions—and in many of the other sparsity-based frictions models it approximates, such as when income opportunities are drawn from a normal distribution around the target income—there is no window outside of which the bunching mass falls to zero. As a result, some excess bunching mass will spill over outside of any particular bunching window—including the window chosen visually or algorithmically when implementing the conventional approach. This spillover mass tends to “pull up” the estimated polynomial fit in the vicinity of the kink, causing the procedure to underestimate the difference between the observed density and the counterfactual, and hence the bunching mass. In our model, in contrast, the distortions due to frictions are endogenously modeled throughout the income distribution, including at points far from the threshold, and so they should not exert an upward pull on the ability density around the threshold.

To explore the role of this factor in producing the bias evident in Figure A7, we note that this source of bias should become more severe when the polynomial fit is allowed to be more flexible. In Figure A8, we reproduce the estimates in Figure A5 with different polynomial degrees of 1 (linear), 3, 5, and 10. Consistent with this story, Figure A8(b) shows that when the polynomial degree is higher, the counterfactual density bends farther up into the bunching mass, and the elasticity estimate is more severely biased downward. In contrast, Figure A8(a) demonstrates that our proposed method continues to estimate $\hat{e}$ close to e_0 across all polynomial degrees, suggesting that this method is robust to misspecification in the shape of the ability density in a way that the conventional approach is not.

Although this first factor appears to play an important role in the downward bias of the conventional bunching estimator, it does not appear to be the sole explanation, because even the linear polynomial specification in Figure A5(a) produces a substantial underestimate of the true elasticity.

The second factor contributing to downward bias in the conventional method relates to the integration constraint imposed when estimating the counterfactual polynomial fit. The logic for such a constraint comes from the observation that any taxpayers bunching around a threshold must come from points to the right of the threshold under the counterfactual, and so the total population under the actual and counterfactual income densities must be the same.³⁰ However, as illustrated by the hollow blue points in Figure A1(a), the presence of a kink may

³⁰Describing the rationale for imposing the integration constraint, Chetty et al. (2011) remarks that an unadjusted polynomial fit “... overestimates [the bunching mass] because it does not account for the fact that the additional individuals at the kink come from points to the right of the kink. That is, it does not satisfy the constraint that the area under the counterfactual must equal the area under the empirical distribution. To account for this problem, we shift the counterfactual distribution to the right of the kink upward until it satisfies the integration constraint.”

induce taxpayers to appear inside the plotted region who were previously outside of it. In other words, although such an integration constraint does apply to the global income density, it need not apply within the particular region over which the bunching estimator is applied.³¹

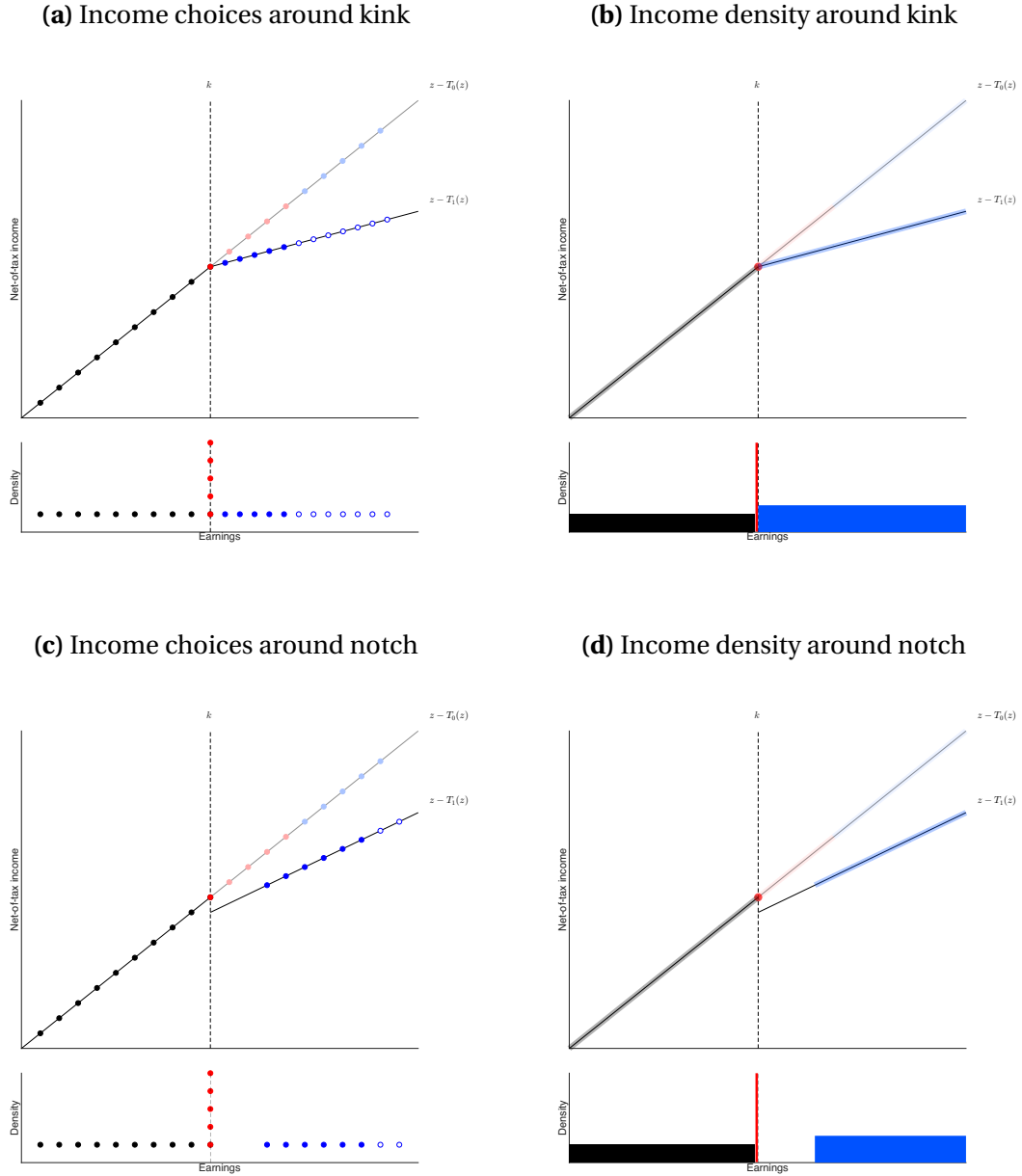
We now compare our estimator to other conventional bunching estimators that differ in how they construct counterfactuals, namely Saez (2010) and Mortenson and Whitten (2016), the working paper that preceded Mortenson and Whitten (2020), with an ultimate focus on the importance of integration constraint. We illustrate the differences in counterfactuals between these approaches in Figure A9(a). The approach developed in Saez (2010) constructs two linear counterfactuals on either side of the kink with the assumption that the densities are uniformly distributed on either side of the threshold. In order to construct the counterfactual, the approach takes the average value of the densities that occur outside of the bunching window and extrapolates that density through to the kink threshold. This is done on either side of the kink resulting in two counterfactuals. An alternative approach is developed in Mortenson and Whitten (2016) who construct a piecewise linear counterfactual on either side of the kink, in a similar vein to Saez (2010), but to allow for that counterfactual to take into account the slope of the observed densities on either side of the kink. Finally, we also consider an implementation of the approach in Chetty et al. (2011) where we do not impose the “integration constraint.” This allows for the possibility that the bunching mass may be reallocated to income bins beyond the region depicted in the histogram, which would cause the total integral under the counterfactual distribution to be smaller than the total integral of population across the empirical distribution. The practical implication of this is that the counterfactual distribution is shifted downward relative to the approach which imposes the integration constraint, as is depicted in Figure A9(a).

Next, we compare the elasticities produced under these four approaches to our estimates for varying values of the lumpiness parameter. We report these results in Figure A9(b). Imposing the integration constraint in the Chetty et al. (2011) approach produces lower elasticities than when the constraint is not imposed. Intuitively, by imposing the constraint, the counterfactual density is shifted upward, which causes the estimate of bunching to fall, leading to a lower elasticity. The Mortenson and Whitten (2016) elasticities are very similar to the Chetty et al. (2011) elasticities without the integration constraint. In that sense, the specification is nearly identical to the counterfactual specification in Mortenson and Whitten (2016), apart from the fact that the latter approach estimates two counterfactuals on either side of the threshold,

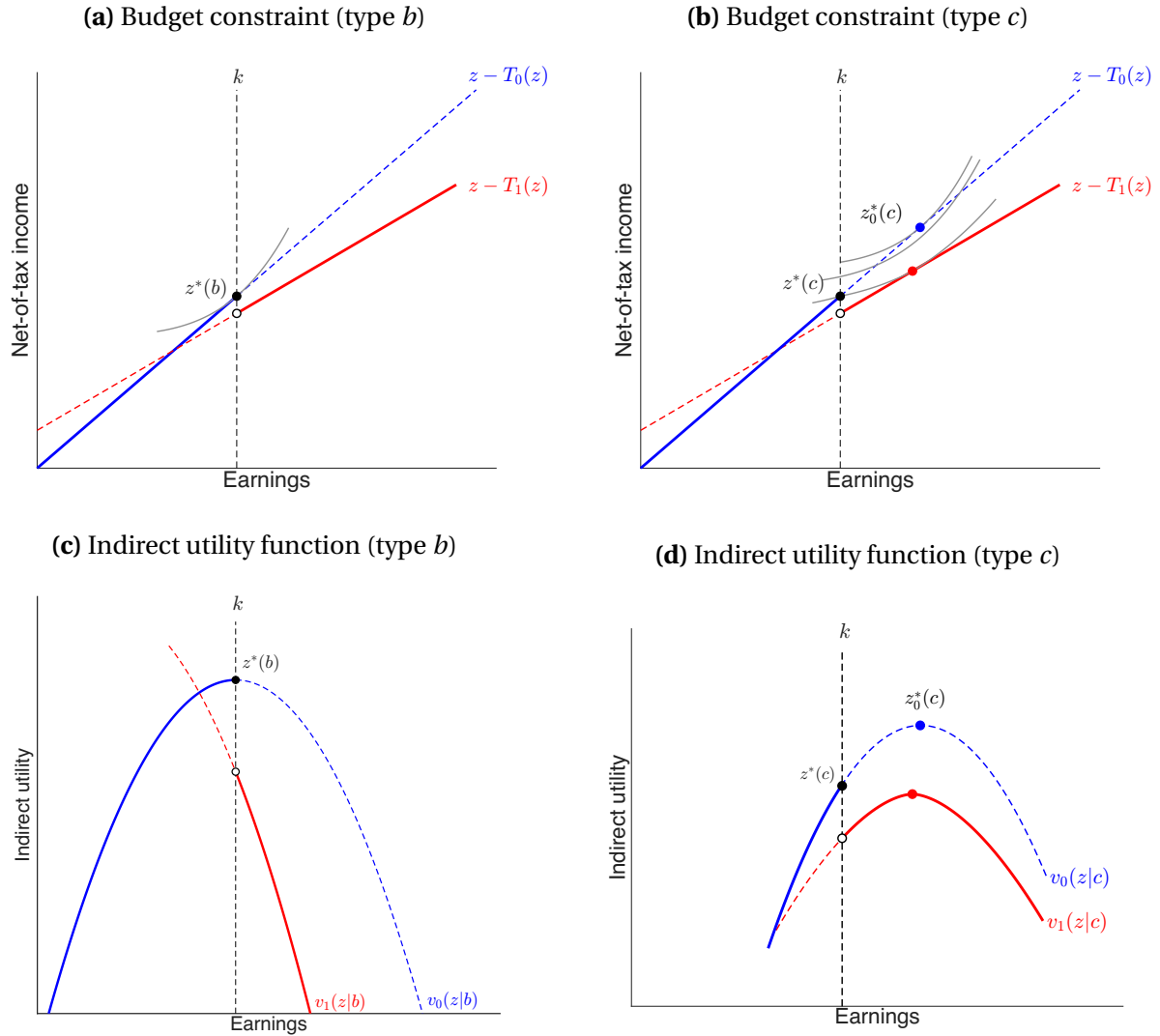
³¹Indeed, Figure A1(b), which illustrates the observed income density in the frictionless model with a continuous uniform type density, demonstrates that the kink may induce both extra mass at the kink *and* higher density at incomes above the kink, in which case the true counterfactual density under T_0 —which extends the uniform density below k to points above it—clearly has a lower integral over the plotted region than the observed density does.

thereby allowing for a different slope on the counterfactual on either side of the kink threshold. The visual similarity between these counterfactuals is evident in Figure A9(a). Out of all of the conventional approaches, the Saez (2010) approach produces the largest elasticities. The reason for this becomes evident when considering the counterfactuals produced in Figure A9(a). Given the empirical distribution slopes downwards, by assuming uniformly distributed densities, the Saez (2010) approach produces a counterfactual that is significantly lower than the other counterfactuals in the bunching region above the kink, leading to a higher measure of bunching, and a higher estimated elasticity.

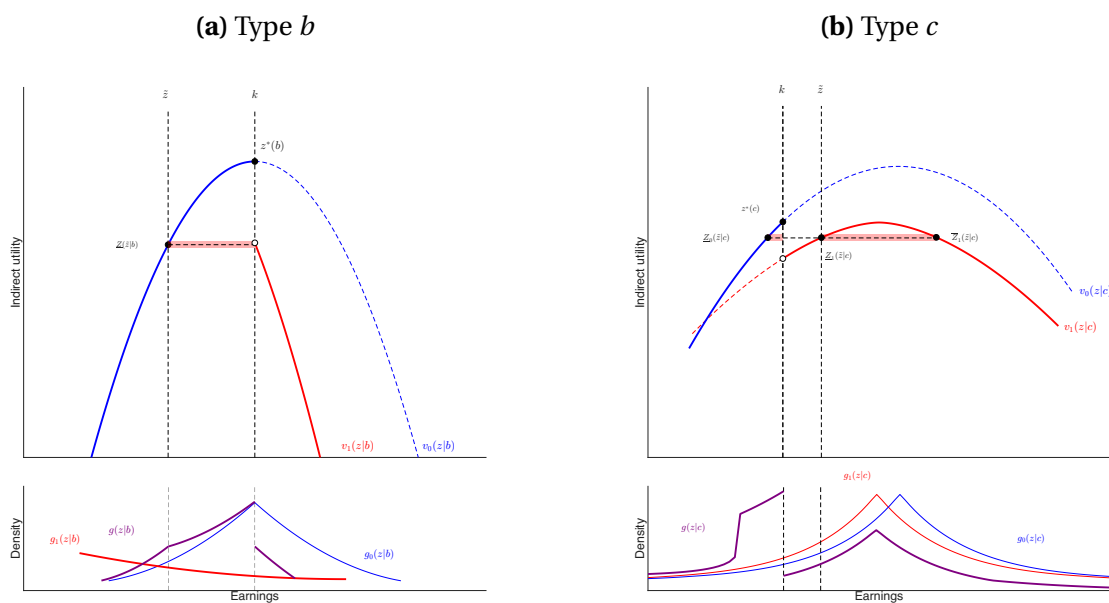
For smaller values of the lumpiness parameter, only Mortenson and Whitten (2016) and the Chetty et al. (2011) approach without the integration constraint can recover the true elasticity. However, for large values of the lumpiness parameter, not even these approaches are able to recover the true elasticity and exhibit a significant downward bias, whereas our approach can consistently recover the elasticity, irrespective of the extent of lumpiness in the observed empirical distribution.

Figure A1: Bunching patterns under a frictionless model

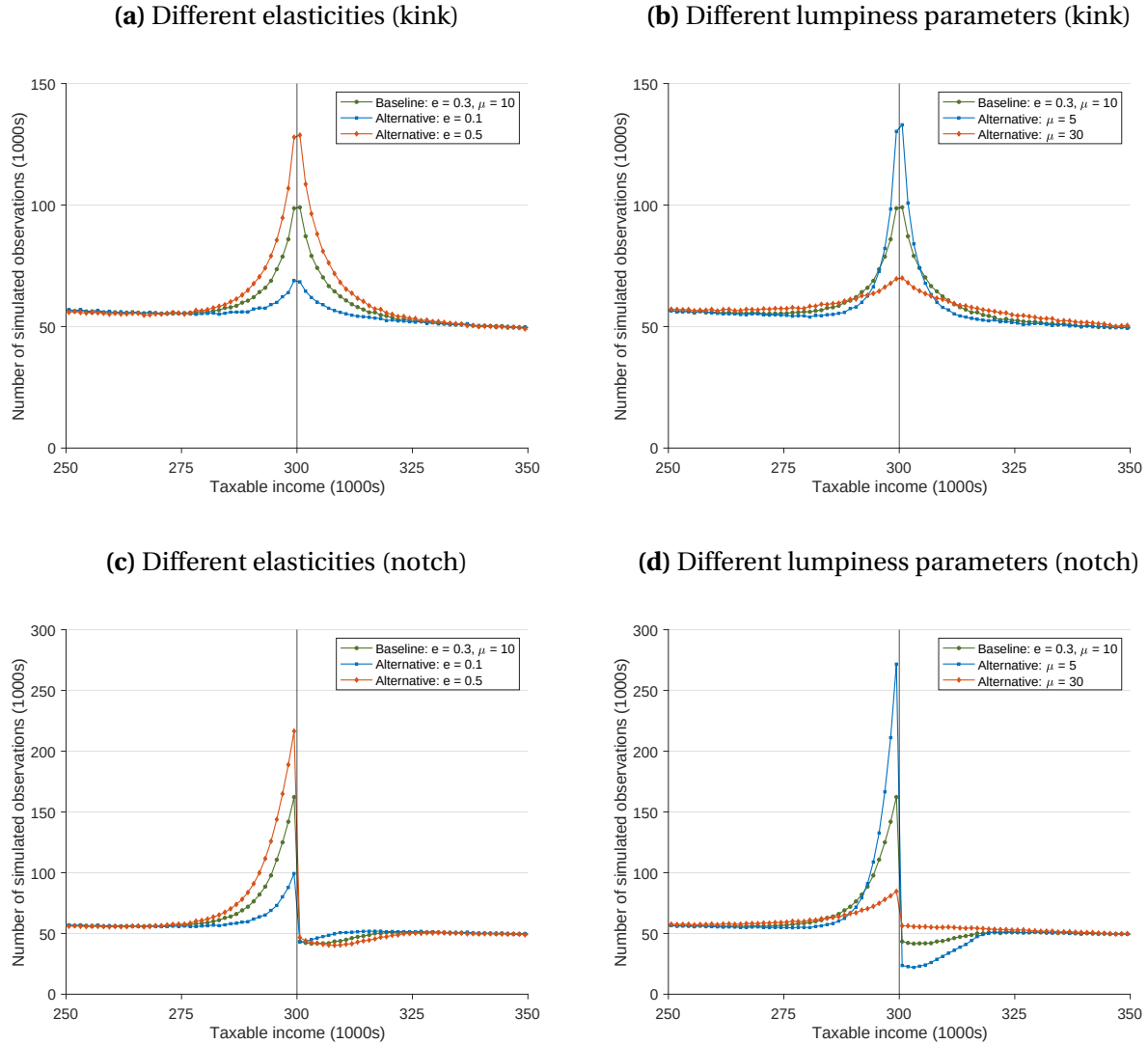
Panel (a) plots income choices for discrete types in the presence of a kink, for a uniform type distribution. Black points denote types who choose the same income under the linear tax $T_0(z)$ and the kinked tax schedule. Red points denote types who bunch at the bracket threshold k under the kinked tax schedule; their counterfactual income choices under $T_0(z)$ are plotted in light red for reference. Blue points denote types who choose incomes above the threshold under the kink. Hollow blue points denote agents whose counterfactual incomes under $T_0(z)$ lie outside the displayed range of incomes. The lower portion of panel (a) displays the observed probability density function from these choices. Panel (b) translates to the case of continuous types, which exhibits an atom of mass at the threshold k and a jump in the density around that threshold, due to the compression of incomes in response to the higher marginal tax rate above the kink. Panels (c) and (d) are the same as panels (a) and (b), but in the presence of a tax notch.

Figure A2: Utility from income choices around a tax notch

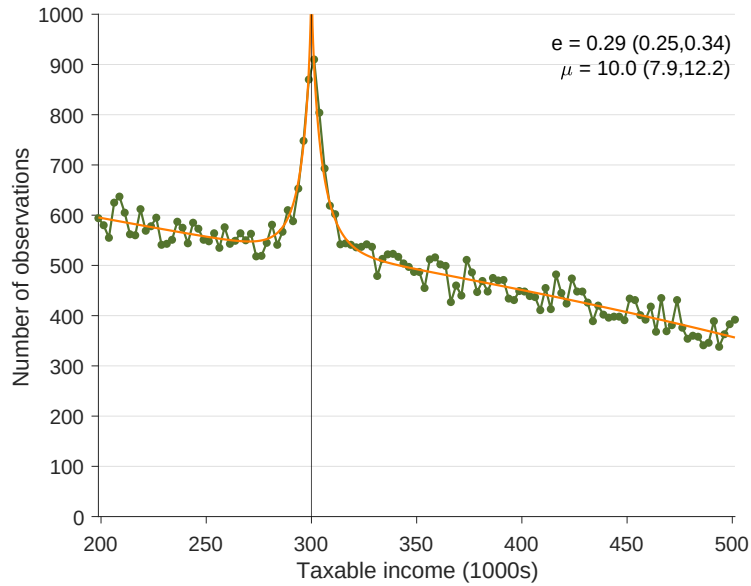
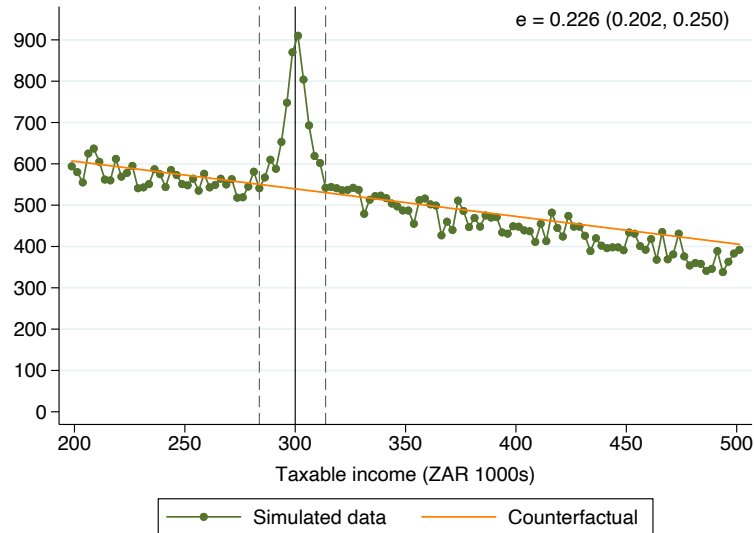
This figure is analogous to Figure 4, but in the presence of a notch, which produces a discontinuity in the indirect utility function (panels (c) and (d)). In the case of type c , the notch produces a non-monotonic indirect utility function with two local maxima (panel (d)).

Figure A3: Type-conditional income density around a notch

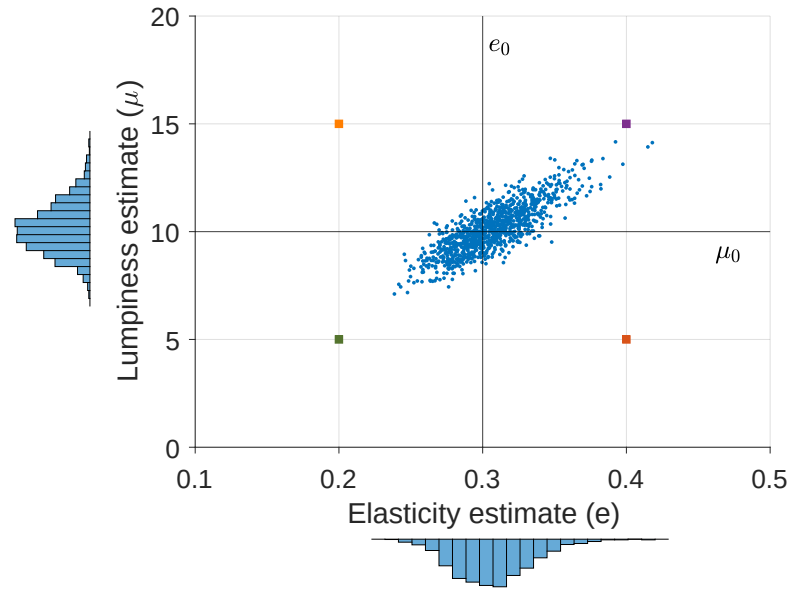
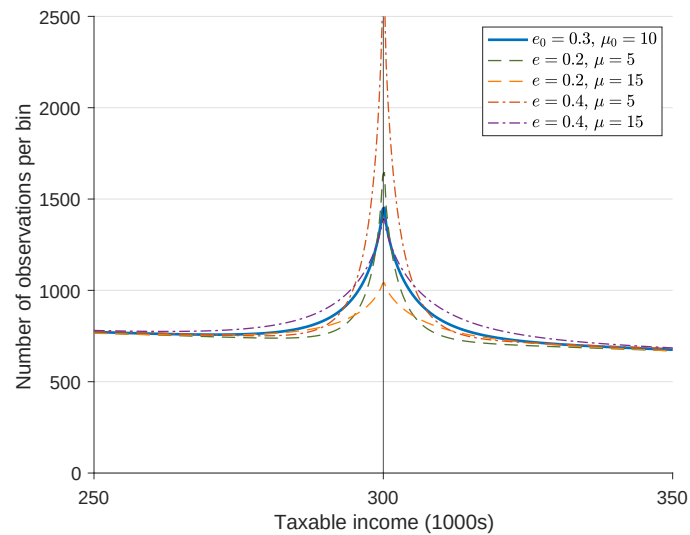
This figure is analogous to Figure 5, but in the presence of a notch. As shown in panel (b), when the indirect utility function has multiple local maxima, the dominating income range may be a disjoint set, in which case the type-conditional density is multimodal.

Figure A4: Simulated effects of parameter variations on income densities

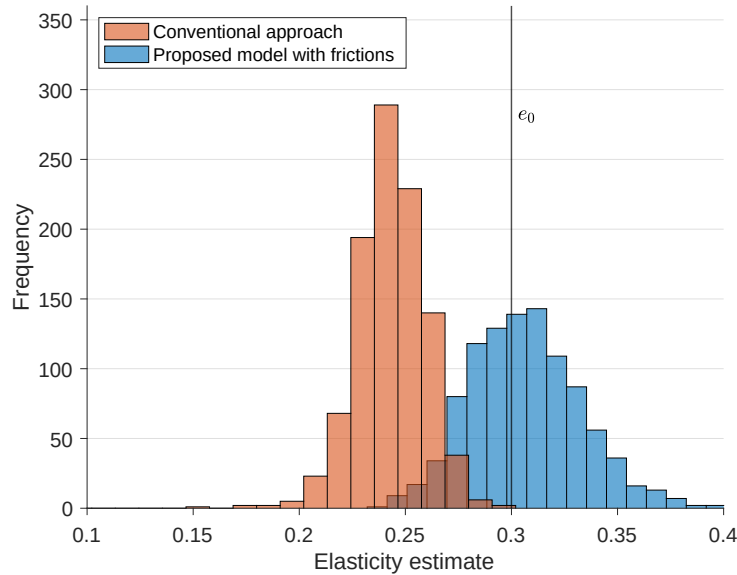
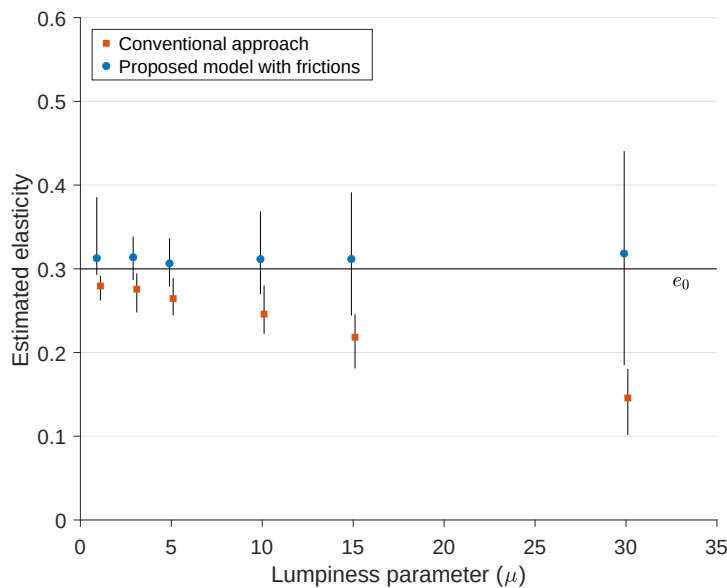
This figure plots income histograms from simulated datasets under the uniform sparsity model of frictions. For each simulation, we draw agents from an ability distribution with a linear density. We assume agents have a homogeneous income elasticity, e_0 , and for each agent we then draw a sparse set of income opportunities from a Poisson process with a specified lumpiness parameter, μ_0 . Each agent chooses the income opportunity that delivers the highest utility. We bin the resulting incomes to construct the income histograms displayed above. Panels (a) and (b) display simulated income histograms around a progressive tax kink for different values of e_0 and μ_0 , respectively. Panels (c) and (d) display histograms around a tax notch. In each case, the marginal tax rate rises from 0.1 to 0.2 at \$300,000, and for the notch simulations in panels (c) and (d), the level of tax liability increases by \$1000.

Figure A5: Parameter estimates from simulated data**(a)** Sparsity-based frictions estimator for a single simulation round**(b)** Conventional bunching estimator for a single simulation round

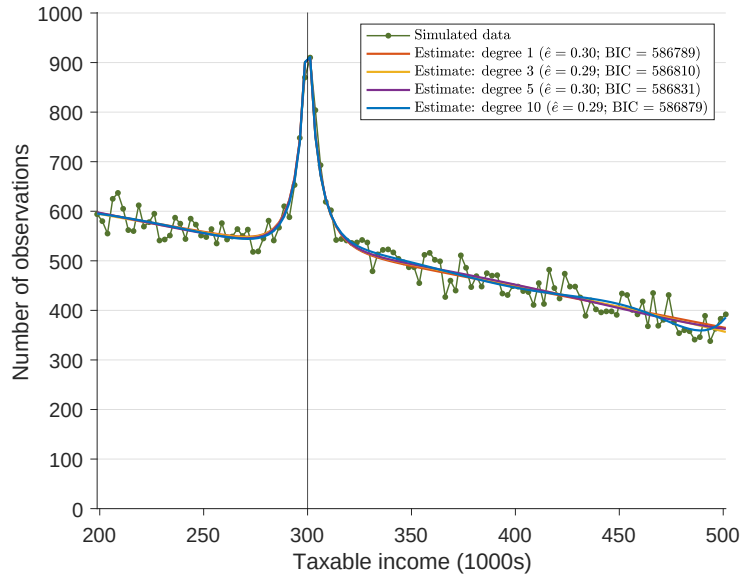
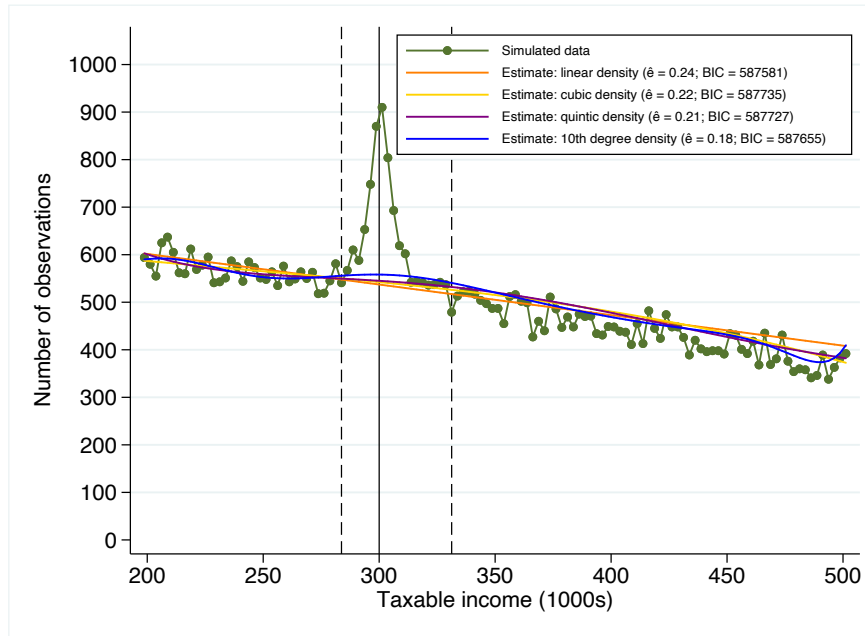
This figure displays the estimation of the maximum likelihood model and the conventional bunching estimator for one round of simulated data. The simulation is constructed as in Figure A4, with a true elasticity of $e_0 = 0.3$ and a lumpiness parameter of $\mu_0 = 10$, but using a smaller number of drawn observations ($M = 100,000$) to produce a level of sampling noise similar to that in our empirical application in Section 4. Panel (a) displays the results of applying our maximum likelihood estimation method described in Section 3.4. Estimates of $\hat{e}$ and $\hat{\mu}$ and their 95 percent confidence intervals are reported in the upper corner. Panel (b) illustrates the conventional bunching estimator, applied to the same round of simulated data, resulting in an elasticity estimate well below the true value $e_0 = 0.3$. The vertical dashed lines display the algorithmically selected bunching window, and the orange line plots the best-fit polynomial to the data points outside the bunching window.

Figure A6: Joint identification of elasticity and lumpiness estimates**(a)** Joint distribution of $\hat{e}$ and $\hat{\mu}$ estimates**(b)** Income densities for different combinations of e and μ 

In Panel (a), each blue point plots the combination of parameter estimates $(\hat{e}, \hat{\mu})$ from one round of simulated data like that in Figure A5(a). Marginal histograms of the estimates are plotted for each axis. Panel (b) plots the model-generated income density under the true parameters of the data-generating process, $e_0 = 0.3$ and $\mu_0 = \$10,000$, as well as under the four different combinations corresponding to the colored square points in panel (a).

Figure A7: Elasticity estimates using the conventional approach**(a)** Distribution of elasticity estimates under each approach**(b)** Elasticity estimates under each approach for different lumpiness parameters

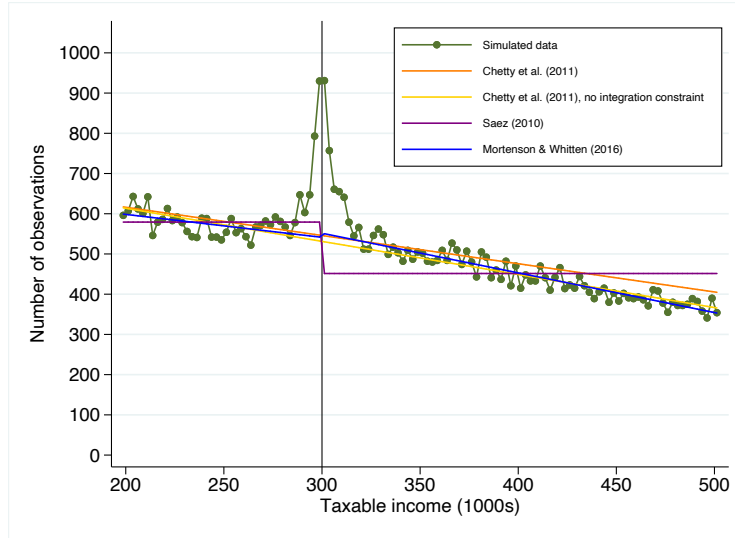
Panel (a) plots the histogram of elasticity estimates under the conventional approach (orange) and the maximum likelihood method allowing for frictions (blue). The vertical line at e_0 locates the true elasticity of the data generating process used to construct the simulated datasets. To construct panel (b), we produce histograms like those in panel (a) using simulated data with several different lumpiness parameters, holding fixed the true elasticity. Panel (b) displays the mean and 95 percent confidence intervals for the distribution of elasticity estimates in each case.

Figure A8: Estimated elasticities assuming different polynomial degrees**(a)** Estimator with frictions**(b)** Conventional estimator

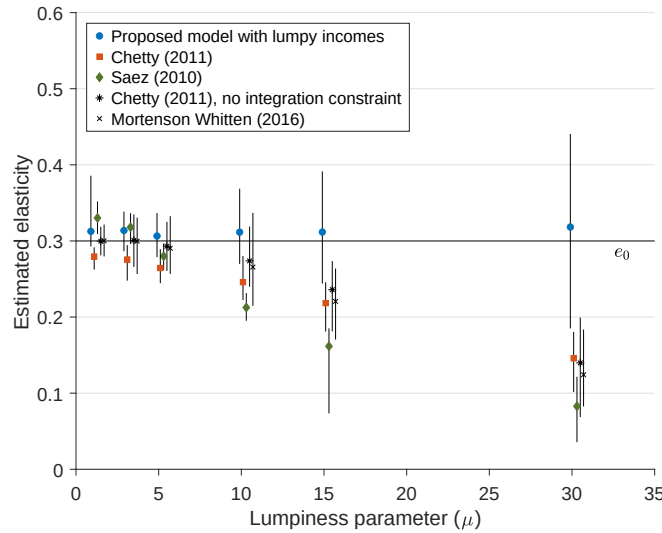
This figure reports the estimated elasticity for one round of simulated data, plotted in green, using both the conventional approach with frictionless income choice (panel a) and our estimation method with lumpy income choice (panel b), assuming different polynomial degrees for the counterfactual (or ability) density. The true ability density of the data-generating process is linear, with a true elasticity value of $e_0 = 0.3$.

Figure A9: Counterfactuals and elasticity estimates using various conventional approaches and our approach, for varying lumpiness parameters

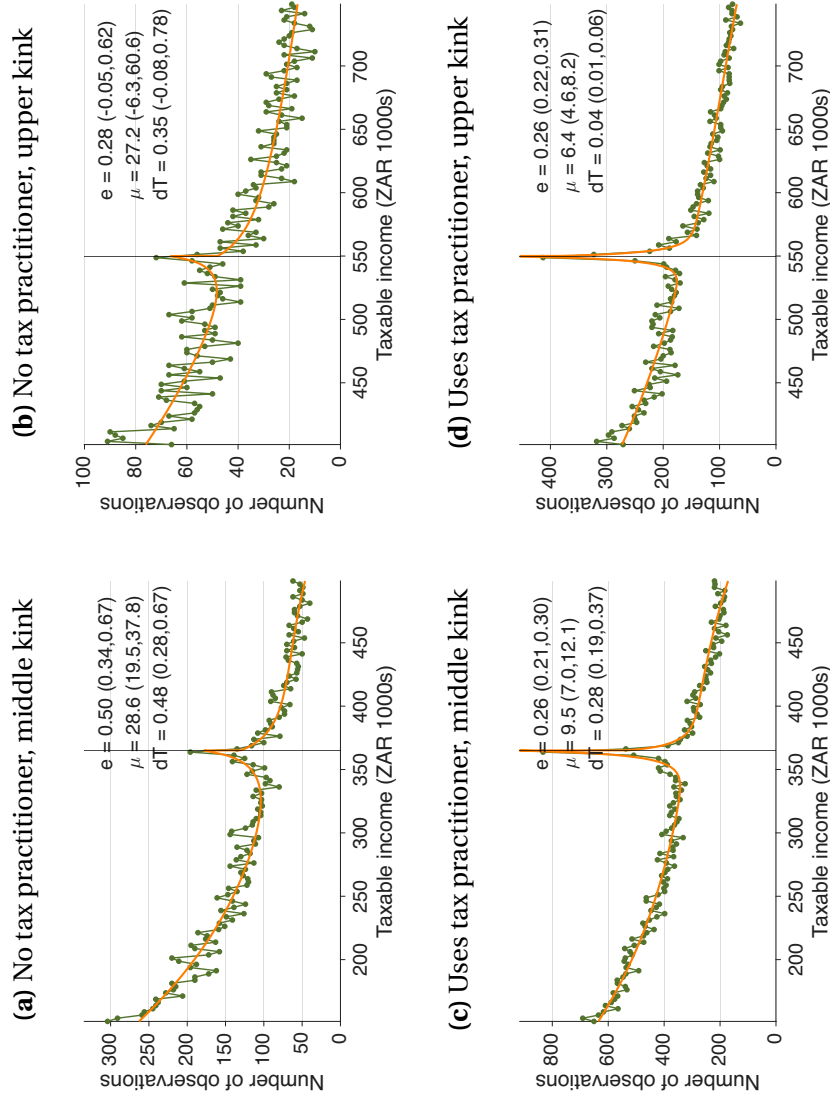
(a) Alternative approaches to constructing counterfactuals



(b) Comparing the elasticity estimates



In panel (a), we illustrate the counterfactuals produced under four different conventional bunching approaches to estimating elasticities for a simulated dataset where $\mu = 10$. In panel (b), we simulate 100 rounds of data using a constant elasticity $e_0 = 0.3$ at each value of the lumpiness parameter μ_0 shown in the plots. We then estimate the elasticity $\hat{e}$ using our estimation approach and four conventional bunching estimators. The vertical lines indicate the 95 percent confidence intervals for the $\hat{e}$ estimates. For the conventional methods, we adapt the automated bunching window approach in Bosch, Dekker and Strohmaier (2020) in order to account for each method's approach to constructing a counterfactual distribution.

Figure A10: Differences in lumpiness by tax practitioner usage, CIT middle/upper kinks

Green points plot the income histograms of South African Small Businesses. In all panels, orange lines plot the best-fit income density under the uniform sparsity model. Figures report parameter estimates e (elasticity of taxable income), μ (average distance between income opportunities in ZAR 1000s), and ΔT (the estimated “as-if” discrete change in tax liability at the bracket threshold, in ZAR 1000s). Numbers in parentheses indicate the 95 percent confidence interval on parameter estimates generated by the MLE method. The sample is split into firms that do not use professional tax practitioners to prepare their tax returns in Panels (a)–(b) and firms that use professional tax practitioners in Panels (c)–(d).